

WILP, BITS Pilani · Mathematical Foundations for Machine Learning (ZC416)

Practice Set

Linear Algebra, Analytic Geometry, Decompositions,
Vector Calculus, Optimisation and Backpropagation

62 exam-style problems across 12 topics · every problem fully solved, step by step

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How to read this document

Each topic opens with a page listing the formulas that topic needs, and then the problems of that topic, one after another. Every problem is laid out the same way:

- **Navy box** — the problem, with the marks for each part.
- **Gold box** — what the question is testing, and the order to attack it in. Read this before the solution and try the problem yourself.
- **Steel STEP badges** — the worked solution, one computation per line; each badge is tagged with the part it answers.
- **Green answer line** — the final result for every part.
- **Grey box** — where marks are actually lost on this kind of problem.
- **Answers at a glance** — the numbers only, for quick self-checking.

Topic map

What each topic drills, and where to find it

#	Topic	What it drills	Problems
1	Linear systems, Gaussian elimination, RREF	REF/RREF, rank, $0/1/\infty$ solutions, free variables, a parameter in the system	1–6
2	Vector spaces, subspaces, independence, basis, dimension	subspace tests and counter-examples, independence, k for dependence, basis of a span and of a solution set, extending a basis	7–15
3	The dot product, norms & inner products	orthogonality and similarity from the dot product, $\ell_1/\ell_2/\ell_\infty$ norms, distances, weighted inner products, projection	16–20
4	Orthogonality, Gram–Schmidt, orthogonal matrices	Gram–Schmidt, orthonormal bases, projection onto a span, $Q^\top Q = I$ preserving lengths and angles	21–23
5	Eigenvalues & eigenvectors	characteristic polynomial, algebraic and geometric multiplicity, eigenvalues in symbols, rotation eigenvalues, $p(A)$, AB vs BA	24–30
6	Eigen decomposition (diagonalization)	$A = PDP^{-1}$ and $A = PDP^\top$, repeated and zero eigenvalues, matrix powers	31–32
7	SVD & singular values	full SVD of a small matrix, singular values from $AA^\top$, rank, spectral and Frobenius norm, SVD against eigendecomposition	33–36
8	Cholesky decomposition	deriving L entry by entry, eigenvalues of L , determinant, solving with L , existence conditions	37–38
9	Positive definiteness & quadratic forms	Sylvester’s criterion with parameters, semidefiniteness, validity of an inner product, sums of squares	39–41
10	Gradient, directional derivative, tangent planes	partial derivatives and gradients, Jacobians, vector gradient identities, directional derivatives, implicit tangent planes, linearization and its error	42–49
11	Taylor expansion, Hessian, critical points	T_2 in one and two variables, the $\frac{1}{2}\mathbf{h}^\top H\mathbf{h}$ form, stationary points, the second-derivative test, reading an expansion backwards	50–57
12	Automatic differentiation & backpropagation	forward pass, local derivatives, backward pass, the fan-out (sum) rule, sigmoid and tanh neurons, one gradient-descent step	58–62

Contents

Topic 1: Linear Systems, Gaussian Elimination & RREF	6
Problem 1: Three unknowns, one answer	6
Problem 2: When the numbers refuse	8
Problem 3: A homogeneous system	9
Problem 4: Rank of a matrix on its own	11
Problem 5: A system with two dials	13
Problem 6: Four unknowns, one parameter: when is the system consistent?	15
Topic 2: Vector Spaces, Subspaces, Linear Independence, Basis & Dimension	18
Problem 7: The plane $x + y + z = 0$	18
Problem 8: The right half-plane is not a subspace	19
Problem 9: Independence of three vectors in $\mathbb{R}^3$	19
Problem 10: The classic dependent triple	20
Problem 11: A non-standard basis of $\mathbb{R}^3$	21
Problem 12: Is this a basis of the plane W ?	22
Problem 13: For which k are three vectors dependent?	23
Problem 14: Basis and dimension of a span	24
Problem 15: A subspace of $\mathbb{R}^4$ from two equations, and extending its basis	25
Topic 3: The Dot Product, Norms & Inner Products	28
Problem 16: The dot product as an orthogonality test	28
Problem 17: The dot product as a similarity score	30
Problem 18: Orthogonal complement, angle and basis in $\mathbb{R}^3$	31
Problem 19: A weighted inner product on $\mathbb{R}^2$	33
Problem 20: Projection in a weighted inner product	34
Topic 4: Orthogonality, Gram–Schmidt & Orthogonal Matrices	36
Problem 21: Gram–Schmidt on three vectors, then verify	36
Problem 22: Gram–Schmidt in $\mathbb{R}^4$ and a projection onto a span	37
Problem 23: An orthogonal matrix: lengths, angles and the rotation it performs	38
Topic 5: Eigenvalues & Eigenvectors	40
Problem 24: Eigen-analysis of a 3×3 matrix	40
Problem 25: Parametric eigen-analysis	41
Problem 26: Is the 3×3 matrix diagonalizable?	42
Problem 27: Powers without brute force: determinant and trace	43
Problem 28: Eigenvalues of a matrix polynomial $p(A)$	44
Problem 29: AB and BA share their eigenvalues	45
Problem 30: Eigenvalues and eigenvectors of a rotation	46
Topic 6: Eigen Decomposition (Diagonalization)	48

Problem 31: Eigendecomposition $A = PDP^{-1}$	48
Problem 32: Eigendecomposition of a rank-deficient symmetric matrix, and a matrix power	49
Topic 7: SVD & Singular Values	51
Problem 33: Full SVD of a 3×2 matrix	51
Problem 34: Full SVD of a 2×3 matrix	52
Problem 35: Capstone: eigendecomposition vs. SVD of one matrix	53
Problem 36: Singular values, rank, norms and the first singular vectors of a 2×3 matrix	54
Topic 8: Cholesky Decomposition	57
Problem 37: Cholesky factor of a 3×3 matrix, and what it buys you	57
Problem 38: When does a Cholesky factor exist? (a parameter)	59
Topic 9: Positive Definiteness & Quadratic Forms	61
Problem 39: Classifying $A(t)$ by eigenvalues	61
Problem 40: Which bilinear forms are inner products?	62
Problem 41: Positive (semi)definiteness with two parameters, and the quadratic form	63
Topic 10: Gradient, Directional Derivative & Tangent Planes	66
Problem 42: Partial derivatives and the gradient from scratch	66
Problem 43: Gradient with exponential and trigonometric terms	67
Problem 44: Jacobian of a vector-valued map	68
Problem 45: Vector gradient identities and a non-symmetric quadratic	70
Problem 46: Linearization (tangent plane) of a two-variable function	71
Problem 47: Gradient, directional derivative and the steepest direction	72
Problem 48: Normal vector and tangent plane of an implicitly defined surface	73
Problem 49: Linearization of a two-variable model and its error	74
Topic 11: Taylor Expansion, Hessian & Critical-Point Classification	76
Problem 50: Second-order Taylor expansion of $e^x \sin y$ at the origin	76
Problem 51: Linear and quadratic estimates of $\sqrt{4.2}$	77
Problem 52: One variable: MVT and a Taylor estimate	78
Problem 53: Second-order Taylor about a non-stationary point	79
Problem 54: A quadratic function: Taylor is exact	81
Problem 55: Four stationary points, all three types	82
Problem 56: A mixed term: determinant route and eigenvalue route	84
Problem 57: Read a Taylor expansion backwards	85
Topic 12: Automatic Differentiation & Backpropagation	87
Problem 58: Pure chain (warm-up)	87
Problem 59: Fan-out in the middle, with a reciprocal	88
Problem 60: Diagnose a recorded backward pass	89
Problem 61: Sigmoid neuron with squared loss and one GD step	91

Problem 62: Tanh neuron with an incoming error signal	92
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Topic 1

Linear Systems, Gaussian Elimination & RREF

Reduce an augmented matrix, state the rank, decide whether there are no solutions, one solution or infinitely many, and write the general solution — with and without a parameter in the system.

Formulas you need for this topic

- Augmented matrix $[A \mid \mathbf{b}]$. Legal row operations: $R_i \leftrightarrow R_j$, $R_i \rightarrow cR_i$ with $c \neq 0$, $R_i \rightarrow R_i + cR_j$.
- rank = number of pivots in any echelon form = number of nonzero rows.
- rank $A < \text{rank}[A \mid \mathbf{b}]$: no solution. rank $A = \text{rank}[A \mid \mathbf{b}] = n$: exactly one. $= r < n$: infinitely many, with $n - r$ free variables.
- rank $A + \text{nullity } A = n$. General solution = particular solution + null-space part.
- A parameter inside A controls the rank; a parameter inside $\mathbf{b}$ controls consistency. Never divide by a parameter without splitting into cases.

Problem 1 · Three unknowns, one answer

Use Gauss–Jordan elimination to solve

$$\begin{aligned}x + y + z &= 6 \\2x - y + z &= 3 \\x + 2y - z &= 2\end{aligned}$$

Confirm your classification with the rank criterion.

How to attempt this (the plan)

What it tests: handling three rows without losing track, and using a row swap to dodge fractions.

Plan: clear column 1 with two operations; swap rows to get a pivot equal to 1; clear column 2 below; read the ranks (classify); then normalise the last pivot and clear upward, column 3 first, then column 2; read the answer and substitute back.

Worked solution

STEP 1 **setup** Augmented matrix. Here $n = 3$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 2 \end{array} \right]$$

STEP 2 **REF** Clear column 1 below the pivot. Two operations, both using R_1 . For $R_2 \rightarrow R_2 - 2R_1$:

$$2 - 2 = 0, \quad -1 - 2 = -3, \quad 1 - 2 = -1, \quad 3 - 12 = -9.$$

For $R_3 \rightarrow R_3 - R_1$:

$$1 - 1 = 0, \quad 2 - 1 = 1, \quad -1 - 1 = -2, \quad 2 - 6 = -4.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & -2 & -4 \end{array} \right]$$

STEP 3 **REF** Swap to get a friendly pivot. Taking -3 as the second pivot would force fractions immediately, so swap $R_2 \leftrightarrow R_3$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & -3 & -1 & -9 \end{array} \right]$$

STEP 4 **REF** Clear below the second pivot with $R_3 \rightarrow R_3 + 3R_2$:

$$-3 + 3(1) = 0, \quad -1 + 3(-2) = -7, \quad -9 + 3(-4) = -21.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & -7 & -21 \end{array} \right]$$

This is REF, with three pivots.

STEP 5 **rank** Classify before finishing: $\text{rank}(A) = 3$, $\text{rank}([A | b]) = 3$, $n = 3$. All equal, so exactly one solution exists.

STEP 6 **RREF** Normalise the third pivot: $R_3 \rightarrow -\frac{1}{7}R_3$ gives $[0 \ 0 \ 1 \mid 3]$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

STEP 7 **RREF** Clear column 3 above the pivot. $R_2 \rightarrow R_2 + 2R_3$:

$$-2 + 2(1) = 0, \quad -4 + 2(3) = 2.$$

Then $R_1 \rightarrow R_1 - R_3$:

$$1 - 1 = 0, \quad 6 - 3 = 3.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

STEP 8 **RREF** Clear column 2 above the pivot with $R_1 \rightarrow R_1 - R_2$:

$$1 - 1 = 0, \quad 3 - 2 = 1.$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

This is the RREF.

STEP 9 **check** Check in the originals:

$$1 + 2 + 3 = 6 \checkmark, \quad 2(1) - 2 + 3 = 3 \checkmark, \quad 1 + 2(2) - 3 = 2 \checkmark.$$

Answer. $x = 1$, $y = 2$, $z = 3$. Unique, since $\text{rank}(A) = \text{rank}([A | b]) = 3 = n$.

Where marks are lost

Going upward in the wrong order: clear column 3 before column 2. If you clean column 2 first, the later column-3 operation puts new entries back into column 2. Doing two operations that

share a source row in one step and losing track of which row was used. Dropping the sign when computing $-1 + 3(-2) = -7$.

Answers at a glance

$x = 1, y = 2, z = 3$; $\text{rank}(A) = \text{rank}([A \mid b]) = 3 = n$, unique.

Problem 2 · When the numbers refuse

Show that the following system has no solution, and identify the exact row of the reduced matrix that proves it.

$$\begin{aligned}x + y + z &= 3 \\2x + 3y + z &= 5 \\3x + 4y + 2z &= 10\end{aligned}$$

How to attempt this (the plan)

What it tests: spotting inconsistency, and saying why in rank language rather than by feel. **Plan:** clear column 1, clear column 2, look for a row $[0 \ 0 \ 0 \mid c]$ with $c \neq 0$; then compare $\text{rank}(A)$ with $\text{rank}([A \mid b])$.

Worked solution

STEP 1 **setup** Augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 3 & 1 & 5 \\ 3 & 4 & 2 & 10 \end{array} \right]$$

STEP 2 **REF** Clear column 1. With $R_2 \rightarrow R_2 - 2R_1$:

$$2 - 2 = 0, \quad 3 - 2 = 1, \quad 1 - 2 = -1, \quad 5 - 6 = -1.$$

With $R_3 \rightarrow R_3 - 3R_1$:

$$3 - 3 = 0, \quad 4 - 3 = 1, \quad 2 - 3 = -1, \quad 10 - 9 = 1.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & -1 & 1 \end{array} \right]$$

Rows 2 and 3 now agree on the left of the bar but disagree on the right.

STEP 3 **REF** Clear below the second pivot with $R_3 \rightarrow R_3 - R_2$:

$$0 - 0 = 0, \quad 1 - 1 = 0, \quad -1 - (-1) = 0, \quad 1 - (-1) = 2.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 2 \end{array} \right]$$

STEP 4 **row** Read the fatal row. Row 3 is $[0 \ 0 \ 0 \mid 2]$. As an equation it says

$$0 \cdot x + 0 \cdot y + 0 \cdot z = 2, \quad \text{that is} \quad 0 = 2,$$

which no choice of x, y, z can make true.

STEP 5 **rank** Say it with ranks. Inside the first three columns there are two pivots, so $\text{rank}(A) = 2$. Across all four columns there are three pivots, because row 3 has its pivot in the b column, so $\text{rank}([A \mid b]) = 3$. Since $\text{rank}(A) = 2 < 3 = \text{rank}([A \mid b])$, the system is inconsistent.

STEP 6 **why** Where it came from. $R_1 + R_2 = [3 \ 4 \ 2]$ on the left, matching row 3 exactly. So the left sides satisfy $\text{eq}_1 + \text{eq}_2 = \text{eq}_3$, but the right sides give $3 + 5 = 8 \neq 10$.

Answer. No solution. The row $[0 \ 0 \ 0 \mid 2]$ proves it; in rank terms $\text{rank}(A) = 2 < \text{rank}([A \mid b]) = 3$.

Where marks are lost

Carrying on to RREF, or trying to write a solution, after a row $[0 \cdots 0 \mid c]$ with $c \neq 0$ has appeared. Saying “inconsistent” without pointing to the row or without naming both ranks. Computing $\text{rank}(A)$ and $\text{rank}([A \mid b])$ from two separate eliminations instead of reading both off one reduced augmented matrix.

Answers at a glance

No solution; row $[0 \ 0 \ 0 \mid 2]$; $\text{rank}(A) = 2 < \text{rank}([A \mid b]) = 3$.

Problem 3 · A homogeneous system

Determine whether the system below has a solution other than $x = y = z = 0$. If it does, find every solution and give a basis for the null space of the coefficient matrix.

$$\begin{aligned}x + 2y + 3z &= 0 \\2x + 5y + 7z &= 0 \\3x + 7y + 10z &= 0\end{aligned}$$

How to attempt this (the plan)

What it tests: whether you realise consistency is automatic here, so the only real question is r versus n . **Plan:** reduce A (the zero column stays zero); read r ; if $r < n$ there are non-trivial solutions; go to RREF, name the free variable, write the solution as $t\mathbf{v}$ and quote $\mathbf{v}$ as the basis; check rank–nullity.

Worked solution

STEP 1 **setup** Augmented matrix, with a zero column on the right.

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 5 & 7 & 0 \\ 3 & 7 & 10 & 0 \end{array} \right]$$

That last column stays zero through every row operation, so we may track only A and remember the 0.

STEP 2 **REF** Clear column 1. $R_2 \rightarrow R_2 - 2R_1$:

$$2 - 2 = 0, \quad 5 - 4 = 1, \quad 7 - 6 = 1.$$

$$R_3 \rightarrow R_3 - 3R_1:$$

$$3 - 3 = 0, \quad 7 - 6 = 1, \quad 10 - 9 = 1.$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right]$$

STEP 3 **REF** Clear below the second pivot. $R_3 \rightarrow R_3 - R_2$ gives a zero row:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

STEP 4 **rank** Classify. $\text{rank}(A) = 2$ and $\text{rank}([A \mid \mathbf{0}]) = 2$; they are equal automatically, so the system is consistent. Since $r = 2 < 3 = n$, there is $3 - 2 = 1$ free variable, hence infinitely many solutions. So non-trivial solutions do exist.

STEP 5 **RREF** Reduce to RREF. Clear column 2 above the pivot with $R_1 \rightarrow R_1 - 2R_2$:

$$1 - 0 = 1, \quad 2 - 2 = 0, \quad 3 - 2 = 1.$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

STEP 6 **solve** Solve. Pivots in columns 1 and 2, so x and y are basic and z is free. Set $z = t$. Row 1 gives $x + z = 0$, so $x = -t$. Row 2 gives $y + z = 0$, so $y = -t$.

STEP 7 **vector** Vector form:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

There is no constant vector out front. A homogeneous system always has $\mathbf{x}_p = \mathbf{0}$, so the answer is pure null space.

STEP 8 **check** Check. Take $t = 1$, giving $(-1, -1, 1)$:

$$-1 + 2(-1) + 3(1) = 0 \checkmark, \quad -2 - 5 + 7 = 0 \checkmark, \quad -3 - 7 + 10 = 0 \checkmark.$$

STEP 9 **check** Rank-nullity check: $\text{rank}(A) = 2$, $\text{nullity}(A) = 1$, and $2 + 1 = 3 = n \checkmark$.

Answer. Non-trivial solutions exist. Every solution is $t(-1, -1, 1)$, $t \in \mathbb{R}$, and a basis for the null space is $\{(-1, -1, 1)^\top\}$.

Where marks are lost

Writing the null space as “ $(-1, -1, 1)$ ” alone: the null space is the whole set $\{t(-1, -1, 1) : t \in \mathbb{R}\}$, and the single vector is a basis for it. Forgetting that the trivial solution is still included ($t = 0$). Wasting time checking consistency: a homogeneous system is never inconsistent.

Answers at a glance

Non-trivial solutions exist; solutions $t(-1, -1, 1)$; null-space basis $\{(-1, -1, 1)^\top\}$; rank = 2, nullity = 1.

Problem 4 · Rank of a matrix on its own

For the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 3 & 5 & 7 \\ 2 & 5 & 8 & 11 \end{bmatrix}$$

- (a) find $\text{rank}(A)$; [2]
- (b) decide whether $A\mathbf{x} = \mathbf{0}$ has a non-trivial solution, and if so find a basis for the null space; [2]
- (c) verify the rank–nullity theorem. [1]

How to attempt this (the plan)

What it tests: rank as a property of a matrix without any b attached, plus the link from rank straight to the null space. **Plan:** (a) clear column 1 with three operations, move the zero row down, clear the last row, count pivots. (b) compare r with $n = 4$; go to RREF, set the two free variables to s, t , and split the solution into two vectors. (c) add rank + nullity.

Worked solution

STEP 1 (a) Look before you compute. Row 2 is exactly $2 \times$ row 1. Also row 1 plus row 3 is $[2 \ 5 \ 8 \ 11]$, which is row 4. So two of the four rows are combinations of the others and the rank should come out as 2.

STEP 2 (a) Clear column 1. Three operations, all using R_1 :

$$R_2 \rightarrow R_2 - 2R_1: [2 - 2, 4 - 4, 6 - 6, 8 - 8] = [0 \ 0 \ 0 \ 0]$$

$$R_3 \rightarrow R_3 - R_1: [1 - 1, 3 - 2, 5 - 3, 7 - 4] = [0 \ 1 \ 2 \ 3]$$

$$R_4 \rightarrow R_4 - 2R_1: [2 - 2, 5 - 4, 8 - 6, 11 - 8] = [0 \ 1 \ 2 \ 3]$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

STEP 3 (a) Move the zero row down. REF requires zero rows at the bottom, so swap $R_2 \leftrightarrow R_3$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

STEP 4 (a) Clear the last row. $R_4 \rightarrow R_4 - R_2$ gives another zero row:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This is REF. Two nonzero rows, so two pivots, in columns 1 and 2. Hence $\text{rank}(A) = 2$.

STEP 5 (b) Decide on non-trivial solutions. Here $n = 4$ and $r = 2$, so $n - r = 2 > 0$. A homogeneous system with rank below the number of unknowns always has non-trivial solutions. So yes.

STEP 6 (b) Reduce to RREF. Both pivots are already 1. Clear column 2 above with $R_1 \rightarrow R_1 - 2R_2$:

$$1 - 0 = 1, \quad 2 - 2 = 0, \quad 3 - 4 = -1, \quad 4 - 6 = -2.$$

$$\begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

STEP 7 (b) Write the equations. Pivot columns 1 and 2, so x_1, x_2 are basic; columns 3 and 4 have no pivots, so x_3, x_4 are free:

$$x_1 - x_3 - 2x_4 = 0, \quad x_2 + 2x_3 + 3x_4 = 0.$$

STEP 8 (b) Introduce parameters. Put $x_3 = s$ and $x_4 = t$. Then

$$x_1 = s + 2t, \quad x_2 = -2s - 3t.$$

STEP 9 (b) Split into two vectors, grouping the s terms and the t terms:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = s \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}.$$

Each vector comes from setting one free variable to 1 and the other to 0; the 1/0 pattern in the bottom two slots also proves the two vectors are independent.

STEP 10 (b) Check both against row 4 of the original, $[2 \ 5 \ 8 \ 11]$:

$$2(1) + 5(-2) + 8(1) + 11(0) = 2 - 10 + 8 = 0 \checkmark$$

$$2(2) + 5(-3) + 8(0) + 11(1) = 4 - 15 + 11 = 0 \checkmark$$

STEP 11 (c) Rank-nullity. $\text{nullity}(A) = 2$ because the basis has two vectors, and

$$\text{rank}(A) + \text{nullity}(A) = 2 + 2 = 4 = n \checkmark.$$

Answer. $\text{rank}(A) = 2$. Non-trivial solutions exist; a null-space basis is $\{(1, -2, 1, 0)^\top, (2, -3, 0, 1)^\top\}$, so $\text{nullity}(A) = 2$ and $2 + 2 = 4$.

Where marks are lost

Counting nonzero entries, or nonzero rows of the original matrix, instead of pivots in echelon form. Leaving the zero row in the middle and calling the matrix REF. Writing only one null-space vector when there are two free variables. Sign slips in $3 - 4 = -1$ and $4 - 6 = -2$ when clearing above the pivot.

Answers at a glance

(a) $\text{rank}(A) = 2$ (b) yes; basis $\{(1, -2, 1, 0)^\top, (2, -3, 0, 1)^\top\}$ (c) $2 + 2 = 4 = n$.

Problem 5 · A system with two dials

Consider the system

$$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 14 \\x + 2y + az &= b\end{aligned}$$

where a and b are real constants. Find all values of a and b for which the system has

- (a) a unique solution, and give the solution in terms of a and b ; [3]
- (b) no solution; [1]
- (c) infinitely many solutions, and give the general solution. [2]

How to attempt this (the plan)

What it tests: eliminating while a pivot might secretly be zero. The classic way to lose marks is to divide by $a - 3$ without checking whether it is zero. **Plan:** reduce with the safe pivots (both equal to 1) until the whole problem sits in one row $[0 \ 0 \ a - 3 \mid b - 14]$; then split into $a \neq 3$ (back substitute), $a = 3$ with $b \neq 14$ (contradiction), $a = 3$ with $b = 14$ (one free variable, go to RREF).

Worked solution

STEP 1 **setup** Augmented matrix. Here $n = 3$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 1 & 2 & a & b \end{array} \right]$$

STEP 2 **REF** Clear column 1. The pivot is a plain 1 and does not depend on a or b , so this step is always safe. $R_2 \rightarrow R_2 - R_1$:

$$1 - 1 = 0, \quad 2 - 1 = 1, \quad 3 - 1 = 2, \quad 14 - 6 = 8.$$

$R_3 \rightarrow R_3 - R_1$:

$$1 - 1 = 0, \quad 2 - 1 = 1, \quad a - 1, \quad b - 6.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 1 & a - 1 & b - 6 \end{array} \right]$$

STEP 3 **REF** Clear below the second pivot. The second pivot is 1, again free of a and b , so $R_3 \rightarrow R_3 - R_2$ is safe:

$$1 - 1 = 0, \quad (a - 1) - 2 = a - 3, \quad (b - 6) - 8 = b - 14.$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & a-3 & b-14 \end{array} \right]$$

STEP 4 (row) Stop and look at row 3. Everything depends on the single row

$$[0 \ 0 \ a-3 \mid b-14], \quad \text{i.e.} \quad (a-3)z = b-14.$$

Whether this row carries a pivot, a contradiction, or vanishes decides the whole problem. Do not divide by $a-3$ yet.

STEP 5 (a) Case $a \neq 3$. Then $a-3 \neq 0$, so row 3 has a pivot in column 3. Three pivots in the first three columns give $\text{rank}(A) = 3$, none in the b column, so $\text{rank}([A \mid b]) = 3$. With $3 = 3 = n$ the solution is unique, whatever b is. Now division is legal:

$$z = \frac{b-14}{a-3}.$$

STEP 6 (a) Back substitute into row 2, $y + 2z = 8$:

$$y = 8 - 2z = 8 - \frac{2(b-14)}{a-3}.$$

STEP 7 (a) Back substitute into row 1, $x + y + z = 6$:

$$\begin{aligned} x &= 6 - y - z \\ &= 6 - \left(8 - \frac{2(b-14)}{a-3}\right) - \frac{b-14}{a-3} \\ &= -2 + \frac{2(b-14)}{a-3} - \frac{b-14}{a-3} \\ &= -2 + \frac{b-14}{a-3}. \end{aligned}$$

STEP 8 (a) Spot-check with numbers. Take $a = 4, b = 15$. Then $z = 1/1 = 1, y = 8 - 2 = 6, x = -2 + 1 = -1$. Test all three originals:

$$-1 + 6 + 1 = 6 \checkmark, \quad -1 + 12 + 3 = 14 \checkmark, \quad -1 + 12 + 4 = 15 \checkmark.$$

STEP 9 (b) Case $a = 3$ and $b \neq 14$. Row 3 becomes $[0 \ 0 \ 0 \mid b-14]$ with $b-14 \neq 0$. That is the equation $0 = b-14$, which is false. Column 3 has lost its pivot, so $\text{rank}(A) = 2$, while the b column has gained one, so $\text{rank}([A \mid b]) = 3$. Since $2 < 3$, there is no solution.

STEP 10 (c) Case $a = 3$ and $b = 14$. Row 3 becomes $[0 \ 0 \ 0 \mid 0]$, which is harmless. Now $\text{rank}(A) = \text{rank}([A \mid b]) = 2$, and $2 < 3 = n$, so there are infinitely many solutions with $3 - 2 = 1$ free variable.

STEP 11 (c) Solve this case. Clear column 2 above the pivot with $R_1 \rightarrow R_1 - R_2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Pivots in columns 1 and 2, so z is free. Put $z = t$. Row 1 gives $x - z = -2$, so $x = -2 + t$. Row 2 gives $y + 2z = 8$, so $y = 8 - 2t$.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 8 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}.$$

STEP 12 (c) Check symbolically. With $a = 3$, $b = 14$ the third equation is identical to the second, so only two need checking:

$$\begin{aligned}(-2 + t) + (8 - 2t) + t &= 6 + (t - 2t + t) = 6 \checkmark \\(-2 + t) + 2(8 - 2t) + 3t &= 14 + (t - 4t + 3t) = 14 \checkmark\end{aligned}$$

Answer. (a) Unique when $a \neq 3$ (any b): $z = \frac{b-14}{a-3}$, $y = 8 - \frac{2(b-14)}{a-3}$, $x = -2 + \frac{b-14}{a-3}$.
(b) No solution when $a = 3$, $b \neq 14$. (c) Infinitely many when $a = 3$, $b = 14$: $(x, y, z) = (-2, 8, 0) + t(1, -2, 1)$, $t \in \mathbb{R}$.

Where marks are lost

Dividing by $a - 3$ before saying $a \neq 3$, and then never treating $a = 3$ as its own case. Forgetting that b only matters on the knife edge $a = 3$: a (left of the bar) controls $\text{rank}(A)$, b (right of the bar) controls consistency. Giving “no solution” for all $a = 3$ and missing the $b = 14$ case. In (a), leaving x as an unsimplified expression with two fractions.

Answers at a glance

(a) $a \neq 3$: $z = \frac{b-14}{a-3}$, $y = 8 - \frac{2(b-14)}{a-3}$, $x = -2 + \frac{b-14}{a-3}$ (b) $a = 3$, $b \neq 14$ (c) $a = 3$, $b = 14$: $(-2, 8, 0) + t(1, -2, 1)$.

Problem 6 · Four unknowns, one parameter: when is the system consistent?

Consider the system of linear equations

$$\begin{aligned}x + y + z + w &= 3, \\x + 2y + 3z + 4w &= 5, \\2x + 3y + 4z + 5w &= 8, \\3x + 5y + 7z + 9w &= k,\end{aligned}$$

where $k \in \mathbb{R}$.

- Find all values of k for which the system is consistent. [1.5]
- For each such k , state the rank of the coefficient matrix and the number of free variables. [1]
- Write the general solution in parametric (vector) form. [1.5]

How to attempt this (the plan)

What it tests: row reduction with a symbol in the right-hand column, reading consistency from a $[0\ 0\ 0\ 0\ | \ c]$ row, and counting free variables as $n - \text{rank}$. **Plan:** reduce the augmented matrix once, keeping k untouched; the last row decides (a); count pivots for (b); go to RREF, name the free variables, write the answer for (c).

Worked solution

STEP 1 (a) Write the augmented matrix ($n = 4$ unknowns).

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 3 \\ 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 8 \\ 3 & 5 & 7 & 9 & k \end{array} \right]$$

STEP 2 (a) Clear column 1 using R_1 (three operations, one at a time).

$$R_2 \rightarrow R_2 - R_1: \quad [0 \ 1 \ 2 \ 3 \ | \ 2]$$

$$R_3 \rightarrow R_3 - 2R_1: \quad [0 \ 1 \ 2 \ 3 \ | \ 2]$$

$$R_4 \rightarrow R_4 - 3R_1: \quad [0 \ 2 \ 4 \ 6 \ | \ k - 9]$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 3 & 2 \\ 0 & 2 & 4 & 6 & k - 9 \end{array} \right]$$

STEP 3 (a) Clear column 2 using R_2 .

$$R_3 \rightarrow R_3 - R_2: \quad [0 \ 0 \ 0 \ 0 \ | \ 0]$$

$$R_4 \rightarrow R_4 - 2R_2: \quad [0 \ 0 \ 0 \ 0 \ | \ k - 13]$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & k - 13 \end{array} \right]$$

STEP 4 (a) Read the last row: it says $0 = k - 13$. This is true only when $k = 13$. For $k \neq 13$ the row is $[0 \ 0 \ 0 \ 0 \ | \ c]$ with $c \neq 0$, so $\text{rank}[A \ | \ b] = 3 > 2 = \text{rank } A$ and there is no solution.

STEP 5 (b) For $k = 13$: two pivots (columns 1 and 2), so $\text{rank } A = \text{rank}[A \ | \ b] = 2$. Free variables: $n - r = 4 - 2 = 2$ (namely z and w , the pivot-free columns).

STEP 6 (c) Go to RREF with $R_1 \rightarrow R_1 - R_2$:

$$[1 \ 1 \ 1 \ 1 \ | \ 3] - [0 \ 1 \ 2 \ 3 \ | \ 2] = [1 \ 0 \ -1 \ -2 \ | \ 1].$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & -2 & 1 \\ 0 & 1 & 2 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

STEP 7 (c) Set $z = s$, $w = t$. Row 1: $x - z - 2w = 1 \Rightarrow x = 1 + s + 2t$. Row 2: $y + 2z + 3w = 2 \Rightarrow y = 2 - 2s - 3t$.

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}, \quad s, t \in \mathbb{R}.$$

STEP 8 (c) Check in equation 4 with $k = 13$: $3(1 + s + 2t) + 5(2 - 2s - 3t) + 7s + 9t = 3 + 10 + (3 - 10 + 7)s + (6 - 15 + 9)t = 13$. ✓

Answer. (a) Consistent iff $k = 13$. (b) $\text{rank } A = 2$, two free variables. (c) $(x, y, z, w) = (1 + s + 2t, 2 - 2s - 3t, s, t), s, t \in \mathbb{R}$.

Where marks are lost

Reducing the coefficient matrix and the augmented matrix separately (do it once, read both ranks off the same picture). Writing " $k = 13$ gives infinitely many solutions" without saying *why* ($r = 2 < 4$). Counting free variables as $n - m = 0$ instead of $n - r = 2$. Dropping the zero rows before you have finished reading the ranks.

Answers at a glance

(a) $k = 13$ only (b) $\text{rank } A = 2$, 2 free variables (c) $(1, 2, 0, 0) + s(1, -2, 1, 0) + t(2, -3, 0, 1)$

Topic 2

Vector Spaces, Subspaces, Linear Independence, Basis & Dimension

Decide whether a set is a subspace, with a reason or a counter-example; test independence and find the parameter value that makes vectors dependent; find a basis and the dimension of a span or of a solution set, and extend a basis to the whole space.

Formulas you need for this topic

- Subspace test: $\mathbf{0} \in W$, and W closed under addition and scalar multiplication (one failure is enough to disprove).
- Independent $\iff c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = \mathbf{0}$ forces every $c_i = 0$.
- n vectors in $\mathbb{R}^n$: independent $\iff \det[\mathbf{v}_1 \dots \mathbf{v}_n] \neq 0$.
- Basis = independent + spanning; $\dim W$ = number of basis vectors. In a d -dimensional space, d independent vectors are automatically a basis.
- Basis of a span: row-reduce $[\mathbf{v}_1 \dots \mathbf{v}_k]$ and keep the *original* vectors sitting in pivot columns.
- Subspace given by equations: solve, then one basis vector per free variable; $\dim = n - \text{rank}$.

Problem 7 · The plane $x + y + z = 0$

Show that $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ is a subspace of $\mathbb{R}^3$.

How to attempt this (the plan)

What it tests: the Subspace Test for a plane through the origin, given by one linear homogeneous equation. **Plan:** check the zero vector; add two generic elements and regroup; scale a generic element and factor out c .

Worked solution

STEP 1 **zero** $0 + 0 + 0 = 0$, so $(0, 0, 0) \in W$. ✓

STEP 2 **add** Take $(x_1, y_1, z_1), (x_2, y_2, z_2) \in W$, so $x_1 + y_1 + z_1 = 0$ and $x_2 + y_2 + z_2 = 0$. The sum is $(x_1 + x_2, y_1 + y_2, z_1 + z_2)$ and

$$(x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = (x_1 + y_1 + z_1) + (x_2 + y_2 + z_2) = 0 + 0 = 0.$$

So the sum is in W . ✓

STEP 3 **scale** Take $c \in \mathbb{R}$ and $(x, y, z) \in W$. Then $c(x, y, z) = (cx, cy, cz)$ and

$$cx + cy + cz = c(x + y + z) = c \cdot 0 = 0.$$

So $c(x, y, z) \in W$. ✓

STEP 4 **conclude** All three tests pass, so W is a subspace of $\mathbb{R}^3$. Geometrically it is a plane through the origin (with normal vector $(1, 1, 1)$).

Answer. W is a subspace of $\mathbb{R}^3$: a plane through the origin.

Where marks are lost

Forgetting the zero-vector check. Regrouping the sum without saying *why* each bracket is 0 (the membership hypothesis). Claiming closure by picking two particular vectors.

Answers at a glance

Subspace (plane through the origin).

Problem 8 · The right half-plane is not a subspace

Show that $W = \{(x, y) \in \mathbb{R}^2 : x \geq 0\}$ is *not* a subspace of $\mathbb{R}^2$.

How to attempt this (the plan)

What it tests: disproving a subspace claim with one concrete counterexample. **Plan:** the condition is an inequality, so try a negative scalar: pick a vector in W , multiply by -1 , and show the result leaves W .

Worked solution

STEP 1 **pick** Take $u = (2, 3)$. Since $2 \geq 0$, $u \in W$.

STEP 2 **scale** Take $c = -1$:

$$cu = (-1)(2, 3) = (-2, -3).$$

STEP 3 **check** The first coordinate is -2 , and $-2 \geq 0$ is false. So $cu \notin W$. ✗

STEP 4 **conclude** W is not closed under scalar multiplication, so W is not a subspace. (It does contain $(0, 0)$ and is closed under addition; one failure is enough.)

Answer. Not a subspace: $(2, 3) \in W$ but $(-1)(2, 3) = (-2, -3) \notin W$.

Where marks are lost

Trying to prove “not a subspace” with a general argument instead of a concrete counterexample: one explicit vector and one explicit scalar earns the marks. Testing only positive scalars and wrongly concluding that closure holds. Checking $0 \in W$ and stopping: that test passes here, so it decides nothing.

Answers at a glance

Not a subspace; fails scalar closure with $c = -1$.

Problem 9 · Independence of three vectors in $\mathbb{R}^3$

Are the vectors $(1, 0, 1)$, $(0, 1, 1)$, $(1, 1, 0)$ linearly independent in $\mathbb{R}^3$?

How to attempt this (the plan)

What it tests: the homogeneous-system test for independence, plus the REF/rank and determinant shortcuts. **Plan:** set $av_1 + bv_2 + cv_3 = 0$, write three equations, show only $a = b = c = 0$ works; confirm by row-reducing the matrix of rows to REF and counting non-zero rows.

Worked solution

STEP 1 **set up** $a(1, 0, 1) + b(0, 1, 1) + c(1, 1, 0) = (0, 0, 0)$ gives

$$a + c = 0, \quad b + c = 0, \quad a + b = 0.$$

STEP 2 **solve** From the first two equations, $a = -c$ and $b = -c$. Substitute into the third:

$$-c + (-c) = 0 \Rightarrow -2c = 0 \Rightarrow c = 0,$$

hence $a = b = 0$ as well. Only the trivial solution exists.

STEP 3 **REF cross-check** Vectors as rows:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

Three non-zero rows, so rank = 3 = number of vectors.

STEP 4 **determinant** $\det \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = 1(0 - 1) - 0 + 1(0 - 1) = -2 \neq 0$, confirming independence.

Answer. Linearly independent (only the trivial solution; rank 3; $\det = -2 \neq 0$).

Where marks are lost

Checking only that no vector is a multiple of another; for three or more vectors that is not enough. Stopping at $a = -c$, $b = -c$ without using the third equation. Row-reducing incorrectly and getting a zero row (a sign slip in R_3 is the usual cause).

Answers at a glance

Independent; rank = 3, $\det = -2$.

Problem 10 · The classic dependent triple

Determine whether $\{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$ is linearly independent in $\mathbb{R}^3$.

How to attempt this (the plan)

What it tests: detecting dependence via a free variable, reading rank from REF, and extracting the explicit dependency relation. **Plan:** write the homogeneous system, eliminate, notice two equations coincide, set the free variable $c = t$ and produce a non-trivial solution; verify it; confirm with REF (rank $2 < 3$) and $\det = 0$.

Worked solution

STEP 1 **set up** $a(1, 2, 3) + b(4, 5, 6) + c(7, 8, 9) = 0$ gives

$$a + 4b + 7c = 0, \quad 2a + 5b + 8c = 0, \quad 3a + 6b + 9c = 0.$$

STEP 2 **eliminate a**

$$R_2 \rightarrow R_2 - 2R_1 : \quad -3b - 6c = 0 \Rightarrow b + 2c = 0,$$

$$R_3 \rightarrow R_3 - 3R_1 : \quad -6b - 12c = 0 \Rightarrow b + 2c = 0 \quad (\text{same equation}).$$

STEP 3 **free variable** Only two independent equations remain, so one unknown is free. Let $c = t$. Then $b = -2t$ and

$$a = -4b - 7c = -4(-2t) - 7t = 8t - 7t = t.$$

With $t = 1$: $(a, b, c) = (1, -2, 1)$.

STEP 4 **verify**

$$1(1, 2, 3) - 2(4, 5, 6) + 1(7, 8, 9) = (1, 2, 3) - (8, 10, 12) + (7, 8, 9) = (0, 0, 0).$$

The relation holds. ✓ A non-trivial combination equals 0, so the set is linearly dependent.

STEP 5 **REF cross-check** Vectors as rows:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 4R_1, R_3 \rightarrow R_3 - 7R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

Only 2 non-zero rows: rank = 2 < 3, hence dependent.

STEP 6 **relation from REF** Track the operations on the third row: $R'_3 = v_3 - 7v_1$, then $R''_3 = R'_3 - 2(v_2 - 4v_1) = v_3 - 7v_1 - 2v_2 + 8v_1 = v_1 - 2v_2 + v_3$. Setting this zero row to 0 gives $v_1 - 2v_2 + v_3 = 0$, the same relation as Step 4. Also $\det = 0$, as it must be.

Answer. Linearly dependent: $v_1 - 2v_2 + v_3 = 0$, i.e. $(1, 2, 3) - 2(4, 5, 6) + (7, 8, 9) = 0$.

Where marks are lost

Stopping at “two equations are the same” without producing an explicit non-trivial solution; the marks are for the relation $(1, -2, 1)$. Sign error in $a = -4b - 7c$ with $b = -2t$. Concluding “independent” from rank = 2 because two rows are non-zero: compare the rank with the *number of vectors*, 3.

Answers at a glance

Dependent; $(a, b, c) = (1, -2, 1)$; rank = 2; $\det = 0$.

Problem 11 · A non-standard basis of $\mathbb{R}^3$

Show that $B = \{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ is a basis of $\mathbb{R}^3$.

How to attempt this (the plan)

What it tests: the counting shortcut: 3 independent vectors in the 3-dimensional space $\mathbb{R}^3$ automatically span it. **Plan:** prove independence by solving the homogeneous system (or REF / determinant); then invoke the size argument.

Worked solution

STEP 1 **set up** $a(1, 1, 0) + b(1, 0, 1) + c(0, 1, 1) = (0, 0, 0)$ gives

$$a + b = 0, \quad a + c = 0, \quad b + c = 0.$$

STEP 2 **solve** $b = -a$ and $c = -a$; substitute into $b + c = 0$:

$$-2a = 0 \Rightarrow a = 0, \quad \text{so } b = c = 0.$$

Only the trivial solution: B is linearly independent.

STEP 3 **REF cross-check** Vectors as rows:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Three non-zero rows, so rank = 3. Equivalently $\det = -2 \neq 0$.

STEP 4 **size argument** $\dim \mathbb{R}^3 = 3$ and B is an independent set of 3 vectors, so B also spans $\mathbb{R}^3$. Hence B is a basis.

Answer. B is a basis of $\mathbb{R}^3$ (independent, and 3 vectors in a 3-dimensional space).

Where marks are lost

Proving independence and forgetting to say why spanning follows (the size argument must be stated). Proving spanning by solving $a(1, 1, 0) + b(1, 0, 1) + c(0, 1, 1) = (x, y, z)$ in general: correct but slow, and errors creep in. Sign slip in $R_3 \rightarrow R_3 + R_2$.

Answers at a glance

Basis: independent ($\det = -2$), size $3 = \dim \mathbb{R}^3$.

Problem 12 · Is this a basis of the plane W ?

Let $W = \{(x, y, z) \in \mathbb{R}^3 : x - 2y - z = 0\}$.

(a) Show that $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$ is a basis of W . [2]

(b) Verify that $\mathbf{p} = (3, 1, 1)^\top$ lies in W and write it in this basis. [1]

How to attempt this (the plan)

What it tests: the three things a basis needs — the vectors are *in* W , they are independent, and there are $\dim W$ of them. **Plan:** substitute each vector into the equation; check non-

proportionality; get $\dim W$ from rank–nullity; then solve a 2-unknown system for (b).

Worked solution

STEP 1 (a) Both vectors satisfy $x - 2y - z = 0$:

$$(1, 0, 1) : 1 - 0 - 1 = 0 \checkmark \quad (2, 1, 0) : 2 - 2 - 0 = 0 \checkmark$$

So both lie in W .

STEP 2 (a) They are linearly independent: neither is a scalar multiple of the other (a multiple of $(1, 0, 1)$ has middle entry 0, but $(2, 1, 0)$ has middle entry 1).

STEP 3 (a) $\dim W = 2$: W is the null space of the 1×3 matrix $\begin{bmatrix} 1 & -2 & -1 \end{bmatrix}$, which has rank 1, so $\dim W = 3 - 1 = 2$ by rank–nullity.

STEP 4 (a) Two independent vectors inside a 2-dimensional space automatically span it. Hence the set is a basis of W .

STEP 5 (b) Membership: $3 - 2(1) - 1 = 0$. $\checkmark$ Now solve $a(1, 0, 1) + b(2, 1, 0) = (3, 1, 1)$:

$$\text{2nd coordinate: } b = 1.$$

$$\text{3rd coordinate: } a = 1.$$

$$\text{1st coordinate check: } a + 2b = 1 + 2 = 3. \checkmark$$

Answer. (a) Both vectors are in W , they are independent, and $\dim W = 2$, so they form a basis. (b) $\mathbf{p} = 1 \cdot (1, 0, 1) + 1 \cdot (2, 1, 0)$, i.e. coordinates $(1, 1)$.

Where marks are lost

Checking independence but forgetting to check the vectors actually satisfy the equation of W . Claiming "they span W " with no argument — the dimension count is the argument. In (b), solving only two coordinates and not checking the third.

Answers at a glance

(a) basis (in W + independent + $\dim W = 2$) (b) $\mathbf{p} = (1, 0, 1) + (2, 1, 0)$, coordinates $(1, 1)$

Problem 13 · For which k are three vectors dependent?

Consider

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ k \\ 5 \end{bmatrix}.$$

- (a) Find all k for which $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent. [1.5]
- (b) For that k , write $\mathbf{v}_3$ as a combination of $\mathbf{v}_1$ and $\mathbf{v}_2$. [1]
- (c) State $\text{rank}[\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]$ for every real k . [0.5]

How to attempt this (the plan)

What it tests: "three vectors in $\mathbb{R}^3$ are dependent $\iff$ the 3×3 determinant is 0", and turning a dependence into an explicit relation. **Plan:** expand the determinant along a row containing k ; solve; then solve $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{v}_3$ for (b); rank is 3 or 2 for (c).

Worked solution

STEP 1 (a) Put the vectors as columns and expand the determinant along the first row:

$$D = \det \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & k \\ 2 & 1 & 5 \end{bmatrix} = 1 \cdot (3 \cdot 5 - k \cdot 1) - 2 \cdot (1 \cdot 5 - k \cdot 2) + 1 \cdot (1 \cdot 1 - 3 \cdot 2).$$

$$D = (15 - k) - 2(5 - 2k) + (1 - 6) = 15 - k - 10 + 4k - 5 = 3k.$$

STEP 2 (a) Dependent $\iff D = 0 \iff 3k = 0 \iff k = 0$.

STEP 3 (b) With $k = 0$, solve $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{v}_3 = (1, 0, 5)^\top$:

$$a + 2b = 1, \quad a + 3b = 0, \quad 2a + b = 5.$$

Subtract the first two: $b = -1$. Then $a = 1 - 2b = 3$.

Check the third: $2(3) + (-1) = 5$. ✓

So $\mathbf{v}_3 = 3\mathbf{v}_1 - \mathbf{v}_2$.

STEP 4 (c) For $k \neq 0$ the determinant is nonzero, so the three columns are independent and the rank is 3. For $k = 0$ the rank drops to 2 (two independent columns $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ in their span).

Answer. (a) $k = 0$. (b) $\mathbf{v}_3 = 3\mathbf{v}_1 - \mathbf{v}_2$. (c) rank = 3 for $k \neq 0$, rank = 2 for $k = 0$.

Where marks are lost

Sign slips in the cofactor expansion (the middle term carries a minus). Stopping at " $k = 0$ " in (b) without producing the actual relation. Saying "rank = 2" for all k — the rank is 3 except at the special value.

Answers at a glance

(a) $k = 0$ (b) $\mathbf{v}_3 = 3\mathbf{v}_1 - \mathbf{v}_2$ (c) rank 3 if $k \neq 0$, rank 2 if $k = 0$

Problem 14 · Basis and dimension of a span

Let

$$W = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix} \right\} \subseteq \mathbb{R}^3.$$

- (a) Find a basis of W chosen from the given vectors. [2]
 (b) State $\dim W$. [0.5]
 (c) Decide whether $(0, 1, 1)^\top \in W$ and, if so, give its coordinates in your basis. [0.5]

How to attempt this (the plan)

What it tests: spotting redundant vectors and confirming with row reduction; pivot columns give the basis. **Plan:** look for obvious multiples/sums first; reduce the matrix whose columns are the vectors; keep the original vectors in pivot columns.

Worked solution

STEP 1 (a) Call the vectors $\mathbf{v}_1, \dots, \mathbf{v}_4$. Look first: $\mathbf{v}_2 = 2\mathbf{v}_1$, and

$$2\mathbf{v}_1 + \mathbf{v}_3 = (2 + 1, 2 + 0, 4 + 1) = (3, 2, 5) = \mathbf{v}_4.$$

So $\mathbf{v}_2$ and $\mathbf{v}_4$ are built from $\mathbf{v}_1$ and $\mathbf{v}_3$.

STEP 2 (a) Confirm by row reduction of $[\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \ \mathbf{v}_4]$:

$$\begin{bmatrix} 1 & 2 & 1 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 1 & 5 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1, \ R_3 \rightarrow R_3 - 2R_1} \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -1 & -1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Pivots in columns 1 and 3.

STEP 3 (a) The original vectors in the pivot columns form a basis:

$$\text{basis of } W = \{(1, 1, 2)^\top, (1, 0, 1)^\top\}.$$

They are independent (not proportional), and steps 1–2 show they span W .

STEP 4 (b) $\dim W = 2$ (two pivots). W is a plane through the origin in $\mathbb{R}^3$.

STEP 5 (c) Solve $a(1, 1, 2) + b(1, 0, 1) = (0, 1, 1)$: second coordinate gives $a = 1$; first gives $1 + b = 0 \Rightarrow b = -1$; third check: $2(1) + (-1) = 1$. ✓ So $(0, 1, 1) \in W$ with coordinates $(1, -1)$.

Answer. (a) $\{(1, 1, 2)^\top, (1, 0, 1)^\top\}$. (b) $\dim W = 2$. (c) Yes: $(0, 1, 1) = \mathbf{v}_1 - \mathbf{v}_3$.

Where marks are lost

Using the nonzero *rows* of the reduced matrix as the basis when the question says "from the given vectors" (rows of an REF are not the original vectors). Forgetting to reduce fully and reporting $\dim W = 3$ because there are three rows.

Answers at a glance

(a) $\{(1, 1, 2), (1, 0, 1)\}$ (b) 2 (c) yes, coordinates $(1, -1)$

Problem 15 · A subspace of $\mathbb{R}^4$ from two equations, and extending its basis

Let

$$W = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 - x_2 + x_4 = 0, \ x_2 + x_3 = 0\}.$$

- (a) Explain why W is a subspace of $\mathbb{R}^4$. [0.5]
- (b) Find a basis of W and $\dim W$. [2]
- (c) Check whether $\mathbf{q} = (2, 1, -1, -1)^\top$ is in W ; if so, give its coordinates in your basis. [0.5]
- (d) Extend your basis of W to a basis of $\mathbb{R}^4$, and prove the extension works. [1]

How to attempt this (the plan)

What it tests: null space = subspace, solving a small homogeneous system for a basis, and completing a basis with standard vectors plus a determinant check. **Plan:** write the equations as $A\mathbf{x} = \mathbf{0}$; free variables x_2, x_4 ; read off two vectors; append two standard basis vectors and show the 4×4 determinant is nonzero.

Worked solution

STEP 1 (a) W is the null space of

$$A = \begin{bmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}, \quad W = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}.$$

A null space always contains $\mathbf{0}$ and is closed under $+$ and scalar multiples ($A(\mathbf{u} + \mathbf{v}) = \mathbf{0}$, $A(c\mathbf{u}) = \mathbf{0}$), so W is a subspace.

STEP 2 (b) A is already in echelon form with pivots in columns 1 and 2, so the free variables are x_3 and x_4 . Put $x_3 = s$, $x_4 = t$. The second equation gives $x_2 = -x_3 = -s$; the first gives $x_1 = x_2 - x_4 = -s - t$:

$$(x_1, x_2, x_3, x_4) = (-s - t, -s, s, t) = s(-1, -1, 1, 0) + t(-1, 0, 0, 1).$$

STEP 3 (b) So $\mathbf{b}_1 = (-1, -1, 1, 0)^\top$, $\mathbf{b}_2 = (-1, 0, 0, 1)^\top$ span W . They are independent: look at coordinates 3 and 4 — $s\mathbf{b}_1 + t\mathbf{b}_2 = \mathbf{0}$ forces $s = 0$ and $t = 0$. Hence a basis is $\{\mathbf{b}_1, \mathbf{b}_2\}$ and $\dim W = 2$ (equivalently $4 - \text{rank } A = 4 - 2 = 2$).

STEP 4 (c) $\mathbf{q} = (2, 1, -1, -1)$: $2 - 1 + (-1) = 0$ ✓, $1 + (-1) = 0$ ✓, so $\mathbf{q} \in W$. Coordinates: $s = x_3 = -1$, $t = x_4 = -1$; check $-\mathbf{b}_1 - \mathbf{b}_2 = (1 + 1, 1 + 0, -1, 0 - 1) = (2, 1, -1, -1)$. ✓

STEP 5 (d) Append $\mathbf{e}_1 = (1, 0, 0, 0)^\top$ and $\mathbf{e}_2 = (0, 1, 0, 0)^\top$ and test the 4×4 determinant (columns $\mathbf{b}_1, \mathbf{b}_2, \mathbf{e}_1, \mathbf{e}_2$):

$$\det \begin{bmatrix} -1 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

Expand along row 3 (only entry: 1 in column 1, sign $(-1)^{3+1} = +$), then along the new row 3 (only entry: 1 in column 1, sign $+$):

$$= 1 \cdot \det \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = 1 \cdot 1 \cdot \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1 \neq 0.$$

So $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{e}_1, \mathbf{e}_2\}$ is a basis of $\mathbb{R}^4$ extending the basis of W .

Answer. (a) $W = \ker A$, a null space. (b) basis $\{(-1, -1, 1, 0), (-1, 0, 0, 1)\}$, $\dim W = 2$. (c) $\mathbf{q} = -\mathbf{b}_1 - \mathbf{b}_2$, coordinates $(-1, -1)$. (d) add $\mathbf{e}_1, \mathbf{e}_2$; determinant $= 1 \neq 0$.

Where marks are lost

Giving only one vector for a 2-dimensional space. Choosing extension vectors that already lie in W (then the determinant is 0). Skipping the determinant (or an equivalent rank argument) in (d) — "these four look independent" earns nothing.

Answers at a glance

(a) null space $\Rightarrow$ subspace (b) $\{(-1, -1, 1, 0), (-1, 0, 0, 1)\}$, $\dim 2$ (c) yes, $(-1, -1)$ (d)
 $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{e}_1, \mathbf{e}_2\}$, $\det = 1$

Topic 3

The Dot Product, Norms & Inner Products

Use the dot product to test orthogonality and to measure similarity; compute norms, distances and cosine similarity; work in a weighted inner product $\mathbf{x}^\top A \mathbf{y}$, test orthogonality there, and project one vector onto another.

Formulas you need for this topic

- $\mathbf{u} \cdot \mathbf{v} = \sum u_i v_i = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$; $\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$.
- Norms: $\|\mathbf{u}\|_1 = \sum |u_i|$, $\|\mathbf{u}\|_2 = \sqrt{\sum u_i^2}$, $\|\mathbf{u}\|_\infty = \max_i |u_i|$.
- Distance $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$; cosine similarity $= \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$ (unchanged if either vector is scaled).
- Weighted inner product $\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^\top A \mathbf{y}$, valid $\iff A$ is symmetric positive definite; $\|\mathbf{x}\|_A = \sqrt{\mathbf{x}^\top A \mathbf{x}}$.
- Projection of $\mathbf{b}$ on the line through $\mathbf{a}$: $\text{proj}_{\mathbf{a}}(\mathbf{b}) = \frac{\langle \mathbf{b}, \mathbf{a} \rangle}{\langle \mathbf{a}, \mathbf{a} \rangle} \mathbf{a}$; the residual $\mathbf{b} - \text{proj}$ is orthogonal to $\mathbf{a}$.
- Cauchy–Schwarz $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$; triangle inequality $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

Problem 16 · The dot product as an orthogonality test

Consider

$$\mathbf{u}_1 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}.$$

- Using dot products, show that $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ are mutually orthogonal, and find the length of each. [1]
- Find the value of k for which $\mathbf{x} = (1, k, 3)^\top$ is orthogonal to $\mathbf{u}_1$. [0.5]
- Verify for the pair $\mathbf{u}_1, \mathbf{u}_2$ that orthogonality gives $\|\mathbf{u}_1 + \mathbf{u}_2\|^2 = \|\mathbf{u}_1\|^2 + \|\mathbf{u}_2\|^2$, and show by expanding $\|\mathbf{u} + \mathbf{v}\|^2$ that this holds for *every* orthogonal pair. [1]
- Write $\mathbf{b} = (3, 1, 2)^\top$ as $c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + c_3 \mathbf{u}_3$ using $c_i = \frac{\mathbf{b} \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i}$, and verify your answer. [1.5]

How to attempt this (the plan)

What it tests: the single fact $\mathbf{u} \cdot \mathbf{v} = 0 \iff \mathbf{u} \perp \mathbf{v}$, and the payoff it brings — in an orthogonal set the coefficients of any vector can be read off one dot product at a time, with no system to solve. **Plan:** (a) three dot products and three lengths; (b) set one dot product to zero and solve the linear equation; (c) expand $(\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v})$ and watch the cross term die; (d) one dot product per coefficient, then add up and check.

Worked solution

STEP 1 (a) The three dot products:

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = (2)(1) + (1)(2) + (-2)(2) = 2 + 2 - 4 = 0,$$

$$\mathbf{u}_1 \cdot \mathbf{u}_3 = (2)(2) + (1)(-2) + (-2)(1) = 4 - 2 - 2 = 0,$$

$$\mathbf{u}_2 \cdot \mathbf{u}_3 = (1)(2) + (2)(-2) + (2)(1) = 2 - 4 + 2 = 0.$$

All three are zero, so the vectors are mutually orthogonal.

STEP 2 (a) The lengths:

$$\|\mathbf{u}_1\| = \sqrt{4+1+4} = 3, \quad \|\mathbf{u}_2\| = \sqrt{1+4+4} = 3, \quad \|\mathbf{u}_3\| = \sqrt{4+4+1} = 3.$$

STEP 3 (b) Orthogonal to $\mathbf{u}_1$ means $\mathbf{x} \cdot \mathbf{u}_1 = 0$:

$$(1)(2) + (k)(1) + (3)(-2) = 2 + k - 6 = k - 4 = 0 \implies k = 4.$$

Check: $(1, 4, 3) \cdot (2, 1, -2) = 2 + 4 - 6 = 0$. ✓

STEP 4 (c) Numerically, $\mathbf{u}_1 + \mathbf{u}_2 = (3, 3, 0)^\top$, so

$$\|\mathbf{u}_1 + \mathbf{u}_2\|^2 = 9 + 9 + 0 = 18, \quad \|\mathbf{u}_1\|^2 + \|\mathbf{u}_2\|^2 = 9 + 9 = 18. \quad \checkmark$$

STEP 5 (c) In general, expand the square as a dot product:

$$\|\mathbf{u} + \mathbf{v}\|^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = \mathbf{u} \cdot \mathbf{u} + 2(\mathbf{u} \cdot \mathbf{v}) + \mathbf{v} \cdot \mathbf{v} = \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2.$$

If $\mathbf{u} \perp \mathbf{v}$ the middle term is 0, leaving $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$. (This is Pythagoras; the cross term is exactly what orthogonality removes.)

STEP 6 (d) Each $\mathbf{u}_i \cdot \mathbf{u}_i = 9$, so every denominator is 9. The three numerators:

$$\mathbf{b} \cdot \mathbf{u}_1 = (3)(2) + (1)(1) + (2)(-2) = 6 + 1 - 4 = 3,$$

$$\mathbf{b} \cdot \mathbf{u}_2 = (3)(1) + (1)(2) + (2)(2) = 3 + 2 + 4 = 9,$$

$$\mathbf{b} \cdot \mathbf{u}_3 = (3)(2) + (1)(-2) + (2)(1) = 6 - 2 + 2 = 6.$$

$$c_1 = \frac{3}{9} = \frac{1}{3}, \quad c_2 = \frac{9}{9} = 1, \quad c_3 = \frac{6}{9} = \frac{2}{3}.$$

STEP 7 (d) Verify by adding the three pieces:

$$\frac{1}{3} \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} + \frac{2}{3} \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} + 1 + \frac{4}{3} \\ \frac{1}{3} + 2 - \frac{4}{3} \\ -\frac{2}{3} + 2 + \frac{2}{3} \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} = \mathbf{b}. \quad \checkmark$$

Note what made this easy: dotting $\mathbf{b} = \sum c_j \mathbf{u}_j$ with $\mathbf{u}_i$ kills every term except one, because $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ for $j \neq i$. The same argument with $\mathbf{b} = \mathbf{0}$ forces every $c_i = 0$, which is why mutually orthogonal nonzero vectors are always linearly independent.

Answer. (a) all three dot products are 0; each length is 3. (b) $k = 4$. (c) $18 = 9 + 9$; in general the cross term $2(\mathbf{u} \cdot \mathbf{v})$ vanishes. (d) $\mathbf{b} = \frac{1}{3}\mathbf{u}_1 + \mathbf{u}_2 + \frac{2}{3}\mathbf{u}_3$.

Where marks are lost

Testing orthogonality with the *cross* product or with lengths instead of the dot product. In (b), writing $\mathbf{x} \cdot \mathbf{u}_1 = 0$ but then solving $2 + k + 6 = 0$ — the third component of $\mathbf{u}_1$ is -2 . In (d), dividing by $\|\mathbf{u}_i\|$ instead of $\mathbf{u}_i \cdot \mathbf{u}_i = \|\mathbf{u}_i\|^2$. Using the shortcut formula for c_i when the set is *not* orthogonal — then a full system must be solved.

Answers at a glance

(a) 0, 0, 0; lengths 3, 3, 3 (b) $k = 4$ (c) $18 = 9 + 9$ (d) $c = (\frac{1}{3}, 1, \frac{2}{3})$

Problem 17 · The dot product as a similarity score

A query is represented by the feature vector $\mathbf{q}$ and three documents by $\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3$:

$$\mathbf{q} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \quad \mathbf{d}_1 = \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}, \quad \mathbf{d}_2 = \begin{bmatrix} 6 \\ 8 \\ 0 \end{bmatrix}, \quad \mathbf{d}_3 = \begin{bmatrix} -4 \\ 3 \\ 0 \end{bmatrix}.$$

- (a) Compute $\|\mathbf{q}\|, \|\mathbf{d}_1\|, \|\mathbf{d}_2\|, \|\mathbf{d}_3\|$ and the dot products $\mathbf{q} \cdot \mathbf{d}_i$. [1]
- (b) Compute the cosine similarity $\cos \theta_i = \frac{\mathbf{q} \cdot \mathbf{d}_i}{\|\mathbf{q}\| \|\mathbf{d}_i\|}$ for each document and rank the three. [1]
- (c) Which pairs among $\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3$ are orthogonal? [0.5]
- (d) Compute the Euclidean distances $d(\mathbf{q}, \mathbf{d}_i) = \|\mathbf{q} - \mathbf{d}_i\|$ and rank the documents by distance. State which of the two scores, cosine or distance, is unchanged when a document is scaled by a positive constant, and justify it from the formula. [1.5]

How to attempt this (the plan)

What it tests: the dot product read as "how much two vectors point the same way", the difference between the raw dot product, the cosine (direction only) and the Euclidean distance (position), and the fact that $\mathbf{u} \cdot \mathbf{v} = 0$ means no similarity at all. **Plan:** norms first, then one dot product per document, then divide; for (d) write out each difference vector before taking its length, and compare the two rankings.

Worked solution

STEP 1 (a) Lengths:

$$\begin{aligned} \|\mathbf{q}\| &= \sqrt{1+4+4} = 3, & \|\mathbf{d}_1\| &= \sqrt{9+16+0} = 5, \\ \|\mathbf{d}_2\| &= \sqrt{36+64+0} = 10, & \|\mathbf{d}_3\| &= \sqrt{16+9+0} = 5. \end{aligned}$$

Dot products:

$$\begin{aligned} \mathbf{q} \cdot \mathbf{d}_1 &= (1)(3) + (2)(4) + (2)(0) = 3 + 8 = 11, \\ \mathbf{q} \cdot \mathbf{d}_2 &= (1)(6) + (2)(8) + (2)(0) = 6 + 16 = 22, \\ \mathbf{q} \cdot \mathbf{d}_3 &= (1)(-4) + (2)(3) + (2)(0) = -4 + 6 = 2. \end{aligned}$$

STEP 2 (b) Divide each by $\|\mathbf{q}\| \|\mathbf{d}_i\|$:

$$\cos \theta_1 = \frac{11}{3 \cdot 5} = \frac{11}{15} \approx 0.733, \quad \cos \theta_2 = \frac{22}{3 \cdot 10} = \frac{22}{30} = \frac{11}{15} \approx 0.733, \quad \cos \theta_3 = \frac{2}{3 \cdot 5} = \frac{2}{15} \approx 0.133.$$

Ranking: $\mathbf{d}_1$ and $\mathbf{d}_2$ tie at the top, $\mathbf{d}_3$ last. Note $\mathbf{d}_2 = 2\mathbf{d}_1$ — twice as long, same direction — so its dot product doubles (22 against 11) but its cosine is identical.

STEP 3 (c) Test each pair:

$$\mathbf{d}_1 \cdot \mathbf{d}_3 = (3)(-4) + (4)(3) = -12 + 12 = 0, \quad \mathbf{d}_2 \cdot \mathbf{d}_3 = (6)(-4) + (8)(3) = -24 + 24 = 0,$$

$$\mathbf{d}_1 \cdot \mathbf{d}_2 = 18 + 32 = 50 \neq 0.$$

So $\mathbf{d}_3 \perp \mathbf{d}_1$ and $\mathbf{d}_3 \perp \mathbf{d}_2$; $\mathbf{d}_1$ and $\mathbf{d}_2$ are not orthogonal (they are parallel).

STEP 4 (d) Difference vectors first, then lengths:

$$\mathbf{q} - \mathbf{d}_1 = (1 - 3, 2 - 4, 2 - 0) = (-2, -2, 2), \quad d(\mathbf{q}, \mathbf{d}_1) = \sqrt{4 + 4 + 4} = \sqrt{12} = 2\sqrt{3} \approx 3.46,$$

$$\mathbf{q} - \mathbf{d}_2 = (1 - 6, 2 - 8, 2 - 0) = (-5, -6, 2), \quad d(\mathbf{q}, \mathbf{d}_2) = \sqrt{25 + 36 + 4} = \sqrt{65} \approx 8.06,$$

$$\mathbf{q} - \mathbf{d}_3 = (1 + 4, 2 - 3, 2 - 0) = (5, -1, 2), \quad d(\mathbf{q}, \mathbf{d}_3) = \sqrt{25 + 1 + 4} = \sqrt{30} \approx 5.48.$$

By distance the order is $\mathbf{d}_1$, then $\mathbf{d}_3$, then $\mathbf{d}_2$ — different from the cosine order, which put $\mathbf{d}_2$ joint first.

STEP 5 (d) The cosine is the scale-invariant one. Replace $\mathbf{d}$ by $c\mathbf{d}$ with $c > 0$:

$$\frac{\mathbf{q} \cdot (c\mathbf{d})}{\|\mathbf{q}\| \|c\mathbf{d}\|} = \frac{c(\mathbf{q} \cdot \mathbf{d})}{c \|\mathbf{q}\| \|\mathbf{d}\|} = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \|\mathbf{d}\|},$$

because the c in the numerator cancels the c in $\|c\mathbf{d}\| = c\|\mathbf{d}\|$. The distance has no such cancellation: $\|\mathbf{q} - c\mathbf{d}\|$ grows with c , which is exactly why $\mathbf{d}_2$ is far from $\mathbf{q}$ yet perfectly aligned with $\mathbf{d}_1$.

Answer. (a) $\|\mathbf{q}\| = 3; 5, 10, 5$; dot products 11, 22, 2. (b) $\frac{11}{15}, \frac{11}{15}, \frac{2}{15}$; $\mathbf{d}_1 = \mathbf{d}_2$ then $\mathbf{d}_3$. (c) $\mathbf{d}_3 \perp \mathbf{d}_1$ and $\mathbf{d}_3 \perp \mathbf{d}_2$. (d) $2\sqrt{3}, \sqrt{65}, \sqrt{30}$; order $\mathbf{d}_1, \mathbf{d}_3, \mathbf{d}_2$; the cosine is scale-invariant.

Where marks are lost

Ranking by the raw dot product and calling $\mathbf{d}_2$ twice as similar as $\mathbf{d}_1$ — the dot product mixes direction with length, which is precisely why it is divided by the two norms. Computing $\|\mathbf{q}\| - \|\mathbf{d}_i\|$ instead of $\|\mathbf{q} - \mathbf{d}_i\|$ for the distance. Forgetting that a zero dot product means the cosine is 0, i.e. an angle of 90° , not "opposite" (that would be a negative dot product).

Answers at a glance

(a) 3; 5, 10, 5; 11, 22, 2 (b) $\frac{11}{15}, \frac{11}{15}, \frac{2}{15}$ (c) $\mathbf{d}_3 \perp \mathbf{d}_1, \mathbf{d}_3 \perp \mathbf{d}_2$ (d) $2\sqrt{3}, \sqrt{65}, \sqrt{30}$; cosine is scale-invariant

Problem 18 · Orthogonal complement, angle and basis in $\mathbb{R}^3$

On $\mathbb{R}^3$ with the standard (Euclidean) inner product, let $u = (1, 1, 0)^\top$ and $v = (0, 1, 1)^\top$.

- Find all vectors $w \in \mathbb{R}^3$ orthogonal to both u and v , and write down the unit vectors of this form. [2]
- Compute the angle ω between u and v . [1]
- Show that $\{u, v, w\}$ is a basis of $\mathbb{R}^3$, and explain geometrically why w is perpendicular to the plane spanned by u and v . [1]

How to attempt this (the plan)

What it tests: turning "orthogonal to" into dot-product equations, the cosine formula for angles, and the determinant test for a basis. **Plan:** (a) write $w = (w_1, w_2, w_3)^\top$, set both dot products to zero, solve the 2×3 homogeneous system, then divide by the length. (b)

$\cos \omega = u^\top v / (\|u\| \|v\|)$. (c) stack u, v, w as columns and check $\det \neq 0$; then argue that w is orthogonal to every combination of u and v .

Worked solution

STEP 1 (a) “Orthogonal to both” means two dot-product equations:

$$w^\top u = w_1 + w_2 = 0, \quad w^\top v = w_2 + w_3 = 0.$$

STEP 2 (a) Solve: $w_1 = -w_2$ and $w_3 = -w_2$. With the free variable $w_2 = t$,

$$w = t(-1, 1, -1)^\top, \quad t \in \mathbb{R}.$$

STEP 3 (a) Normalise. $\|(-1, 1, -1)^\top\| = \sqrt{1+1+1} = \sqrt{3}$, so the unit vectors are

$$w = \pm \frac{1}{\sqrt{3}}(-1, 1, -1)^\top.$$

STEP 4 (b) Compute the pieces of the cosine formula:

$$u^\top v = 0 + 1 + 0 = 1, \quad \|u\| = \sqrt{2}, \quad \|v\| = \sqrt{2}.$$

$$\cos \omega = \frac{u^\top v}{\|u\| \|v\|} = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2} \implies \omega = 60^\circ.$$

STEP 5 (c) Three vectors in $\mathbb{R}^3$ form a basis exactly when their determinant is nonzero. With $w = (-1, 1, -1)^\top$,

$$\det [u \ v \ w] = \det \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} = 1(-1-1) - 0 + (-1)(1-0) = -3 \neq 0,$$

so $\{u, v, w\}$ is a basis.

STEP 6 (c) Geometry: u and v span a plane. Any vector in that plane is $\alpha u + \beta v$, and

$$w^\top (\alpha u + \beta v) = \alpha w^\top u + \beta w^\top v = 0 + 0 = 0,$$

so w is orthogonal to every vector in the plane: w is the normal direction of the plane.

Answer. (a) $w = t(-1, 1, -1)^\top$, unit vectors $\pm \frac{1}{\sqrt{3}}(-1, 1, -1)^\top$. (b) $\omega = 60^\circ$. (c) $\det[u \ v \ w] = -3 \neq 0$, so a basis; w is the plane normal.

Where marks are lost

Giving only one vector w in (a) instead of the whole line $t(-1, 1, -1)^\top$, or forgetting the $\pm$ on the unit vectors. Dividing by $\sqrt{2}$ once instead of twice in (b) (the denominator is $\|u\| \|v\| = 2$). In (c), saying “three vectors so a basis” without checking the determinant.

Answers at a glance

(a) $w = t(-1, 1, -1)^\top$; unit: $\pm \frac{1}{\sqrt{3}}(-1, 1, -1)^\top$ (b) $\omega = 60^\circ$ (c) $\det = -3 \neq 0$; $w \perp$ plane of u, v .

Problem 19 · A weighted inner product on $\mathbb{R}^2$

Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and define a new inner product on $\mathbb{R}^2$ by $\langle x, y \rangle_A = x^\top A y$.

- (a) Show A is symmetric positive definite, so $\langle \cdot, \cdot \rangle_A$ is a genuine inner product. [1]
- (b) The vectors $x = (1, 0)^\top$ and $y = (0, 1)^\top$ are orthogonal in the dot product. Are they still orthogonal under $\langle \cdot, \cdot \rangle_A$? Find the angle between them in this new inner product. [2]
- (c) Find a vector $y' = (a, 1)^\top$ that is orthogonal to x with respect to $\langle \cdot, \cdot \rangle_A$. [1]

How to attempt this (the plan)

What it tests: Sylvester's criterion for positive definiteness, and that orthogonality and angles depend on the inner product. **Plan:** (a) check $A^\top = A$ and the two leading minors. (b) compute $x^\top A y$; if nonzero, compute the A -norms and use the cosine formula. (c) set $x^\top A y' = 0$ and solve for a .

Worked solution

STEP 1 (a) A is symmetric since $A^\top = A$. Sylvester's criterion (all leading principal minors positive):

$$D_1 = 2 > 0, \quad D_2 = \det A = 2 \cdot 2 - 1 \cdot 1 = 3 > 0.$$

So A is positive definite. A symmetric positive-definite matrix always makes $x^\top A y$ a valid inner product: it is bilinear, symmetric, and $x^\top A x > 0$ for $x \neq 0$.

STEP 2 (b) Test orthogonality. First $A y = A(0, 1)^\top = (1, 2)^\top$, hence

$$\langle x, y \rangle_A = x^\top (A y) = (1, 0) \cdot (1, 2) = 1 \neq 0.$$

They are *not* orthogonal under $\langle \cdot, \cdot \rangle_A$.

STEP 3 (b) For the angle we need the A -norms:

$$\|x\|_A = \sqrt{x^\top A x} = \sqrt{A_{11}} = \sqrt{2}, \quad \|y\|_A = \sqrt{y^\top A y} = \sqrt{A_{22}} = \sqrt{2}.$$

$$\cos \theta = \frac{\langle x, y \rangle_A}{\|x\|_A \|y\|_A} = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2} \implies \theta = 60^\circ.$$

Takeaway: orthogonality depends on which inner product you use.

STEP 4 (c) We need $\langle x, y' \rangle_A = 0$. Note $x^\top A = (1, 0)A = (2, 1)$, so

$$\langle x, y' \rangle_A = (2, 1) \cdot (a, 1) = 2a + 1 = 0 \implies a = -\frac{1}{2}.$$

Thus $y' = (-\frac{1}{2}, 1)^\top$.

Answer. (a) $A^\top = A$, $D_1 = 2 > 0$, $D_2 = 3 > 0$: SPD. (b) $\langle x, y \rangle_A = 1 \neq 0$, not orthogonal; $\theta = 60^\circ$. (c) $y' = (-\frac{1}{2}, 1)^\top$.

Where marks are lost

Checking only $\det A > 0$ in (a) and forgetting $D_1 > 0$ (a negative-definite matrix also has positive determinant). Using the Euclidean norms $\|x\| = \|y\| = 1$ in (b) instead of the A -norms

$\sqrt{2}$. In (c), computing Ay' and dotting with x is fine, but a sign slip in $2a + 1 = 0$ gives $a = +\frac{1}{2}$.

Answers at a glance

(a) SPD ($D_1 = 2, D_2 = 3$) (b) $\langle x, y \rangle_A = 1, \theta = 60^\circ$ (c) $y' = (-\frac{1}{2}, 1)^\top$.

Problem 20 · Projection in a weighted inner product

On $\mathbb{R}^3$ use the weighted inner product $\langle x, y \rangle_A = x^\top Ay$ with

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ (SPD).}$$

Let $a = (1, -1, 0)^\top$ and $b = (1, 0, 2)^\top$.

- (a) Compute $\langle b, a \rangle_A$, $\|a\|_A$, and $\|b\|_A$. [2]
- (b) Find the orthogonal projection p of b onto the line spanned by a , in this inner product. [2]
- (c) Verify that the residual $b - p$ is A -orthogonal to a (this is what makes p the orthogonal projection). [1]

How to attempt this (the plan)

What it tests: computing with a weighted inner product and the one-dimensional projection formula with a general inner product. **Plan:** (a) compute Aa and Ab once, then every quantity is a dot product with one of them. (b) $p = \frac{\langle b, a \rangle_A}{\langle a, a \rangle_A} a$. (c) form $r = b - p$ and check $r^\top (Aa) = 0$.

Worked solution

STEP 1 (a) Compute the matrix–vector product once:

$$Aa = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

$$\langle b, a \rangle_A = b^\top (Aa) = (1, 0, 2) \cdot (1, -1, 0) = 1, \quad \|a\|_A^2 = a^\top (Aa) = (1, -1, 0) \cdot (1, -1, 0) = 2.$$

STEP 2 (a) For $\|b\|_A$ use

$$Ab = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}, \quad \|b\|_A^2 = b^\top (Ab) = (1, 0, 2) \cdot (2, 1, 2) = 6.$$

So $\langle b, a \rangle_A = 1$, $\|a\|_A = \sqrt{2}$, $\|b\|_A = \sqrt{6}$.

STEP 3 (b) Projection formula (same as usual, but every inner product is the A -one):

$$p = \frac{\langle b, a \rangle_A}{\langle a, a \rangle_A} a = \frac{1}{2} (1, -1, 0)^\top = \left(\frac{1}{2}, -\frac{1}{2}, 0\right)^\top.$$

STEP 4 (c) The residual is

$$r = b - p = (1, 0, 2)^\top - \left(\frac{1}{2}, -\frac{1}{2}, 0\right)^\top = \left(\frac{1}{2}, \frac{1}{2}, 2\right)^\top.$$

STEP 5 (c) Check A -orthogonality using the same $Aa = (1, -1, 0)^\top$:

$$\langle r, a \rangle_A = r^\top (Aa) = \left(\frac{1}{2}, \frac{1}{2}, 2\right) \cdot (1, -1, 0) = \frac{1}{2} - \frac{1}{2} + 0 = 0. \checkmark$$

Since $r \perp_A a$, p is indeed the orthogonal projection of b onto the line through a .

Answer. (a) $\langle b, a \rangle_A = 1$, $\|a\|_A = \sqrt{2}$, $\|b\|_A = \sqrt{6}$. (b) $p = (\frac{1}{2}, -\frac{1}{2}, 0)^\top$. (c) $r = (\frac{1}{2}, \frac{1}{2}, 2)^\top$, $\langle r, a \rangle_A = 0$.

Where marks are lost

Using the Euclidean formula $p = \frac{b^\top a}{a^\top a} a$ (here $b^\top a = 1$, $a^\top a = 2$ happen to agree, but the method is wrong and is marked down). Checking the residual with the ordinary dot product $r^\top a$ instead of $r^\top Aa$. Forgetting the square root when reporting $\|a\|_A$ and $\|b\|_A$.

Answers at a glance

(a) $1, \sqrt{2}, \sqrt{6}$ (b) $p = (\frac{1}{2}, -\frac{1}{2}, 0)^\top$ (c) $\langle b - p, a \rangle_A = 0 \checkmark$.

Topic 4

Orthogonality, Gram–Schmidt & Orthogonal Matrices

Run Gram–Schmidt and normalise, verify orthonormality, project a vector onto a span using an orthonormal basis, and check that an orthogonal matrix preserves lengths, angles and inner products.

Formulas you need for this topic

- Gram–Schmidt: $\mathbf{u}_1 = \mathbf{v}_1$, and $\mathbf{u}_k = \mathbf{v}_k - \sum_{i < k} \frac{\mathbf{v}_k \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i} \mathbf{u}_i$. Then $\mathbf{e}_i = \mathbf{u}_i / \|\mathbf{u}_i\|$.
- Scaling any $\mathbf{u}_k$ by a nonzero constant keeps the set orthogonal — do it to clear fractions, then normalise at the end.
- With an orthonormal $\{\mathbf{q}_i\}$: projection onto their span is $\sum_i (\mathbf{b} \cdot \mathbf{q}_i) \mathbf{q}_i$, and the residual is orthogonal to every $\mathbf{q}_i$.
- Q orthogonal $\iff Q^\top Q = I \iff Q^{-1} = Q^\top$. Then $\|Q\mathbf{x}\| = \|\mathbf{x}\|$, $(Q\mathbf{x}) \cdot (Q\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$, and $\det Q = \pm 1$ (+1 rotation, -1 reflection).

Problem 21 · Gram–Schmidt on three vectors, then verify

Three direction vectors are

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

- (a) Use the Gram–Schmidt process to build mutually orthogonal vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$. [1.5]
 (b) Normalise them to an orthonormal basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ of $\mathbb{R}^3$. [0.5]
 (c) Verify orthonormality: all three pairwise dot products and all three lengths. [1]

How to attempt this (the plan)

What it tests: the projection formula $\text{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$ applied in order, and the habit of checking. **Plan:** $\mathbf{u}_1 = \mathbf{v}_1$; $\mathbf{u}_2 = \mathbf{v}_2 - \text{proj}_{\mathbf{u}_1} \mathbf{v}_2$; $\mathbf{u}_3 = \mathbf{v}_3 - \text{proj}_{\mathbf{u}_1} \mathbf{v}_3 - \text{proj}_{\mathbf{u}_2} \mathbf{v}_3$; scale any $\mathbf{u}_k$ to clear fractions (scaling keeps orthogonality); divide by lengths; check.

Worked solution

STEP 1 (a) $\mathbf{u}_1 = \mathbf{v}_1 = (1, 2, 2)^\top$, with $\mathbf{u}_1 \cdot \mathbf{u}_1 = 1 + 4 + 4 = 9$.

STEP 2 (a) Second vector. $\mathbf{v}_2 \cdot \mathbf{u}_1 = 1 + 0 + 2 = 3$.

$$\mathbf{u}_2 = \mathbf{v}_2 - \frac{3}{9} \mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix}.$$

Scale by 3 (allowed — direction is all that matters): $\mathbf{u}_2 = (2, -2, 1)^\top$, $\mathbf{u}_2 \cdot \mathbf{u}_2 = 4 + 4 + 1 = 9$.

STEP 3 (a) Third vector. $\mathbf{v}_3 \cdot \mathbf{u}_1 = 0 + 2 + 2 = 4$ and $\mathbf{v}_3 \cdot \mathbf{u}_2 = 0 - 2 + 1 = -1$.

$$\mathbf{u}_3 = \mathbf{v}_3 - \frac{4}{9}\mathbf{u}_1 - \frac{-1}{9}\mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{4}{9} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} + \frac{1}{9} \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}.$$

Coordinate by coordinate:

$$x: 0 - \frac{4}{9} + \frac{2}{9} = -\frac{2}{9}, \quad y: 1 - \frac{8}{9} - \frac{2}{9} = -\frac{1}{9}, \quad z: 1 - \frac{8}{9} + \frac{1}{9} = \frac{2}{9}.$$

Scale by 9: $\mathbf{u}_3 = (-2, -1, 2)^\top$, $\mathbf{u}_3 \cdot \mathbf{u}_3 = 4 + 1 + 4 = 9$.

STEP 4 (b) All three have length $\sqrt{9} = 3$:

$$\mathbf{e}_1 = \frac{1}{3}(1, 2, 2)^\top, \quad \mathbf{e}_2 = \frac{1}{3}(2, -2, 1)^\top, \quad \mathbf{e}_3 = \frac{1}{3}(-2, -1, 2)^\top.$$

STEP 5 (c) Pairwise dot products (the factor $\frac{1}{9}$ multiplies each):

$$\mathbf{e}_1 \cdot \mathbf{e}_2 = \frac{1}{9}(2 - 4 + 2) = 0, \quad \mathbf{e}_1 \cdot \mathbf{e}_3 = \frac{1}{9}(-2 - 2 + 4) = 0, \quad \mathbf{e}_2 \cdot \mathbf{e}_3 = \frac{1}{9}(-4 + 2 + 2) = 0.$$

Lengths: $\|\mathbf{e}_k\|^2 = \frac{1}{9} \cdot 9 = 1$ for each k . ✓

Answer. (a) $\mathbf{u}_1 = (1, 2, 2)$, $\mathbf{u}_2 = (2, -2, 1)$, $\mathbf{u}_3 = (-2, -1, 2)$. (b) $\mathbf{e}_k = \mathbf{u}_k/3$. (c) all dot products 0, all lengths 1.

Where marks are lost

Using $\mathbf{v}_2$ instead of $\mathbf{u}_2$ when projecting $\mathbf{v}_3$ (you must subtract projections onto the *new* orthogonal vectors). Dividing by $\|\mathbf{u}\|$ instead of $\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2$ in the projection formula. Normalising too early and then carrying square roots through the subtraction.

Answers at a glance

(a) $(1, 2, 2)$, $(2, -2, 1)$, $(-2, -1, 2)$ (b) each divided by 3 (c) 0, 0, 0 and 1, 1, 1

Problem 22 · Gram–Schmidt in $\mathbb{R}^4$ and a projection onto a span

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}.$$

- (a) Apply Gram–Schmidt to $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ to obtain an orthonormal set $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$. [2]
 (b) Using $\sum_i (\mathbf{b} \cdot \mathbf{q}_i) \mathbf{q}_i$, find the orthogonal projection of $\mathbf{b} = (1, 2, 3, 4)^\top$ onto $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, and verify that the residual is orthogonal to $\mathbf{q}_1$. [1]

How to attempt this (the plan)

What it tests: Gram–Schmidt in the "normalise as you go" form, and projection onto a subspace as a sum of one-dimensional projections. **Plan:** normalise $\mathbf{v}_1$; subtract $(\mathbf{v}_2 \cdot \mathbf{q}_1) \mathbf{q}_1$ and normalise; repeat with two subtractions for $\mathbf{v}_3$; then dot $\mathbf{b}$ with each $\mathbf{q}_i$ and add up the pieces.

Worked solution

STEP 1 (a) $\|\mathbf{v}_1\| = \sqrt{4} = 2$, so $\mathbf{q}_1 = \frac{1}{2}(1, 1, 1, 1)^\top$.

STEP 2 (a) $\mathbf{v}_2 \cdot \mathbf{q}_1 = \frac{1}{2}(2 + 0 + 2 + 0) = 2$.

$$\mathbf{u}_2 = \mathbf{v}_2 - 2\mathbf{q}_1 = (2, 0, 2, 0) - (1, 1, 1, 1) = (1, -1, 1, -1), \quad \|\mathbf{u}_2\| = 2, \quad \mathbf{q}_2 = \frac{1}{2}(1, -1, 1, -1)^\top.$$

STEP 3 (a) $\mathbf{v}_3 \cdot \mathbf{q}_1 = \frac{1}{2}(0 + 1 + 1 + 0) = 1$ and $\mathbf{v}_3 \cdot \mathbf{q}_2 = \frac{1}{2}(0 - 1 + 1 - 0) = 0$.

$$\mathbf{u}_3 = \mathbf{v}_3 - 1 \cdot \mathbf{q}_1 - 0 \cdot \mathbf{q}_2 = (0, 1, 1, 0) - \frac{1}{2}(1, 1, 1, 1) = \frac{1}{2}(-1, 1, 1, -1), \quad \|\mathbf{u}_3\| = \frac{1}{2}\sqrt{4} = 1.$$

So $\mathbf{q}_3 = \frac{1}{2}(-1, 1, 1, -1)^\top$. (Quick check: $\mathbf{q}_2 \cdot \mathbf{q}_3 = \frac{1}{4}(-1 - 1 + 1 + 1) = 0$. ✓)

STEP 4 (b) Coefficients:

$$\mathbf{b} \cdot \mathbf{q}_1 = \frac{1}{2}(1 + 2 + 3 + 4) = 5, \quad \mathbf{b} \cdot \mathbf{q}_2 = \frac{1}{2}(1 - 2 + 3 - 4) = -1, \quad \mathbf{b} \cdot \mathbf{q}_3 = \frac{1}{2}(-1 + 2 + 3 - 4) = 0.$$

$$\text{proj}(\mathbf{b}) = 5\mathbf{q}_1 - 1 \cdot \mathbf{q}_2 + 0 \cdot \mathbf{q}_3 = \frac{5}{2}(1, 1, 1, 1) - \frac{1}{2}(1, -1, 1, -1) = (2, 3, 2, 3)^\top.$$

STEP 5 (b) Residual $\mathbf{r} = \mathbf{b} - \text{proj}(\mathbf{b}) = (1 - 2, 2 - 3, 3 - 2, 4 - 3) = (-1, -1, 1, 1)$, and $\mathbf{r} \cdot \mathbf{q}_1 = \frac{1}{2}(-1 - 1 + 1 + 1) = 0$. ✓

Answer. (a) $\mathbf{q}_1 = \frac{1}{2}(1, 1, 1, 1)$, $\mathbf{q}_2 = \frac{1}{2}(1, -1, 1, -1)$, $\mathbf{q}_3 = \frac{1}{2}(-1, 1, 1, -1)$. (b) $\text{proj}(\mathbf{b}) = (2, 3, 2, 3)$, residual $(-1, -1, 1, 1)$, orthogonal to $\mathbf{q}_1$.

Where marks are lost

Projecting onto the original $\mathbf{v}_i$ instead of the orthonormal $\mathbf{q}_i$ — the simple sum formula only works with an orthonormal set. Forgetting to normalise $\mathbf{u}_2$ before using it to build $\mathbf{u}_3$. Dividing by $\|\mathbf{u}\|$ instead of $\mathbf{u} \cdot \mathbf{u}$ when the projection is written in the unnormalised form.

Answers at a glance

(a) $\frac{1}{2}(1, 1, 1, 1)$, $\frac{1}{2}(1, -1, 1, -1)$, $\frac{1}{2}(-1, 1, 1, -1)$ (b) $(2, 3, 2, 3)$, residual $(-1, -1, 1, 1)$

Problem 23 · An orthogonal matrix: lengths, angles and the rotation it performs

A transformation of 2-D feature vectors is given by

$$Q = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

- (a) Verify $Q^\top Q = I$, and compute $Q\mathbf{x}$ and $Q\mathbf{y}$. [1]
- (b) Verify $\|Q\mathbf{x}\| = \|\mathbf{x}\|$ and $\|Q\mathbf{y}\| = \|\mathbf{y}\|$. [0.5]
- (c) Compute $\cos$ of the angle between $\mathbf{x}$ and $\mathbf{y}$, and between $Q\mathbf{x}$ and $Q\mathbf{y}$; conclude. [1]
- (d) Find $\det Q$ and the angle θ by which Q rotates the plane; write down Q^{-1} without computing an inverse. [0.5]

How to attempt this (the plan)

What it tests: the definition $Q^T Q = I$ and its two consequences (lengths and angles are preserved), plus recognising a rotation matrix $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$. **Plan:** multiply $Q^T Q$ entry by entry; do the two matrix–vector products; compute dot products and norms before and after; read $\cos \theta = \frac{3}{5}$ from Q .

Worked solution

STEP 1 (a)

$$Q^T Q = \frac{1}{25} \begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix} = \frac{1}{25} \begin{bmatrix} 9+16 & -12+12 \\ -12+12 & 16+9 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I. \checkmark$$

$$Q\mathbf{x} = \frac{1}{5} \begin{bmatrix} 3 \cdot 4 - 4 \cdot 3 \\ 4 \cdot 4 + 3 \cdot 3 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 0 \\ 25 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}, \quad Q\mathbf{y} = \frac{1}{5} \begin{bmatrix} 3 - 8 \\ 4 + 6 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} -5 \\ 10 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

STEP 2 (b)

$$\|\mathbf{x}\| = \sqrt{16+9} = 5 = \sqrt{0+25} = \|Q\mathbf{x}\|, \quad \|\mathbf{y}\| = \sqrt{1+4} = \sqrt{5} = \sqrt{1+4} = \|Q\mathbf{y}\|. \checkmark$$

STEP 3 (c)

Before: $\mathbf{x}^T \mathbf{y} = 4 + 6 = 10$, so $\cos \theta = \frac{10}{5\sqrt{5}} = \frac{2}{\sqrt{5}}$. After: $(Q\mathbf{x})^T (Q\mathbf{y}) = 0 \cdot (-1) + 5 \cdot 2 = 10$, norms unchanged, so $\cos \theta' = \frac{2}{\sqrt{5}}$ as well. The angle is preserved ($\theta = \theta' \approx 26.6^\circ$).

STEP 4 (d)

$\det Q = \frac{1}{25}(9+16) = 1$, so Q is a rotation (not a reflection). Matching $Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ gives $\cos \theta = \frac{3}{5}$, $\sin \theta = \frac{4}{5}$, so $\theta = \arctan \frac{4}{3} \approx 53.13^\circ$ anticlockwise. For an orthogonal matrix $Q^{-1} = Q^T = \frac{1}{5} \begin{bmatrix} 3 & 4 \\ -4 & 3 \end{bmatrix}$ (rotation by $-\theta$).

Answer. (a) $Q^T Q = I$; $Q\mathbf{x} = (0, 5)$, $Q\mathbf{y} = (-1, 2)$. (b) lengths 5 and $\sqrt{5}$ both preserved. (c) $\cos \theta = 2/\sqrt{5}$ before and after. (d) $\det Q = 1$, $\theta \approx 53.13^\circ$, $Q^{-1} = Q^T$.

Where marks are lost

Forgetting the $\frac{1}{5}$ (then $Q^T Q = 25I$ and nothing is preserved). Computing the angle only after the transformation and calling it "preserved". Confusing $\det = -1$ (reflection) with $\det = 1$ (rotation).

Answers at a glance

(a) I ; $(0, 5)$, $(-1, 2)$ (b) 5, $\sqrt{5}$ (c) $2/\sqrt{5}$ both (d) 1; $\approx 53.13^\circ$; Q^T

Topic 5

Eigenvalues & Eigenvectors

Characteristic polynomial, eigenvalues with algebraic and geometric multiplicity, eigenvectors and eigenspace dimensions, eigenvalues in symbols, complex eigenvalues of a rotation, eigenvalues of $p(A)$, and short proofs.

Formulas you need for this topic

- Eigenvalues from $\det(A - \lambda I) = 0$; eigenvectors from $(A - \lambda I)\mathbf{x} = \mathbf{0}$.
- Algebraic multiplicity = multiplicity as a root; geometric multiplicity = $n - \text{rank}(A - \lambda I) = \dim$ of the eigenspace; always $1 \leq \text{gm} \leq \text{am}$.
- Diagonalizable $\iff \text{gm} = \text{am}$ for every eigenvalue. Distinct eigenvalues $\Rightarrow$ diagonalizable.
- $\sum \lambda_i = \text{tr } A$ and $\prod \lambda_i = \det A$ — use both as a check.
- Spectral mapping: if $A\mathbf{v} = \lambda\mathbf{v}$ then $p(A)\mathbf{v} = p(\lambda)\mathbf{v}$; in particular $A^k\mathbf{v} = \lambda^k\mathbf{v}$.
- Triangular matrix: the eigenvalues are the diagonal entries.

Problem 24 · Eigen-analysis of a 3×3 matrix

Consider

$$A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{bmatrix}.$$

- Find the characteristic polynomial and all eigenvalues with their algebraic multiplicities.
- For every eigenvalue find a basis of its eigenspace and state the geometric multiplicity.
- Hence decide, with reason, whether A is diagonalizable.

How to attempt this (the plan)

What it tests: the full eigen-workflow on a matrix with a repeated eigenvalue. **Plan:** expand $\det(A - \lambda I)$ (row operations first); solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for each root and count free variables; compare the two multiplicities.

Worked solution

STEP 1 (a) Note $A = 3I + J$ with J the all-ones matrix. Subtract row 1 from rows 2 and 3 of $A - \lambda I$ and expand:

$$\det(A - \lambda I) = -(\lambda - 6)(\lambda - 3)^2.$$

So $\lambda = 6$ has algebraic multiplicity 1 and $\lambda = 3$ has algebraic multiplicity 2.

STEP 2 (b) $\lambda = 6$: $(A - 6I)\mathbf{v} = \mathbf{0}$ reads

$$-2x + y + z = 0, \quad x - 2y + z = 0, \quad x + y - 2z = 0,$$

whose solution is $x = y = z$. Basis $\{(1, 1, 1)^\top\}$, geometric multiplicity 1.

STEP 3 (b) $\lambda = 3$: every row of $A - 3I$ is $(1, 1, 1)$, so the system collapses to $x + y + z = 0$.

Two free variables:

$$v = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad \text{geometric multiplicity 2.}$$

STEP 4 (c) For each eigenvalue $\text{gm} = \text{am}$ ($1 = 1$ and $2 = 2$), so the three eigenvectors form a basis of $\mathbb{R}^3$ and A is diagonalizable. (A is symmetric, so the spectral theorem guarantees this anyway.)

Answer. $\det(A - \lambda I) = -(\lambda - 6)(\lambda - 3)^2$. $\lambda = 6$: am 1, eigenvector $(1, 1, 1)^\top$. $\lambda = 3$: am 2, eigenspace $\{x + y + z = 0\}$ with basis $(-1, 1, 0)^\top, (-1, 0, 1)^\top$, gm 2. A is diagonalizable.

Where marks are lost

Giving only one eigenvector for $\lambda = 3$ and then wrongly concluding “not diagonalizable”. Sign slips when expanding the 3×3 determinant (check: $\text{tr } A = 12 = 6 + 3 + 3$). Writing the eigenspace as a single vector instead of a basis of two.

Answers at a glance

(a) $-(\lambda - 6)(\lambda - 3)^2$; $\lambda = 6$ (am 1), $\lambda = 3$ (am 2) (b) $(1, 1, 1)^\top; (-1, 1, 0)^\top, (-1, 0, 1)^\top$; gm 1, 2
(c) diagonalizable.

Problem 25 · Parametric eigen-analysis

Let $a, b, c \in \mathbb{R}$ with $b \neq 0$ and consider the upper-triangular matrix

$$A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}.$$

- (a) Find the eigenvalues and an eigenvector for each (in terms of a, b, c).
- (b) State, with justification, exactly when A is diagonalizable.

How to attempt this (the plan)

What it tests: eigen-analysis in symbols and the role of a repeated eigenvalue. **Plan:** eigenvalues are the diagonal entries; solve $(A - \lambda I)v = 0$ for each; then split into the cases $a \neq c$ and $a = c$.

Worked solution

STEP 1 (a) Triangular, so $\lambda_1 = a, \lambda_2 = c$.

STEP 2 (a) $\lambda_1 = a$:

$$A - aI = \begin{bmatrix} 0 & b \\ 0 & c - a \end{bmatrix} \Rightarrow by = 0, (c - a)y = 0.$$

Since $b \neq 0, y = 0$ and x is free: $v_1 = (1, 0)^\top$.

STEP 3 (a) $\lambda_2 = c$ (assume $c \neq a$):

$$A - cI = \begin{bmatrix} a-c & b \\ 0 & 0 \end{bmatrix} \Rightarrow (a-c)x + by = 0 \Rightarrow x = -\frac{b}{a-c}y.$$

Taking $y = a - c$: $v_2 = (-b, a - c)^\top$.

STEP 4 (b) Case $a \neq c$: two distinct eigenvalues give two independent eigenvectors, so A is diagonalizable.

STEP 5 (b) Case $a = c$: $\lambda = a$ has algebraic multiplicity 2, but

$$A - aI = \begin{bmatrix} 0 & b \\ 0 & 0 \end{bmatrix} \text{ has rank 1 } (b \neq 0),$$

so the eigenspace is one-dimensional. Geometric multiplicity $1 < 2$: not diagonalizable.

Answer. $\lambda_1 = a$, $v_1 = (1, 0)^\top$; $\lambda_2 = c$, $v_2 = (-b, a - c)^\top$. A is diagonalizable $\iff a \neq c$ (when $a = c$ and $b \neq 0$ it is defective).

Where marks are lost

Dividing by $a - c$ without noting it may be zero. Writing $v_2 = (b, a - c)^\top$ (sign error). Claiming “always diagonalizable because triangular” — the case $a = c$ is a Jordan block.

Answers at a glance

(a) $a, (1, 0)^\top$; $c, (-b, a - c)^\top$ (b) diagonalizable iff $a \neq c$.

Problem 26 · Is the 3×3 matrix diagonalizable?

Decide whether

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

is diagonalizable. Justify using eigenvalues, algebraic and geometric multiplicities.

How to attempt this (the plan)

What it tests: the multiplicity test on a matrix with one repeated eigenvalue. **Plan:** read the eigenvalues off the diagonal; compute $\text{rank}(A - 2I)$; compare gm with am for $\lambda = 2$.

Worked solution

STEP 1 (λ) A is upper-triangular, so $\lambda = 2$ (algebraic multiplicity 2) and $\lambda = 3$ (algebraic multiplicity 1).

STEP 2 (gm) Eigenspace of the repeated eigenvalue:

$$A - 2I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{rank} = 2.$$

Null space has dimension $3 - 2 = 1$: $y = 0, z = 0, x$ free, so the eigenspace is $\text{span}\{(1, 0, 0)^\top\}$ and $\text{gm} = 1$.

STEP 3 **test** For $\lambda = 2$, $\text{gm} = 1 < 2 = \text{am}$. So A cannot have a basis of eigenvectors.

Answer. A is not diagonalizable: for $\lambda = 2$, geometric multiplicity $1 < \text{algebraic multiplicity } 2$ (only one independent eigenvector $(1, 0, 0)^\top$). The block $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ is a defective Jordan block.

Where marks are lost

Counting three eigenvalues $(2, 2, 3)$ and calling that three eigenvectors. Miscounting $\text{rank}(A - 2I)$ as 1. Forgetting to state the comparison $\text{gm} < \text{am}$ explicitly — that is the justification asked for.

Answers at a glance

$\lambda = 2$ (am 2, gm 1), $\lambda = 3$ (am 1, gm 1) not diagonalizable.

Problem 27 · Powers without brute force: determinant and trace

Consider

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 3 \\ 1 & -1 & 4 \end{bmatrix}, \quad \text{whose characteristic polynomial is } (\lambda - 1)(\lambda - 2)(\lambda - 3).$$

Without forming any matrix power explicitly, compute

- (a) $\det(A^8)$,
- (b) $\text{tr}(A^6)$,
- (c) $\text{tr}(A^6 - 7A^2)$.

How to attempt this (the plan)

What it tests: $\det(A^k) = (\det A)^k$ and $\text{tr}(A^k) = \sum \lambda_i^k$. **Plan:** read the eigenvalues from the factored polynomial, then work only with the scalars 1, 2, 3.

Worked solution

STEP 1 **(λ)** Eigenvalues from the factored polynomial: $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$. Sanity check: $\text{tr } A = 2 + 0 + 4 = 6 = 1 + 2 + 3$ and $\det A = 1 \cdot 2 \cdot 3 = 6$.

STEP 2 **(a)** A^k has eigenvalues λ_i^k , so $\det(A^k) = (\det A)^k$:

$$\det(A^8) = (\det A)^8 = 6^8 = 1\,679\,616.$$

STEP 3 **(b)** Trace of a power:

$$\text{tr}(A^6) = \sum_i \lambda_i^6 = 1^6 + 2^6 + 3^6 = 1 + 64 + 729 = 794.$$

STEP 4 (c) Trace is linear, and A^2 has eigenvalues λ_i^2 :

$$\operatorname{tr}(A^6 - 7A^2) = \sum_i (\lambda_i^6 - 7\lambda_i^2) = 794 - 7(1 + 4 + 9) = 794 - 98 = 696.$$

Answer. $\det(A^8) = 6^8 = 1\,679\,616$; $\operatorname{tr}(A^6) = 794$; $\operatorname{tr}(A^6 - 7A^2) = 696$.

Where marks are lost

Writing $\operatorname{tr}(A^6) = (\operatorname{tr} A)^6$ — trace does not behave like that; only the determinant is multiplicative. Forgetting the 7 multiplies the whole sum $\sum \lambda_i^2$. Multiplying out A^6 by hand, which the question forbids and which wastes time.

Answers at a glance

(a) 1 679 616 (b) 794 (c) 696.

Problem 28 · Eigenvalues of a matrix polynomial $p(A)$

Let A be a 3×3 matrix whose eigenvalues are 1, 2, 4, and define

$$B = A^2 - 5A + 6I.$$

- Show that if λ is an eigenvalue of A (with eigenvector v), then $\lambda^2 - 5\lambda + 6$ is an eigenvalue of B (with the same eigenvector).
- Hence find the eigenvalues of B , then $\det(B)$ and $\operatorname{tr}(B)$, and state whether B is invertible.
- For which eigenvalues of A would B fail to be invertible?

How to attempt this (the plan)

What it tests: the spectral mapping $\lambda \mapsto p(\lambda)$. **Plan:** apply B to v and pull out scalars; evaluate p at 1, 2, 4; multiply/add for $\det$ and tr ; find the roots of p for part (c).

Worked solution

STEP 1 (a) Suppose $Av = \lambda v$. Then $A^2v = A(\lambda v) = \lambda^2v$, so

$$Bv = (A^2 - 5A + 6I)v = \lambda^2v - 5\lambda v + 6v = (\lambda^2 - 5\lambda + 6)v.$$

Thus v is still an eigenvector, now with eigenvalue $p(\lambda) = \lambda^2 - 5\lambda + 6$.

STEP 2 (b) Apply p to each eigenvalue of A :

$$p(1) = 1 - 5 + 6 = 2, \quad p(2) = 4 - 10 + 6 = 0, \quad p(4) = 16 - 20 + 6 = 2.$$

So B has eigenvalues $\{2, 0, 2\}$.

STEP 3 (b) Hence

$$\det(B) = 2 \cdot 0 \cdot 2 = 0, \quad \operatorname{tr}(B) = 2 + 0 + 2 = 4.$$

Because 0 is an eigenvalue, B is not invertible — even though A is (none of its eigenvalues is 0).

STEP 4 (c) B is singular exactly when $p(\lambda) = 0$ for some eigenvalue λ of A . Since $p(\lambda) = (\lambda - 2)(\lambda - 3)$, this happens iff A has 2 or 3 as an eigenvalue. Here $\lambda = 2$ occurs, which is why B is singular.

STEP 5 (note) Remark (Cayley–Hamilton). If instead $A = \begin{bmatrix} 4 & 5 \\ 0 & 1 \end{bmatrix}$, with eigenvalues 1, 4 and characteristic polynomial $\lambda^2 - 5\lambda + 4$, then $A^2 - 5A + 4I = 0$ and so $A^2 - 5A + 6I = 2I$: $p(A)$ collapses to a multiple of I .

Answer. $p(\lambda) = \lambda^2 - 5\lambda + 6$ maps each eigenvalue; eigenvalues of B are 2, 0, 2. $\det B = 0$, $\text{tr } B = 4$, so B is not invertible. B is singular iff A has eigenvalue 2 or 3 (the roots of p).

Where marks are lost

Forgetting the $6I$ term when applying B to v (it contributes $6v$, not 6). Concluding “ A invertible, so B invertible”. In (c), answering only “ $\lambda = 2$ ” and missing the second root $\lambda = 3$.

Answers at a glance

(a) $Bv = p(\lambda)v$ (b) 2, 0, 2; $\det B = 0$, $\text{tr } B = 4$, singular (c) $\lambda = 2$ or 3.

Problem 29 · AB and BA share their eigenvalues

Let $A, B \in \mathbb{R}^{n \times n}$.

- (a) Prove: if $\lambda \neq 0$ is an eigenvalue of AB with eigenvector $\mathbf{v}$, then λ is an eigenvalue of BA , and give its eigenvector. [1.5]
- (b) Prove that if 0 is an eigenvalue of AB then 0 is an eigenvalue of BA . [0.5]
- (c) Verify both statements numerically for $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$: compute AB , BA , both characteristic polynomials and the eigenvalues. [1]

How to attempt this (the plan)

What it tests: a two-line eigenvector argument (multiply the eigen-equation by B) and the determinant route for $\lambda = 0$. **Plan:** start from $AB\mathbf{v} = \lambda\mathbf{v}$, apply B , group as $(BA)(B\mathbf{v})$; explain why $B\mathbf{v} \neq \mathbf{0}$; for (b) use $\det(BA) = \det(AB)$; then compute in (c).

Worked solution

STEP 1 (a) Suppose $AB\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, $\lambda \neq 0$. Multiply on the left by B :

$$B(AB\mathbf{v}) = \lambda B\mathbf{v} \implies (BA)(B\mathbf{v}) = \lambda(B\mathbf{v}).$$

STEP 2 (a) The vector $\mathbf{w} = B\mathbf{v}$ is nonzero: if $B\mathbf{v} = \mathbf{0}$ then $AB\mathbf{v} = \mathbf{0} = \lambda\mathbf{v}$, forcing $\lambda = 0$ (since $\mathbf{v} \neq \mathbf{0}$), a contradiction. Hence $\mathbf{w} = B\mathbf{v}$ is an eigenvector of BA with eigenvalue λ .

STEP 3 (b) 0 is an eigenvalue of $AB \iff \det(AB) = 0$. But $\det(BA) = \det B \det A = \det A \det B = \det(AB) = 0$, so 0 is an eigenvalue of BA too.

STEP 4 (c)

$$AB = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1+4 & 0+2 \\ 0+2 & 0+1 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}, \quad BA = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4+1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}.$$

$$\det(AB - \lambda I) = (5 - \lambda)(1 - \lambda) - 4 = \lambda^2 - 6\lambda + 1,$$

$$\det(BA - \lambda I) = (1 - \lambda)(5 - \lambda) - 4 = \lambda^2 - 6\lambda + 1.$$

Same polynomial, so the same eigenvalues:

$$\lambda = \frac{6 \pm \sqrt{36 - 4}}{2} = 3 \pm 2\sqrt{2}.$$

Consistency check: $\text{tr}(AB) = \text{tr}(BA) = 6$ and $\det(AB) = \det(BA) = 1$. ✓

Answer. (a) $BA(B\mathbf{v}) = \lambda(B\mathbf{v})$ with $B\mathbf{v} \neq \mathbf{0}$. (b) $\det(BA) = \det(AB) = 0$. (c) $AB = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$,
 $BA = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$, both with $\lambda^2 - 6\lambda + 1$, eigenvalues $3 \pm 2\sqrt{2}$.

Where marks are lost

Forgetting to show $B\mathbf{v} \neq \mathbf{0}$ (an eigenvector must be nonzero — that is where $\lambda \neq 0$ is used). Claiming $AB = BA$ — they differ here, only the eigenvalues agree. Expanding the characteristic polynomial with a sign slip in the -4 .

Answers at a glance

(a) eigenvector $B\mathbf{v}$ (b) via $\det(AB) = \det(BA)$ (c) $\lambda^2 - 6\lambda + 1$, $\lambda = 3 \pm 2\sqrt{2}$ for both

Problem 30 · Eigenvalues and eigenvectors of a rotation

Let $A(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$.

- (a) For $\theta = 60^\circ$, find all eigenvalues of $A(\theta)$. [1]
 (b) Find the corresponding eigenvectors. [1.5]
 (c) Use the eigenvalues to show $A(60^\circ)^6 = I$. For $\theta = 180^\circ$, state the eigenvalues and the eigenspace(s). [0.5]

How to attempt this (the plan)

What it tests: a real matrix with complex eigenvalues (a rotation moves every real vector, so no real eigenvector exists), solving $(A - \lambda I)\mathbf{x} = \mathbf{0}$ with complex entries, and λ^6 from the polar form. **Plan:** write $A(60^\circ)$ with $\cos = \frac{1}{2}$, $\sin = \frac{\sqrt{3}}{2}$; characteristic equation $\lambda^2 - 2\cos \theta \lambda + 1 = 0$; solve; take one row of $(A - \lambda I)$ for each eigenvector; write $\lambda = e^{\pm i\pi/3}$.

Worked solution

STEP 1 (a) $A = A(60^\circ) = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$.

$$\det(A - \lambda I) = \left(\frac{1}{2} - \lambda\right)^2 + \frac{3}{4} = \lambda^2 - \lambda + 1 = 0.$$

$$\lambda = \frac{1 \pm \sqrt{1 - 4}}{2} = \frac{1 \pm i\sqrt{3}}{2}.$$

In polar form $\lambda_{1,2} = \cos 60^\circ \pm i \sin 60^\circ = e^{\pm i\pi/3}$; both have modulus 1.

STEP 2 (b) For $\lambda_1 = \frac{1+i\sqrt{3}}{2}$: $A - \lambda_1 I = \begin{bmatrix} -\frac{i\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{i\sqrt{3}}{2} \end{bmatrix}$. Row 1 gives

$$-\frac{i\sqrt{3}}{2}x_1 - \frac{\sqrt{3}}{2}x_2 = 0 \implies x_2 = -ix_1, \quad \mathbf{x}^{(1)} = \alpha \begin{bmatrix} 1 \\ -i \end{bmatrix}, \alpha \neq 0.$$

Check with A directly: $A \begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} \frac{1}{2} + \frac{i\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} - \frac{i}{2} \end{bmatrix}$ and $\lambda_1 \begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} \frac{1+i\sqrt{3}}{2} \\ \frac{-i+i\sqrt{3}}{2} \end{bmatrix}$. ✓

STEP 3 (b) For $\lambda_2 = \overline{\lambda_1}$ take the conjugate eigenvector:

$$\mathbf{x}^{(2)} = \beta \begin{bmatrix} 1 \\ i \end{bmatrix}, \beta \neq 0.$$

(Row 1 of $A - \lambda_2 I$: $\frac{i\sqrt{3}}{2}x_1 - \frac{\sqrt{3}}{2}x_2 = 0 \implies x_2 = ix_1$.)

STEP 4 (c) $\lambda^6 = (e^{\pm i\pi/3})^6 = e^{\pm 2\pi i} = 1$ for both eigenvalues. A is diagonalizable over $\mathbb{C}$ (two distinct eigenvalues), $A = PDP^{-1}$, so $A^6 = PD^6P^{-1} = PIP^{-1} = I$. (Geometrically: six rotations by 60° make a full turn.)

For $\theta = 180^\circ$: $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$, so $\lambda = -1$ with algebraic multiplicity 2; every nonzero vector is an eigenvector, and the eigenspace is all of $\mathbb{R}^2$ (dimension 2).

Answer. (a) $\lambda = \frac{1 \pm i\sqrt{3}}{2} = e^{\pm i\pi/3}$. (b) $\alpha(1, -i)^\top$ for λ_1 , $\beta(1, i)^\top$ for λ_2 . (c) $\lambda^6 = 1 \implies A^6 = I$; at 180° : $\lambda = -1$ (twice), eigenspace $\mathbb{R}^2$.

Where marks are lost

Stopping at "no real eigenvalues" — the exam wants the complex ones. Dropping the $+\frac{3}{4}$ when expanding $(\frac{1}{2} - \lambda)^2 + \frac{3}{4}$. Pairing the wrong eigenvector with the wrong eigenvalue (always check one product $A\mathbf{x} = \lambda\mathbf{x}$).

Answers at a glance

(a) $\frac{1 \pm i\sqrt{3}}{2}$ (b) $(1, -i)$ and $(1, i)$ (c) $A^6 = I$; $\theta = 180^\circ$: $\lambda = -1, -1$, eigenspace $\mathbb{R}^2$

Topic 6

Eigen Decomposition (Diagonalization)

Compute P and D with $A = PDP^{-1}$, and for a symmetric matrix an orthogonal P with $A = PDP^T$, including repeated and zero eigenvalues; then use the decomposition for matrix powers.

Formulas you need for this topic

- $A = PDP^{-1}$ with the eigenvectors as the columns of P , in the *same order* as the eigenvalues in $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.
- Symmetric A : eigenvectors for different eigenvalues are automatically orthogonal, so P can be taken orthogonal and $A = PDP^T = \sum_i \lambda_i \mathbf{p}_i \mathbf{p}_i^T$.
- A repeated eigenvalue needs as many independent eigenvectors as its multiplicity; for an orthogonal P , choose them orthogonal to each other.
- $A^n = PD^nP^{-1}$ with $D^n = \text{diag}(\lambda_1^n, \dots, \lambda_n^n)$.

Problem 31 · Eigendecomposition $A = PDP^{-1}$

Find an eigendecomposition $A = PDP^{-1}$ of the non-symmetric matrix

$$A = \begin{bmatrix} 5 & -1 \\ 2 & 2 \end{bmatrix},$$

giving P , D , P^{-1} explicitly, and verify the factorization.

How to attempt this (the plan)

What it tests: building P and D from eigenpairs and inverting a 2×2 . **Plan:** characteristic polynomial; one eigenvector per root; stack as columns of P ; invert with the adjugate formula; multiply back.

Worked solution

STEP 1 λ Eigenvalues:

$$\det(A - \lambda I) = (5 - \lambda)(2 - \lambda) + 2 = \lambda^2 - 7\lambda + 12 = (\lambda - 3)(\lambda - 4),$$

so $\lambda_1 = 3$, $\lambda_2 = 4$.

STEP 2 v Eigenvectors.

$$\lambda_1 = 3: \quad A - 3I = \begin{bmatrix} 2 & -1 \\ 2 & -1 \end{bmatrix} \Rightarrow 2x = y, \quad p_1 = (1, 2)^T.$$

$$\lambda_2 = 4: \quad A - 4I = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix} \Rightarrow x = y, \quad p_2 = (1, 1)^T.$$

STEP 3 P, D Assemble:

$$P = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}, \quad \det P = -1, \quad P^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}.$$

STEP 4 check Verify:

$$PD = \begin{bmatrix} 3 & 4 \\ 6 & 4 \end{bmatrix},$$

$$PDP^{-1} = \begin{bmatrix} 3 & 4 \\ 6 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 5 & -1 \\ 2 & 2 \end{bmatrix} = A. \checkmark$$

Answer. $A = PDP^{-1}$ with $P = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$, $P^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$.

Where marks are lost

Putting the eigenvalues in D in an order that does not match the columns of P . Forgetting the $1/\det P$ factor (here -1) in P^{-1} . Using $P^\top$ instead of P^{-1} — A is not symmetric, so P is not orthogonal.

Answers at a glance

$$\lambda = 3, 4 \quad P = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}, D = \text{diag}(3, 4), P^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}.$$

Problem 32 · Eigendecomposition of a rank-deficient symmetric matrix, and a matrix power

Consider the data matrix

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

- (a) Compute P and D with $A = PDP^{-1}$, choosing P orthogonal (so $P^{-1} = P^\top$). [2.5]
 (b) State rank A from the eigenvalues. [0.5]
 (c) Show that $A^n = 5^{n-1}A$ for every $n \geq 1$, and hence write A^{10} . [1]

How to attempt this (the plan)

What it tests: the characteristic polynomial of a block matrix, a repeated eigenvalue whose eigenspace is 2-dimensional (so the matrix is still diagonalizable), orthonormal eigenvectors for a symmetric matrix, and using D^n . **Plan:** expand $\det(A - \lambda I)$ along the third row; solve the 2×2 block; find one eigenvector per eigenvalue, normalise; for the repeated eigenvalue pick two orthogonal ones; then $A^n = PD^nP^\top$.

Worked solution**STEP 1** (a) Characteristic polynomial (expand along row 3):

$$\det(A - \lambda I) = (5 - \lambda) \det \begin{bmatrix} 1 - \lambda & 2 \\ 2 & 4 - \lambda \end{bmatrix} = (5 - \lambda) [(1 - \lambda)(4 - \lambda) - 4].$$

$$(1 - \lambda)(4 - \lambda) - 4 = \lambda^2 - 5\lambda = \lambda(\lambda - 5).$$

$$\det(A - \lambda I) = -(\lambda - 5)^2\lambda = 0 \implies \lambda = 0, 5, 5.$$

STEP 2 (a) $\lambda = 0$: solve $A\mathbf{x} = \mathbf{0}$: rows give $x_1 + 2x_2 = 0$ and $5x_3 = 0$, so $\mathbf{x} = t(2, -1, 0)^\top$.

$$\mathbf{p}_1 = \frac{1}{\sqrt{5}}(2, -1, 0)^\top.$$

STEP 3 (a) $\lambda = 5$: $A - 5I = \begin{bmatrix} -4 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Only one independent equation, $-4x_1 + 2x_2 = 0 \Rightarrow x_2 = 2x_1$; x_3 is free. Eigenspace = $\{(s, 2s, t)\}$, dimension 2. Two orthogonal choices:

$$(1, 2, 0)^\top \text{ and } (0, 0, 1)^\top \quad (\text{dot product } 0).$$

$$\mathbf{p}_2 = \frac{1}{\sqrt{5}}(1, 2, 0)^\top, \quad \mathbf{p}_3 = (0, 0, 1)^\top.$$

Note $\mathbf{p}_1 \cdot \mathbf{p}_2 = \frac{1}{5}(2 - 2 + 0) = 0$ — eigenvectors of a symmetric matrix for different eigenvalues are automatically orthogonal.

STEP 4 (a)

$$P = \begin{bmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} & 0 \\ -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad D = \text{diag}(0, 5, 5), \quad A = PDP^\top.$$

Spot check: $PDP^\top = 5\mathbf{p}_2\mathbf{p}_2^\top + 5\mathbf{p}_3\mathbf{p}_3^\top = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix} = A. \checkmark$

STEP 5 (b) Two nonzero eigenvalues $\Rightarrow \text{rank } A = 2$ (the eigenvalue 0 has a 1-dimensional eigenspace = null space).

STEP 6 (c) $A^n = PD^nP^\top$ with $D^n = \text{diag}(0, 5^n, 5^n) = 5^{n-1} \text{diag}(0, 5, 5) = 5^{n-1}D$. Hence

$$A^n = P(5^{n-1}D)P^\top = 5^{n-1}PDP^\top = 5^{n-1}A.$$

$$A^{10} = 5^9 A = 1953125 \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Check $n = 2$ directly: $A^2 = \begin{bmatrix} 1+4 & 2+8 & 0 \\ 2+8 & 4+16 & 0 \\ 0 & 0 & 25 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 0 \\ 10 & 20 & 0 \\ 0 & 0 & 25 \end{bmatrix} = 5A. \checkmark$

Answer. (a) $\lambda = 0, 5, 5$; $P = \left[\frac{(2, -1, 0)}{\sqrt{5}} \quad \frac{(1, 2, 0)}{\sqrt{5}} \quad (0, 0, 1) \right]$, $D = \text{diag}(0, 5, 5)$. (b) $\text{rank } A = 2$. (c) $A^n = 5^{n-1}A$, $A^{10} = 5^9 A$.

Where marks are lost

Giving only one eigenvector for $\lambda = 5$ (the eigenspace is 2-dimensional — you need two, and for an orthogonal P they must be orthogonal to each other). Listing eigenvectors in an order that does not match the diagonal of D . Normalising $(0, 0, 1)$ "by $\sqrt{5}$ " out of habit.

Answers at a glance

(a) $\lambda = 0, 5, 5$; $\mathbf{p}_1 = \frac{(2, -1, 0)}{\sqrt{5}}$, $\mathbf{p}_2 = \frac{(1, 2, 0)}{\sqrt{5}}$, $\mathbf{p}_3 = (0, 0, 1)$ (b) 2 (c) $A^{10} = 5^9 A$

Topic 7

SVD & Singular Values

Eigenvalues of $AA^\top$ or $A^\top A$ to singular values to rank and norms, the full decomposition $A = U\Sigma V^\top$ of a small matrix, the link $\mathbf{v}_1 = \frac{1}{\sigma_1} A^\top \mathbf{u}_1$, and how the SVD relates to the eigendecomposition.

Formulas you need for this topic

- $\sigma_i = \sqrt{\lambda_i(A^\top A)} = \sqrt{\lambda_i(AA^\top)}$ — use whichever of the two is the smaller matrix.
- $\text{rank } A = \text{number of nonzero singular values}$; $\|A\|_2 = \sigma_1$; $\|A\|_F^2 = \sum_i \sigma_i^2 = \sum_{i,j} a_{ij}^2$.
- $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$, hence $\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$ and $\mathbf{v}_i = \frac{1}{\sigma_i} A^\top \mathbf{u}_i$.
- $A = U\Sigma V^\top$ with U ($m \times m$) and V ($n \times n$) orthogonal; extra columns of U or V come from orthogonal completion of a basis.
- Eigenvalues can be negative, singular values never; for a symmetric A , $\sigma_i = |\lambda_i|$.

Problem 33 · Full SVD of a 3×2 matrix

Find a full singular value decomposition $A = U\Sigma V^\top$ of

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix},$$

including the third column of U obtained by orthogonal completion.

How to attempt this (the plan)

What it tests: the standard SVD recipe for a tall matrix. **Plan:** form $A^\top A$ (2×2); its eigenvalues give σ_i and eigenvectors give $\mathbf{v}_i$; then $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$; complete U with a unit vector orthogonal to $\mathbf{u}_1, \mathbf{u}_2$; assemble Σ as 3×2 .

Worked solution

STEP 1 σ, V $A^\top A$, singular values, right singular vectors:

$$A^\top A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

$$\det \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = (2-\lambda)^2 - 1 = 0 \Rightarrow \lambda = 3, 1, \quad \sigma_1 = \sqrt{3}, \sigma_2 = 1.$$

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)^\top \quad (\lambda = 3), \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)^\top \quad (\lambda = 1).$$

STEP 2 u_1, u_2 Left singular vectors $\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i$:

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}} A \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \frac{1}{1} A \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}.$$

STEP 3 u_3 Orthogonal completion $u_3 = (p, q, r)^\top$ with $u_3 \perp u_1, u_2$: from $u_2, q - r = 0 \Rightarrow q = r$; from $u_1, 2p + q + r = 0 \Rightarrow p = -q$. So

$$u_3 = \frac{1}{\sqrt{3}}(-1, 1, 1)^\top.$$

STEP 4 $U\Sigma V^\top$ Assemble:

$$U = \begin{bmatrix} \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad V^\top = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Answer. $\sigma_1 = \sqrt{3}, \sigma_2 = 1; v_1 = \frac{1}{\sqrt{2}}(1, 1)^\top, v_2 = \frac{1}{\sqrt{2}}(1, -1)^\top; u_1 = \frac{1}{\sqrt{6}}(2, 1, 1)^\top, u_2 = \frac{1}{\sqrt{2}}(0, 1, -1)^\top, u_3 = \frac{1}{\sqrt{3}}(-1, 1, 1)^\top; \Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$

Where marks are lost

Writing $\sigma_i = \lambda_i$ instead of $\sqrt{\lambda_i}$. Forgetting to divide Av_i by σ_i (then u_1 is not a unit vector). Making Σ square 2×2 — for a full SVD it must be 3×2 with a zero row. Leaving u_3 unnormalised.

Answers at a glance

$$\sigma = \sqrt{3}, 1 \quad V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad u_1 = \frac{1}{\sqrt{6}}(2, 1, 1), u_2 = \frac{1}{\sqrt{2}}(0, 1, -1), u_3 = \frac{1}{\sqrt{3}}(-1, 1, 1).$$

Problem 34 · Full SVD of a 2×3 matrix

Find the SVD of

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

(Hint: it is quicker to get U and the singular values from the 2×2 matrix $AA^\top$, then recover V from $v_i = \frac{1}{\sigma_i}A^\top u_i$; the missing right vector spans $\ker(A^\top A)$.)

How to attempt this (the plan)

What it tests: the SVD of a wide matrix, using the smaller Gram matrix. **Plan:** form $AA^\top$; eigenpairs give σ_i and u_i ; $v_i = A^\top u_i / \sigma_i$; the third v spans $\ker A$; Σ is 2×3 .

Worked solution

STEP 1 σ, U $AA^\top$, singular values, left singular vectors:

$$AA^\top = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \Rightarrow \lambda = 3, 1, \quad \sigma_1 = \sqrt{3}, \sigma_2 = 1.$$

$$u_1 = \frac{1}{\sqrt{2}}(1, 1)^\top (\lambda = 3), \quad u_2 = \frac{1}{\sqrt{2}}(1, -1)^\top (\lambda = 1).$$

STEP 2 v_1, v_2 Right singular vectors $v_i = \frac{1}{\sigma_i} A^\top u_i$:

$$v_1 = \frac{1}{\sqrt{3}} A^\top \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad v_2 = \frac{1}{1} A^\top \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

STEP 3 v_3 Third right vector from the null space. A is 2×3 , so V needs a third column v_3 with singular value 0, spanning $\ker A (= \ker A^\top A)$. Solving $Av = 0$: $x + y = 0$, $y + z = 0$ gives

$$v_3 = \frac{1}{\sqrt{3}}(1, -1, 1)^\top.$$

STEP 4 $U\Sigma V^\top$ Assemble:

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad V = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{bmatrix}.$$

The single zero column of Σ reflects that A has rank 2 but 3 columns.

Answer. $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$; $U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $\Sigma = \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $v_1 = \frac{1}{\sqrt{6}}(1, 2, 1)^\top$, $v_2 = \frac{1}{\sqrt{2}}(1, 0, -1)^\top$, $v_3 = \frac{1}{\sqrt{3}}(1, -1, 1)^\top$.

Where marks are lost

Computing the 3×3 matrix $A^\top A$ and then struggling with a cubic — use the hint. Forgetting v_3 altogether, leaving V as 3×2 (then V is not orthogonal). Sign of v_3 : any unit vector in $\ker A$ is fine, but it must satisfy $Av_3 = 0$.

Answers at a glance

$$\sigma = \sqrt{3}, 1 \quad U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad v_1 = \frac{1}{\sqrt{6}}(1, 2, 1), v_2 = \frac{1}{\sqrt{2}}(1, 0, -1), v_3 = \frac{1}{\sqrt{3}}(1, -1, 1).$$

Problem 35 · Capstone: eigendecomposition vs. SVD of one matrix

For $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$:

- find its eigendecomposition (note the negative eigenvalue);
- find its SVD;
- explain precisely how the two are related.

How to attempt this (the plan)

What it tests: how EVD and SVD relate for a symmetric matrix with a negative eigenvalue. **Plan:** EVD via the quadratic and orthonormal eigenvectors; SVD via $A^\top A$, computing $u_i = Av_i/\sigma_i$

and watching the sign; then state $\sigma_i = |\lambda_i|$ and $u_i = \pm v_i$.

Worked solution

STEP 1 (a) Eigendecomposition:

$$\det(A - \lambda I) = (1 - \lambda)^2 - 4 = (\lambda - 3)(\lambda + 1) \Rightarrow \lambda_1 = 3, \lambda_2 = -1.$$

Orthonormal eigenvectors $q_1 = \frac{1}{\sqrt{2}}(1, 1)^\top$ (for 3) and $q_2 = \frac{1}{\sqrt{2}}(1, -1)^\top$ (for -1). With $Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$,

$$A = Q \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix} Q^\top.$$

STEP 2 (b) SVD. $A^\top A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ has eigenvalues 9 and 1, so $\sigma_1 = 3, \sigma_2 = 1$ — note $\sigma_i = |\lambda_i|$.

The right singular vectors are the same eigenvectors $v_1 = \frac{1}{\sqrt{2}}(1, 1)^\top, v_2 = \frac{1}{\sqrt{2}}(1, -1)^\top$, and

$$u_1 = \frac{1}{3}Av_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = v_1, \quad u_2 = \frac{1}{1}Av_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -v_2.$$

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \quad V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

STEP 3 (c) For a symmetric matrix the singular values are the absolute values of the eigenvalues, $\sigma_i = |\lambda_i|$, and the right singular vectors are the eigenvectors. Where an eigenvalue is positive, $u_i = v_i$; where it is negative (here $\lambda_2 = -1$) the minus sign is absorbed into the left vector, $u_i = -v_i$. Thus EVD and SVD coincide exactly when every eigenvalue is non-negative (a positive-semidefinite matrix); otherwise they differ only by these sign flips between U and V .

Answer. EVD: $A = Q \operatorname{diag}(3, -1) Q^\top, Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. SVD: $\Sigma = \operatorname{diag}(3, 1) = \operatorname{diag}(|\lambda_i|)$,
 $V = Q, U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ (the negative eigenvalue flips the sign of u_2 relative to v_2).

Where marks are lost

Writing $\Sigma = \operatorname{diag}(3, -1)$ — singular values are never negative. Setting $U = V$ without computing $u_2 = Av_2/\sigma_2$, which misses the sign flip. Claiming “EVD = SVD for every symmetric matrix” — only for positive-semidefinite ones.

Answers at a glance

(a) $\lambda = 3, -1, Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ (b) $\sigma = 3, 1, V = Q, U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ (c) $\sigma_i = |\lambda_i|$,
 $u_i = \operatorname{sign}(\lambda_i)v_i$.

Problem 36 · Singular values, rank, norms and the first singular vectors of a 2×3 matrix

Let

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 1 & -2 \end{bmatrix}.$$

- (a) Compute $AA^\top$ and its eigenvalues; hence the singular values of A . [1.5]
 (b) State $\text{rank } A$, the spectral norm $\|A\|_2$, and the Frobenius norm $\|A\|_F$; check $\|A\|_F^2 = \sigma_1^2 + \sigma_2^2$. [1]
 (c) Find the first left singular vector $\mathbf{u}_1$ and, using $\mathbf{v}_1 = \frac{1}{\sigma_1} A^\top \mathbf{u}_1$, the first right singular vector $\mathbf{v}_1$. [1]

How to attempt this (the plan)

What it tests: $\sigma_i = \sqrt{\lambda_i(AA^\top)}$, $\text{rank} = \text{number of nonzero singular values}$, $\|A\|_2 = \sigma_1$, $\|A\|_F^2 = \sum \sigma_i^2 = \sum a_{ij}^2$, and the link $A^\top \mathbf{u}_1 = \sigma_1 \mathbf{v}_1$. **Plan:** the small matrix $AA^\top$ is 2×2 — use it; eigenvector for the largest eigenvalue; multiply by $A^\top$; divide by σ_1 .

Worked solution**STEP 1 (a)**

$$AA^\top = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 1+1+4 & 1+1-4 \\ 1+1-4 & 1+1+4 \end{bmatrix} = \begin{bmatrix} 6 & -2 \\ -2 & 6 \end{bmatrix}.$$

$$\det(AA^\top - \lambda I) = (6 - \lambda)^2 - 4 = 0 \implies 6 - \lambda = \pm 2 \implies \lambda = 8, 4.$$

$$\sigma_1 = \sqrt{8} = 2\sqrt{2}, \quad \sigma_2 = \sqrt{4} = 2.$$

($A^\top A$ is 3×3 with eigenvalues 8, 4, 0; the extra 0 is not a singular value of a 2×3 matrix of rank 2.)

STEP 2 (b) Two nonzero singular values $\implies \text{rank } A = 2$.

$$\|A\|_2 = \sigma_1 = 2\sqrt{2} \approx 2.83, \quad \|A\|_F = \sqrt{1+1+4+1+1+4} = \sqrt{12} = 2\sqrt{3}.$$

$$\sigma_1^2 + \sigma_2^2 = 8 + 4 = 12 = \|A\|_F^2. \checkmark$$

STEP 3 (c) Eigenvector of $AA^\top$ for $\lambda = 8$: $(AA^\top - 8I)\mathbf{u} = \begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix} \mathbf{u} = \mathbf{0} \implies u_1 = -u_2$.

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, -1)^\top.$$

$$A^\top \mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}, \quad \mathbf{v}_1 = \frac{1}{\sigma_1} A^\top \mathbf{u}_1 = \frac{1}{2\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Check: $A\mathbf{v}_1 = (2, -2)^\top = 2\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, -1)^\top = \sigma_1 \mathbf{u}_1. \checkmark$

Answer. (a) $AA^\top = \begin{bmatrix} 6 & -2 \\ -2 & 6 \end{bmatrix}$, $\lambda = 8, 4$, $\sigma = 2\sqrt{2}, 2$. (b) $\text{rank } 2$, $\|A\|_2 = 2\sqrt{2}$, $\|A\|_F = 2\sqrt{3}$.
 (c) $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, -1)$, $\mathbf{v}_1 = (0, 0, 1)$.

Where marks are lost

Reporting the eigenvalues 8, 4 as the singular values (take square roots). Using $A^\top A$ (a 3×3 determinant) when $AA^\top$ is 2×2 — more work and more slips. Forgetting to divide by σ_1 when converting $\mathbf{u}_1$ to $\mathbf{v}_1$.

Answers at a glance

(a) $\lambda = 8, 4; \sigma_1 = 2\sqrt{2}, \sigma_2 = 2$ (b) rank 2; $2\sqrt{2}; 2\sqrt{3}$ (c) $\mathbf{u}_1 = \frac{(1, -1)}{\sqrt{2}}, \mathbf{v}_1 = (0, 0, 1)$

Topic 8

Cholesky Decomposition

Derive the lower-triangular L with $B = LL^T$ entry by entry, read off the eigenvalues of L and the determinant of B , solve a system using L alone, and know exactly when the factorization exists.

Formulas you need for this topic

- $B = LL^T$ with L lower triangular and positive diagonal. Match entries column by column:
- $l_{11} = \sqrt{b_{11}}, \quad l_{i1} = b_{i1}/l_{11}, \quad l_{jj} = \sqrt{b_{jj} - \sum_{k<j} l_{jk}^2}, \quad l_{ij} = \frac{b_{ij} - \sum_{k<j} l_{ik}l_{jk}}{l_{jj}} \text{ for } i > j.$
- Such an L exists $\iff B$ is symmetric positive definite. The algorithm breaks (a zero or a negative under the root) exactly when it is not.
- L is triangular, so its eigenvalues are its diagonal entries and $\det B = (\det L)^2$.
- Solving $Bx = b$: forward-substitute $Ly = b$ (top down), then back-substitute $L^T x = y$ (bottom up).

Problem 37 · Cholesky factor of a 3×3 matrix, and what it buys you

Consider

$$B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 13 & 8 \\ 1 & 8 & 9 \end{bmatrix}.$$

- Derive the lower-triangular L with $B = LL^T$ (positive diagonal). [2]
- Write down the eigenvalues of L and compute $\det B$ from L . [0.5]
- Solve $Bx = b$ for $b = (4, 23, 18)^T$ using L only (forward then backward substitution). [1]
- Explain from L why B is positive definite. [0.5]

How to attempt this (the plan)

What it tests: the six Cholesky formulas in order, the triangular-matrix facts (eigenvalues on the diagonal, $\det B = (\det L)^2$), and the two-triangular-solves trick. **Plan:** match entries of LL^T with B column by column: l_{11}, l_{21}, l_{31} from column 1, then l_{22}, l_{32} , then l_{33} ; solve $Ly = b$ top-down, then $L^T x = y$ bottom-up.

Worked solution

STEP 1 (a) Write $L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$. Entry by entry of $LL^T = B$:

$$l_{11}^2 = 1 \Rightarrow l_{11} = 1.$$

$$l_{21}l_{11} = 2 \Rightarrow l_{21} = 2.$$

$$l_{31}l_{11} = 1 \Rightarrow l_{31} = 1.$$

$$l_{21}^2 + l_{22}^2 = 13 \Rightarrow l_{22} = \sqrt{13 - 4} = 3.$$

$$l_{31}l_{21} + l_{32}l_{22} = 8 \Rightarrow l_{32} = \frac{8 - 2}{3} = 2.$$

$$l_{31}^2 + l_{32}^2 + l_{33}^2 = 9 \Rightarrow l_{33} = \sqrt{9 - 1 - 4} = 2.$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & 2 & 2 \end{bmatrix}.$$

Check the $(3,2)$ entry of $LL^\top$: $1 \cdot 2 + 2 \cdot 3 + 2 \cdot 0 = 8$. ✓

STEP 2 (b) L is triangular, so its eigenvalues are its diagonal entries: 1, 3, 2.

$$\det B = \det L \cdot \det L^\top = (1 \cdot 3 \cdot 2)^2 = 36.$$

STEP 3 (c) Forward substitution $Ly = \mathbf{b}$:

$$y_1 = 4.$$

$$2y_1 + 3y_2 = 23 \Rightarrow y_2 = \frac{23 - 8}{3} = 5.$$

$$y_1 + 2y_2 + 2y_3 = 18 \Rightarrow y_3 = \frac{18 - 4 - 10}{2} = 2.$$

Backward substitution $L^\top \mathbf{x} = \mathbf{y}$, with $L^\top = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{bmatrix}$:

$$2x_3 = 2 \Rightarrow x_3 = 1.$$

$$3x_2 + 2x_3 = 5 \Rightarrow x_2 = 1.$$

$$x_1 + 2x_2 + x_3 = 4 \Rightarrow x_1 = 1.$$

Check: $B(1,1,1)^\top = (1 + 2 + 1, 2 + 13 + 8, 1 + 8 + 9) = (4, 23, 18)$. ✓

STEP 4 (d) For any $\mathbf{x} \neq \mathbf{0}$, $\mathbf{x}^\top B \mathbf{x} = \mathbf{x}^\top L L^\top \mathbf{x} = \|L^\top \mathbf{x}\|^2 \geq 0$, and it is 0 only if $L^\top \mathbf{x} = \mathbf{0}$; since $L^\top$ is triangular with nonzero diagonal it is invertible, so $\mathbf{x} = \mathbf{0}$. Hence $\mathbf{x}^\top B \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$: B is positive definite. (Equivalently: a Cholesky factor with positive diagonal exists $\iff B$ is symmetric positive definite.)

Answer. (a) $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & 2 & 2 \end{bmatrix}$. (b) eigenvalues of L : 1, 3, 2; $\det B = 36$. (c) $\mathbf{x} = (1, 1, 1)^\top$. (d) $\mathbf{x}^\top B \mathbf{x} = \|L^\top \mathbf{x}\|^2 > 0$.

Where marks are lost

Computing l_{32} as $8/l_{22}$ and forgetting to subtract $l_{31}l_{21}$ first. Claiming the eigenvalues of B are 1, 3, 2 (those are eigenvalues of L , not of B ; B 's eigenvalues are different, only their product 36 is easy). Solving $L^\top \mathbf{x} = \mathbf{y}$ top-down instead of bottom-up.

Answers at a glance

(a) $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & 2 & 2 \end{bmatrix}$ (b) 1, 3, 2; $\det B = 36$ (c) $(1, 1, 1)$ (d) $\|L^\top \mathbf{x}\|^2 > 0$

Problem 38 · When does a Cholesky factor exist? (a parameter)

Let $A(t) = \begin{bmatrix} 4 & 6 \\ 6 & t \end{bmatrix}$, $t \in \mathbb{R}$.

- (a) Find all t for which $A(t)$ has a Cholesky factorization $A = LL^\top$ with positive diagonal. [1]
 (b) For $t = 13$ derive L and verify $LL^\top = A$. [1.5]
 (c) What happens to the algorithm at $t = 9$? Classify $A(9)$ and exhibit a nonzero $\mathbf{v}$ with $\mathbf{v}^\top A(9)\mathbf{v} = 0$. [0.5]

How to attempt this (the plan)

What it tests: the equivalence "Cholesky with positive diagonal exists $\iff$ symmetric positive definite", Sylvester's criterion with a parameter, and reading the breakdown of the algorithm ($l_{22}^2 \leq 0$). **Plan:** leading minors $4 > 0$ and $4t - 36 > 0$; run the formulas at $t = 13$; at $t = 9$ compute l_{22} .

Worked solution

STEP 1 (a) $A(t)$ is symmetric. Leading principal minors:

$$\Delta_1 = 4 > 0, \quad \Delta_2 = \det A(t) = 4t - 36.$$

Positive definite $\iff \Delta_2 > 0 \iff t > 9$. So a Cholesky factorization with positive diagonal exists exactly when $t > 9$.

STEP 2 (b) With $t = 13$:

$$l_{11} = \sqrt{4} = 2, \quad l_{21} = \frac{6}{l_{11}} = 3, \quad l_{22} = \sqrt{13 - l_{21}^2} = \sqrt{13 - 9} = 2.$$

$$L = \begin{bmatrix} 2 & 0 \\ 3 & 2 \end{bmatrix}, \quad LL^\top = \begin{bmatrix} 2 & 0 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 9+4 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 13 \end{bmatrix} = A(13). \checkmark$$

STEP 3 (c) At $t = 9$: $l_{22} = \sqrt{9 - 9} = 0$ — the algorithm produces a zero on the diagonal (and for $t < 9$ it would need the square root of a negative number). $A(9) = \begin{bmatrix} 4 & 6 \\ 6 & 9 \end{bmatrix}$ has $\Delta_2 = 0$, eigenvalues $\lambda = 0$ and $\lambda = 13$ (trace 13, determinant 0): positive *semidefinite*, singular. Its null vector: $4v_1 + 6v_2 = 0 \Rightarrow \mathbf{v} = (3, -2)^\top$, and

$$\mathbf{v}^\top A(9)\mathbf{v} = (3, -2) \begin{bmatrix} 4 & 6 \\ 6 & 9 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = (3, -2) \cdot (12 - 12, 18 - 18) = 0.$$

Answer. (a) $t > 9$. (b) $L = \begin{bmatrix} 2 & 0 \\ 3 & 2 \end{bmatrix}$. (c) $l_{22} = 0$; $A(9)$ is positive semidefinite (eigenvalues 0, 13); $\mathbf{v} = (3, -2)^\top$ gives $\mathbf{v}^\top A\mathbf{v} = 0$.

Where marks are lost

Answering $t \geq 9$ for (a) — at $t = 9$ the diagonal entry is 0, not positive. Writing $l_{21} = 6$ (must divide by l_{11}). Calling $A(9)$ "not positive definite" without saying what it *is* (semidefinite) when asked to classify.

Answers at a glance

(a) $t > 9$ (b) $L = \begin{bmatrix} 2 & 0 \\ 3 & 2 \end{bmatrix}$ (c) $l_{22} = 0$; PSD, $\lambda = 0, 13$; $\mathbf{v} = (3, -2)$

Topic 9

Positive Definiteness & Quadratic Forms

Conditions on parameters for positive definiteness and semidefiniteness via principal minors or eigenvalues, whether $\mathbf{x}^\top M \mathbf{y}$ is a valid inner product, and showing a quadratic form is positive by writing it as a sum of squares.

Formulas you need for this topic

- Leading principal minors $\Delta_1, \Delta_2, \dots, \Delta_n$ (top-left $k \times k$ determinants).
- Symmetric A is positive definite $\iff$ every $\Delta_k > 0 \iff$ every eigenvalue > 0 .
- Positive semidefinite $\iff$ every principal minor ≥ 0 (not just the leading ones) $\iff$ every eigenvalue ≥ 0 .
- $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top M \mathbf{y}$ is an inner product $\iff M$ is symmetric and positive definite; a non-symmetric M fails at once.
- To show $Q(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} > 0$, complete squares one variable at a time and check the sum is zero only at $\mathbf{x} = \mathbf{0}$.

Problem 39 · Classifying $A(t)$ by eigenvalues

Consider the symmetric matrix $A(t) = \begin{bmatrix} t & 3 \\ 3 & t \end{bmatrix}, t \in \mathbb{R}$.

- Classify $A(t)$ for all real t as positive definite, positive semidefinite, indefinite, negative semidefinite, or negative definite. [3]
- For which t does $\langle x, y \rangle = x^\top A(t) y$ define a valid inner product? At the borderline value $t = 3$, exhibit a nonzero vector v with $\langle v, v \rangle = 0$. [2]

How to attempt this (the plan)

What it tests: definiteness is read off the eigenvalue signs of a symmetric matrix; an inner product needs strict positive definiteness. **Plan:** (a) solve $\det(A - \lambda I) = 0$ to get $\lambda = t \pm 3$; tabulate the signs over the five ranges of t . (b) require both eigenvalues positive; at $t = 3$ use the eigenvector of $\lambda = 0$.

Worked solution

STEP 1 (a) A symmetric matrix is classified by the signs of its eigenvalues. Solve $\det(A(t) - \lambda I) = 0$:

$$(t - \lambda)^2 - 9 = 0 \implies t - \lambda = \pm 3 \implies \lambda = t - 3 \text{ or } \lambda = t + 3.$$

STEP 2 (a) Read off the signs of $t - 3$ and $t + 3$:

range of t	eigenvalues	classification
$t > 3$	both positive	positive definite
$t = 3$	$\lambda = 6, 0$	positive semidefinite
$-3 < t < 3$	one +, one -	indefinite
$t = -3$	$\lambda = 0, -6$	negative semidefinite
$t < -3$	both negative	negative definite

STEP 3 (b) An inner product needs a positive-definite matrix, so we need $t > 3$.

STEP 4 (b) At $t = 3$ the matrix is only semidefinite, which breaks the rule $\langle v, v \rangle > 0$ for $v \neq 0$. Take $v = (1, -1)^\top$ with $A(3) = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$:

$$A(3)v = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$\langle v, v \rangle = v^\top A(3) v = (1, -1) \cdot (0, 0) = 0,$$

even though $v \neq 0$. So $t = 3$ does not give an inner product.

Answer. (a) $\lambda = t \pm 3$: PD for $t > 3$; PSD at $t = 3$; indefinite for $-3 < t < 3$; NSD at $t = -3$; ND for $t < -3$. (b) inner product iff $t > 3$; at $t = 3$, $v = (1, -1)^\top$ gives $v^\top A(3)v = 0$.

Where marks are lost

Using $\det A(t) = t^2 - 9 > 0$ alone to claim positive definiteness: that also holds for $t < -3$, where the matrix is negative definite. Missing the two borderline cases $t = \pm 3$. In (b), answering $t \geq 3$ instead of $t > 3$; semidefinite is not enough for an inner product.

Answers at a glance

(a) PD: $t > 3$; PSD: $t = 3$; indefinite: $-3 < t < 3$; NSD: $t = -3$; ND: $t < -3$ (b) $t > 3$; $v = (1, -1)^\top$ at $t = 3$.

Problem 40 · Which bilinear forms are inner products?

For each matrix M below, decide whether $\langle x, y \rangle = x^\top M y$ defines a valid inner product on $\mathbb{R}^2$. Justify using the three requirements (bilinear, symmetric, positive-definite); when it fails, exhibit a witness.

(a) $M = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ [2]

(b) $M = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ [2]

(c) $M = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ [2]

How to attempt this (the plan)

What it tests: the three axioms of an inner product and how each can fail for $x^\top M y$. **Plan:** bilinearity is automatic for every matrix. Check symmetry ($M^\top = M$) first; if it holds, run Sylvester (D_1, D_2). For a failure, give a concrete vector (for positivity) or a concrete pair (for symmetry).

Worked solution

STEP 1 (all) Every $x^\top M y$ is automatically bilinear, so only symmetry and positive-definiteness can fail.

STEP 2 (a) M is symmetric, so symmetry holds. Sylvester: $D_1 = 1 > 0$ but

$$D_2 = \det M = 1 - 4 = -3 < 0. \text{ ✗}$$

A witness: $x = (1, -1)^\top$ gives

$$Mx = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad x^\top Mx = (1, -1) \cdot (-1, 1) = -2 < 0.$$

So $\langle x, x \rangle < 0$ is possible — not an inner product.

STEP 3 (b) M is symmetric. Sylvester:

$$D_1 = 2 > 0, \quad D_2 = \det M = 6 - 1 = 5 > 0, \text{ ✓}$$

so M is positive definite. All three requirements hold — this is a valid inner product.

STEP 4 (c) M is not symmetric ($M_{12} = 1 \neq 0 = M_{21}$), which breaks the symmetry requirement directly. Test on the basis vectors:

$$\langle e_1, e_2 \rangle = e_1^\top M e_2 = M_{12} = 1, \quad \langle e_2, e_1 \rangle = e_2^\top M e_1 = M_{21} = 0.$$

Since $\langle e_1, e_2 \rangle \neq \langle e_2, e_1 \rangle$, symmetry fails — not an inner product. (The quadratic form $x^\top Mx = x_1^2 + x_1x_2 + x_2^2$ happens to be positive, but symmetry alone already rules it out.)

Answer. (a) No: $D_2 = -3 < 0$; witness $x = (1, -1)^\top$, $x^\top Mx = -2$. (b) Yes: symmetric, $D_1 = 2$, $D_2 = 5$. (c) No: not symmetric, $\langle e_1, e_2 \rangle = 1 \neq 0 = \langle e_2, e_1 \rangle$.

Where marks are lost

Stopping at “ $D_1 > 0$ ” in (a) and calling M positive definite. Checking positivity of $x^\top Mx$ in (c) and declaring it an inner product without ever checking symmetry. Giving no witness when a form fails: “ $\det < 0$ ” is a reason, but the question asks for a concrete vector or pair.

Answers at a glance

(a) not an inner product ($x = (1, -1)^\top$ gives -2) (b) valid inner product (c) not an inner product (not symmetric).

Problem 41 · Positive (semi)definiteness with two parameters, and the quadratic form

Let

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & a & 1 \\ 0 & 1 & b \end{bmatrix}, \quad a, b \in \mathbb{R}.$$

- (a) Find the conditions on a, b for A to be positive definite. [1.5]
- (b) Find the conditions on a, b for A to be positive semidefinite. [1]
- (c) For $a = b = 2$, write $Q(x, y, z) = \mathbf{x}^\top A \mathbf{x}$ as a sum of squares and hence show $Q > 0$ for all $(x, y, z) \neq (0, 0, 0)$. [1]
- (d) For $a = b = 0$, exhibit a vector with $Q < 0$. [0.5]

How to attempt this (the plan)

What it tests: Sylvester (leading minors) for definiteness versus *all* principal minors for semidefiniteness, completing squares, and finding a witness for indefiniteness. **Plan:** compute $\Delta_1, \Delta_2, \Delta_3$; for (b) also the $(2, 2)$, $(3, 3)$ and the $\{1, 3\}$, $\{2, 3\}$ minors; for (c) group $x^2 + 2xy$ into $(x + y)^2$ and continue; for (d) try a vector that makes the cross term negative.

Worked solution

STEP 1 (a) Leading principal minors:

$$\Delta_1 = 1, \quad \Delta_2 = \det \begin{bmatrix} 1 & 1 \\ 1 & a \end{bmatrix} = a - 1,$$

$$\Delta_3 = \det A = 1 \cdot (ab - 1) - 1 \cdot (b - 0) + 0 = ab - b - 1.$$

Positive definite $\iff \Delta_1, \Delta_2, \Delta_3 > 0$:

$$a > 1 \quad \text{and} \quad ab - b - 1 > 0 \quad (\text{equivalently } b(a - 1) > 1).$$

($\Delta_1 = 1 > 0$ always; $b > 0$ follows from the two conditions.)

STEP 2 (b) For semidefiniteness every principal minor must be ≥ 0 . Principal minors of size 1: $1, a, b$. Size 2: rows/cols $\{1, 2\}$: $a - 1$; $\{1, 3\}$: $1 \cdot b - 0 = b$; $\{2, 3\}$: $ab - 1$. Size 3: $ab - b - 1$. So

$$a \geq 1, \quad b \geq 0, \quad ab \geq 1, \quad ab - b - 1 \geq 0.$$

(None is redundant in general; e.g. $a = 1, b = 0$ passes the first two but fails $ab \geq 1$.)

STEP 3 (c) With $a = b = 2$: $\Delta_2 = 1 > 0$, $\Delta_3 = 4 - 2 - 1 = 1 > 0$, so PD. Expand the form:

$$Q = \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x^2 + 2xy + 2y^2 + 2yz + 2z^2.$$

Complete squares one variable at a time:

$$x^2 + 2xy + y^2 = (x + y)^2, \quad \text{leaving } y^2 + 2yz + 2z^2,$$

$$y^2 + 2yz + z^2 = (y + z)^2, \quad \text{leaving } z^2.$$

$$Q = (x + y)^2 + (y + z)^2 + z^2.$$

This is ≥ 0 , and $Q = 0$ forces $z = 0$, then $y = 0$, then $x = 0$. Hence $Q > 0$ for every nonzero (x, y, z) .

STEP 4 (d) With $a = b = 0$: $Q = x^2 + 2xy + 2yz$. Take $(x, y, z) = (1, -1, 0)$:

$$Q = 1 + 2(1)(-1) + 0 = -1 < 0.$$

(Consistent with (a): $\Delta_2 = -1 < 0$, so A is indefinite.)

Answer. (a) $a > 1$ and $ab - b - 1 > 0$. (b) $a \geq 1, b \geq 0, ab \geq 1, ab - b - 1 \geq 0$. (c) $Q = (x + y)^2 + (y + z)^2 + z^2 > 0$. (d) $(1, -1, 0)$ gives $Q = -1$.

Where marks are lost

Using only *leading* minors ≥ 0 for semidefiniteness (wrong — e.g. $\text{diag}(0, -1)$ has leading minors $0, 0$ but is not PSD). Sign errors in the cofactor expansion of Δ_3 . In (c), writing $Q \geq 0$ and stopping — you must show $Q = 0$ only at the origin.

Answers at a glance

(a) $a > 1, ab - b - 1 > 0$ (b) $a \geq 1, b \geq 0, ab \geq 1, ab - b - 1 \geq 0$ (c) $(x + y)^2 + (y + z)^2 + z^2$
(d) $(1, -1, 0), Q = -1$

Topic 10

Gradient, Directional Derivative & Tangent Planes

Partial derivatives and the gradient at a point, Jacobians of vector-valued maps, the identities for $\nabla(\mathbf{x}^\top A \mathbf{x})$, directional derivatives and the steepest direction, the tangent plane to an implicit surface, and linear approximation with its error.

Formulas you need for this topic

- $\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$; Jacobian $J_{ij} = \frac{\partial f_i}{\partial x_j}$ (one row per output, one column per input).
- Directional derivative $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u}$ with $\|\mathbf{u}\| = 1$ — normalise the direction first.
- Fastest increase along $\nabla f / \|\nabla f\|$ at rate $\|\nabla f\|$; zero rate in any direction orthogonal to ∇f .
- Level surface $F(x, y, z) = 0$: normal $\mathbf{N} = \nabla F(P_0)$, tangent plane $\nabla F(P_0) \cdot (\mathbf{x} - P_0) = 0$, normal line $P_0 + t\nabla F(P_0)$.
- Linearization $L(\mathbf{x}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^\top (\mathbf{x} - \mathbf{x}_0)$; error $= |f - L|$, of second order in the step.
- $\nabla(\mathbf{b}^\top \mathbf{x}) = \mathbf{b}^\top$, $\nabla(\mathbf{x}^\top \mathbf{x}) = 2\mathbf{x}^\top$, $\nabla(\mathbf{x}^\top A \mathbf{x}) = \mathbf{x}^\top (A + A^\top)$ (which is $2\mathbf{x}^\top A$ only when A is symmetric).

Problem 42 · Partial derivatives and the gradient from scratch

Let $f(x, y) = x^2y + 3xy^2 - y^3$.

(a) Compute $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, and write the gradient $\nabla f(x, y)$ as a row vector.

(b) Evaluate ∇f at the point $(2, -1)$.

(c) Verify $\frac{\partial f}{\partial x}$ at $(2, -1)$ directly from the limit definition $\lim_{h \rightarrow 0} \frac{f(2+h, -1) - f(2, -1)}{h}$.

How to attempt this (the plan)

What it tests: the one rule of partial differentiation (hold the other variable fixed) and the limit definition behind it. **Plan:** (a) differentiate term by term twice, once in x and once in y ; (b) substitute $x = 2$, $y = -1$, keeping track of $y^2 = 1$; (c) freeze $y = -1$ to get a one-variable polynomial, expand $f(2+h, -1)$, form the difference quotient, let $h \rightarrow 0$.

Worked solution

STEP 1 (a) $\frac{\partial f}{\partial x}$: hold y fixed and differentiate each term in x only.

$$\frac{\partial}{\partial x}(x^2y) = y \cdot 2x = 2xy, \quad \frac{\partial}{\partial x}(3xy^2) = 3y^2, \quad \frac{\partial}{\partial x}(-y^3) = 0.$$

$$\frac{\partial f}{\partial x} = 2xy + 3y^2.$$

STEP 2 (a) $\frac{\partial f}{\partial y}$: hold x fixed.

$$\frac{\partial}{\partial y}(x^2y) = x^2, \quad \frac{\partial}{\partial y}(3xy^2) = 3x \cdot 2y = 6xy, \quad \frac{\partial}{\partial y}(-y^3) = -3y^2.$$

$$\frac{\partial f}{\partial y} = x^2 + 6xy - 3y^2.$$

STEP 3 (a) Assemble the gradient as a row vector:

$$\nabla f(x, y) = [2xy + 3y^2, x^2 + 6xy - 3y^2].$$

STEP 4 (b) Substitute $x = 2, y = -1$ (note $y^2 = 1$):

$$\left. \frac{\partial f}{\partial x} \right|_{(2, -1)} = 2(2)(-1) + 3(1) = -4 + 3 = -1,$$

$$\left. \frac{\partial f}{\partial y} \right|_{(2, -1)} = (2)^2 + 6(2)(-1) - 3(1) = 4 - 12 - 3 = -11.$$

$$\nabla f(2, -1) = [-1, -11].$$

STEP 5 (c) Reduce to one variable. Fix $y = -1$:

$$f(x, -1) = x^2(-1) + 3x(1) - (-1) = -x^2 + 3x + 1.$$

STEP 6 (c) Form $f(2 + h, -1)$ and $f(2, -1)$:

$$f(2 + h, -1) = -(2 + h)^2 + 3(2 + h) + 1 = -(4 + 4h + h^2) + 6 + 3h + 1 = 3 - h - h^2,$$

$$f(2, -1) = -4 + 6 + 1 = 3.$$

STEP 7 (c) Difference quotient and limit:

$$\frac{f(2 + h, -1) - f(2, -1)}{h} = \frac{(3 - h - h^2) - 3}{h} = \frac{-h - h^2}{h} = -1 - h \xrightarrow{h \rightarrow 0} -1.$$

This matches $\frac{\partial f}{\partial x}(2, -1) = -1$ from part (b). ✓

Answer. (a) $\nabla f = [2xy + 3y^2, x^2 + 6xy - 3y^2]$; (b) $\nabla f(2, -1) = [-1, -11]$; (c) the limit is -1 , agreeing with (b).

Where marks are lost

Treating y as a variable when differentiating in x (e.g. writing $\frac{\partial}{\partial x}(3xy^2) = 3y^2 + 6xy$). Sign slips at $y = -1$: $y^2 = +1$ but $y^3 = -1$. In (c), forgetting to subtract $f(2, -1)$ or cancelling h before expanding the square.

Answers at a glance

(a) $\nabla f = [2xy + 3y^2, x^2 + 6xy - 3y^2]$ (b) $[-1, -11]$ (c) limit = -1 ✓

Problem 43 · Gradient with exponential and trigonometric terms

Let

$$f(x, y, z) = x^2 e^y + y \cos z.$$

(a) Compute $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$.(b) Write ∇f and evaluate it at $(1, 0, 0)$.

How to attempt this (the plan)

What it tests: partials in three variables where each term contains only some of the variables.

Plan: for each variable, identify which of the two terms actually contains it; the other term differentiates to 0. Use $\frac{d}{dy}e^y = e^y$ and $\frac{d}{dz}\cos z = -\sin z$. Then substitute $e^0 = 1$, $\cos 0 = 1$, $\sin 0 = 0$.

Worked solution

STEP 1 (a) $\frac{\partial f}{\partial x}$: e^y and $y \cos z$ contain no x , so only x^2 varies.

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2 e^y) + \underbrace{\frac{\partial}{\partial x}(y \cos z)}_{=0} = e^y \cdot 2x + 0 = 2xe^y.$$

STEP 2 (a) $\frac{\partial f}{\partial y}$: now x^2 and $\cos z$ are constants.

$$\frac{\partial f}{\partial y} = x^2 \cdot \frac{d}{dy}e^y + \cos z \cdot \frac{d}{dy}(y) = x^2 e^y + \cos z.$$

STEP 3 (a) $\frac{\partial f}{\partial z}$: $x^2 e^y$ has no z ; y is a constant multiplier on $\cos z$.

$$\frac{\partial f}{\partial z} = \underbrace{\frac{\partial}{\partial z}(x^2 e^y)}_{=0} + y \cdot \frac{d}{dz}\cos z = 0 + y(-\sin z) = -y \sin z.$$

STEP 4 (b) Assemble and substitute $(1, 0, 0)$ with $e^0 = 1$, $\cos 0 = 1$, $\sin 0 = 0$:

$$\nabla f = [2xe^y, x^2 e^y + \cos z, -y \sin z],$$

$$\nabla f(1, 0, 0) = [2(1)(1), (1)(1) + 1, -(0)(0)] = [2, 2, 0].$$

STEP 5 (b) Check by linearity, $\nabla(f + g) = \nabla f + \nabla g$:

$$\nabla(x^2 e^y) = [2xe^y, x^2 e^y, 0], \quad \nabla(y \cos z) = [0, \cos z, -y \sin z].$$

Adding entry by entry reproduces ∇f . ✓

Answer. (a) $f_x = 2xe^y$, $f_y = x^2 e^y + \cos z$, $f_z = -y \sin z$; (b) $\nabla f(1, 0, 0) = [2, 2, 0]$.

Where marks are lost

Forgetting the $+\cos z$ in $\frac{\partial f}{\partial y}$ because the second term "looks like a z term". Sign slip $\frac{d}{dz}\cos z = -\sin z$. Writing $e^0 = 0$ when substituting.

Answers at a glance

(a) $[2xe^y, x^2 e^y + \cos z, -y \sin z]$ (b) $\nabla f(1, 0, 0) = [2, 2, 0]$

Problem 44 · Jacobian of a vector-valued map

Consider $\mathbf{f} : \mathbb{R}^2 \rightarrow \mathbb{R}^3$,

$$\mathbf{f}(x, y) = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = \begin{bmatrix} x^2 - y^2 \\ x e^y \\ \ln(1 + x^2 + y^2) \end{bmatrix}.$$

- (a) Compute the Jacobian $J_{\mathbf{f}}(x, y)$ and state its dimensions.
- (b) Evaluate it at $(1, 0)$.
- (c) Say in words what the rows and columns represent.

How to attempt this (the plan)

What it tests: building an $m \times n$ Jacobian with entry $(i, j) = \partial f_i / \partial x_j$. **Plan:** one row per output, one column per input, so 3×2 ; compute the six partials row by row (row 3 needs the chain rule on $\ln$); substitute $e^0 = 1$ and $1 + x^2 + y^2 = 2$ entry by entry; then explain rows = gradients of components, columns = response to one input.

Worked solution

STEP 1 (a) Dimensions: 3 outputs, 2 inputs, so $J_{\mathbf{f}} \in \mathbb{R}^{3 \times 2}$ with entry $(i, j) = \frac{\partial f_i}{\partial x_j}$.

STEP 2 (a) Row 1, $f_1 = x^2 - y^2$:

$$\frac{\partial f_1}{\partial x} = 2x, \quad \frac{\partial f_1}{\partial y} = -2y.$$

STEP 3 (a) Row 2, $f_2 = x e^y$: treating y constant, $\frac{\partial f_2}{\partial x} = e^y$; treating x constant, $\frac{\partial f_2}{\partial y} = x e^y$.

STEP 4 (a) Row 3, $f_3 = \ln(1 + x^2 + y^2)$: with inner $u = 1 + x^2 + y^2$ and $\frac{\partial}{\partial x} \ln u = \frac{1}{u} \frac{\partial u}{\partial x}$,

$$\frac{\partial f_3}{\partial x} = \frac{2x}{1 + x^2 + y^2}, \quad \frac{\partial f_3}{\partial y} = \frac{2y}{1 + x^2 + y^2}.$$

STEP 5 (a) Assemble:

$$J_{\mathbf{f}}(x, y) = \begin{bmatrix} 2x & -2y \\ e^y & x e^y \\ \frac{2x}{1 + x^2 + y^2} & \frac{2y}{1 + x^2 + y^2} \end{bmatrix} \in \mathbb{R}^{3 \times 2}.$$

STEP 6 (b) Substitute $x = 1, y = 0$ (so $e^0 = 1$ and $1 + x^2 + y^2 = 2$), entry by entry:

$$J_{\mathbf{f}}(1, 0) = \begin{bmatrix} 2(1) & -2(0) \\ e^0 & (1)e^0 \\ \frac{2(1)}{2} & \frac{2(0)}{2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

STEP 7 (c) Row i is the gradient ∇f_i of the i -th output: a 1×2 row telling how output i responds to each input. Column j is the vector $\partial \mathbf{f} / \partial x_j$: how the whole output vector responds when input x_j alone changes. Here column 1 is $\partial \mathbf{f} / \partial x$ and column 2 is $\partial \mathbf{f} / \partial y$.

STEP 8 (a) Check by stacking gradients: $\nabla f_1 = [2x, -2y]$, $\nabla f_2 = [e^y, x e^y]$, $\nabla f_3 = [\frac{2x}{1+x^2+y^2}, \frac{2y}{1+x^2+y^2}]$; stacking gives the same 3×2 matrix. ✓

Answer. (a) $J_f = \begin{bmatrix} 2x & -2y \\ e^y & xe^y \\ \frac{2x}{1+x^2+y^2} & \frac{2y}{1+x^2+y^2} \end{bmatrix}$, size 3×2 ; (b) $J_f(1,0) = \begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$; (c) rows = ∇f_i , columns = $\partial f / \partial x_j$.

Where marks are lost

Writing the transpose (2×3) because "two variables come first". Putting $\frac{\partial f_2}{\partial y} = e^y$ and forgetting the factor x . Dropping the inner derivative $2x$ or $2y$ in row 3. Evaluating in itself at $(1,0)$ instead of its derivative.

Answers at a glance

(a) 3×2 matrix with rows $[2x, -2y]$, $[e^y, xe^y]$, $[\frac{2x}{1+x^2+y^2}, \frac{2y}{1+x^2+y^2}]$ (b) $\begin{bmatrix} 2 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$ (c) rows = gradients of f_i , columns = sensitivity to x_j

Problem 45 · Vector gradient identities and a non-symmetric quadratic

For $\mathbf{x} \in \mathbb{R}^n$, three identities are worth memorising:

$$\nabla(\mathbf{b}^\top \mathbf{x}) = \mathbf{b}^\top, \quad \nabla(\mathbf{x}^\top \mathbf{x}) = 2\mathbf{x}^\top, \quad \nabla(\mathbf{x}^\top A \mathbf{x}) = \mathbf{x}^\top (A + A^\top).$$

Now let

$$f(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} - \mathbf{b}^\top \mathbf{x}, \quad A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \text{ (not symmetric),} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

- Write f explicitly in components (x_1, x_2) .
- Apply the identities to get ∇f . Why does $A + A^\top$ (not $2A$) appear?
- Evaluate ∇f at $(1,1)$.

How to attempt this (the plan)

What it tests: the standard vector-calculus identities and the reason $A + A^\top$ appears for a non-symmetric A . **Plan:** (a) compute $A\mathbf{x}$, then $\mathbf{x}^\top(A\mathbf{x})$, subtract $\mathbf{b}^\top \mathbf{x}$; (b) form $A + A^\top$, apply the identities, and explain via the double sum $\sum_{i,j} A_{ij}x_i x_j$ that x_k appears in two places; (c) substitute $x_1 = x_2 = 1$ and confirm against direct differentiation of (a).

Worked solution

STEP 1 (a) Compute $A\mathbf{x}$:

$$A\mathbf{x} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 \\ 3x_2 \end{bmatrix}.$$

STEP 2 (a) Compute $\mathbf{x}^\top A\mathbf{x}$:

$$\mathbf{x}^\top(A\mathbf{x}) = [x_1, x_2] \begin{bmatrix} x_1 + 2x_2 \\ 3x_2 \end{bmatrix} = x_1(x_1 + 2x_2) + x_2(3x_2) = x_1^2 + 2x_1x_2 + 3x_2^2.$$

STEP 3 (a) Subtract $\mathbf{b}^\top \mathbf{x} = x_1 + x_2$:

$$f(x_1, x_2) = x_1^2 + 2x_1x_2 + 3x_2^2 - x_1 - x_2.$$

STEP 4 (b) Form $A + A^\top$:

$$A^\top = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}, \quad A + A^\top = \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix}.$$

STEP 5 (b) Apply $\nabla(\mathbf{x}^\top A \mathbf{x}) = \mathbf{x}^\top (A + A^\top)$ and $\nabla(\mathbf{b}^\top \mathbf{x}) = \mathbf{b}^\top$:

$$\begin{aligned} \nabla f &= \mathbf{x}^\top (A + A^\top) - \mathbf{b}^\top = [x_1, x_2] \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix} - [1, 1] \\ &= [2x_1 + 2x_2 - 1, 2x_1 + 6x_2 - 1]. \end{aligned}$$

STEP 6 (b) Why $A + A^\top$? Write $\mathbf{x}^\top A \mathbf{x} = \sum_{i,j} A_{ij}x_i x_j$. Differentiating with respect to x_k , the variable x_k appears both as the " $i = k$ " factor (contributing $\sum_j A_{kj}x_j$, the k -th entry of $\mathbf{x}^\top A^\top$) and as the " $j = k$ " factor (contributing $\sum_i A_{ik}x_i$, the k -th entry of $\mathbf{x}^\top A$). Adding gives the k -th entry of $\mathbf{x}^\top (A + A^\top)$. Only when $A = A^\top$ does $A + A^\top = 2A$.

STEP 7 (c) Substitute $x_1 = x_2 = 1$:

$$\nabla f(1, 1) = [2 + 2 - 1, 2 + 6 - 1] = [3, 7].$$

STEP 8 (c) Check by differentiating the component form from Step 3 directly:

$$\frac{\partial f}{\partial x_1} = 2x_1 + 2x_2 - 1, \quad \frac{\partial f}{\partial x_2} = 2x_1 + 6x_2 - 1,$$

identical to Step 5; at $(1, 1)$ this is $[3, 7]$. ✓

Answer. (a) $f = x_1^2 + 2x_1x_2 + 3x_2^2 - x_1 - x_2$; (b) $\nabla f = \mathbf{x}^\top (A + A^\top) - \mathbf{b}^\top = [2x_1 + 2x_2 - 1, 2x_1 + 6x_2 - 1]$, because x_k appears in both slots of $\sum_{i,j} A_{ij}x_i x_j$; (c) $\nabla f(1, 1) = [3, 7]$.

Where marks are lost

Using $\nabla(\mathbf{x}^\top A \mathbf{x}) = 2\mathbf{x}^\top A$ for a non-symmetric A , which gives $[2x_1 - 1, 4x_1 + 6x_2 - 1]$ and the wrong value $[1, 9]$ at $(1, 1)$. Writing ∇f as a column when the identities are stated for row gradients (be consistent). Forgetting the $-\mathbf{b}^\top$ term after expanding the quadratic.

Answers at a glance

(a) $x_1^2 + 2x_1x_2 + 3x_2^2 - x_1 - x_2$ (b) $[2x_1 + 2x_2 - 1, 2x_1 + 6x_2 - 1]$, since x_k appears twice in the double sum (c) $[3, 7]$

Problem 46 · Linearization (tangent plane) of a two-variable function

A model's output is $f(w_1, w_2) = \ln(w_1) + w_2^2$.

(a) Find the linear approximation $L(w_1, w_2)$ of f about $w^{(0)} = (1, 2)$.

- (b) Compute the actual value $f(1.1, 2.1)$ and the approximate value $L(1.1, 2.1)$.
 (c) Compute the absolute error $|f(1.1, 2.1) - L(1.1, 2.1)|$.
 (Use $\ln 1.1 = 0.09531$.)

How to attempt this (the plan)

What it tests: first-order Taylor in two variables = value + gradient · step. **Plan:** value at the point, two partials at the point, assemble $L = f_0 + f_{w_1}(w_1 - 1) + f_{w_2}(w_2 - 2)$, substitute.

Worked solution

STEP 1 (a) Value at the base point:

$$f(1, 2) = \ln 1 + 2^2 = 0 + 4 = 4.$$

STEP 2 (a) Partial derivatives at the base point:

$$f_{w_1} = \frac{1}{w_1} \Rightarrow f_{w_1}(1, 2) = 1, \quad f_{w_2} = 2w_2 \Rightarrow f_{w_2}(1, 2) = 4.$$

STEP 3 (a) Assemble:

$$L(w_1, w_2) = 4 + 1 \cdot (w_1 - 1) + 4 \cdot (w_2 - 2) = w_1 + 4w_2 - 5.$$

STEP 4 (b) Actual and approximate values:

$$f(1.1, 2.1) = \ln 1.1 + (2.1)^2 = 0.09531 + 4.41 = 4.50531,$$

$$L(1.1, 2.1) = 1.1 + 4(2.1) - 5 = 1.1 + 8.4 - 5 = 4.5.$$

STEP 5 (c) Error:

$$|4.50531 - 4.5| = 0.00531.$$

Answer. (a) $L(w_1, w_2) = 4 + (w_1 - 1) + 4(w_2 - 2) = w_1 + 4w_2 - 5$. (b) $f(1.1, 2.1) = 4.50531$, $L(1.1, 2.1) = 4.5$. (c) Absolute error ≈ 0.0053 .

Where marks are lost

Adding a Hessian term when the question says linear. Evaluating $f_{w_1} = 1/w_1$ at the new point (1.1) instead of the base point (1). Sign slips in $(w_2 - 2)$.

Answers at a glance

(a) $w_1 + 4w_2 - 5$ (b) 4.50531, 4.5 (c) 0.0053

Problem 47 · Gradient, directional derivative and the steepest direction

Let $f(x, y) = x^3 - 3xy^2$.

(a) Compute ∇f at the point $(2, 1)$.

[1]

(b) Find the directional derivative of f at $(2, 1)$ in the direction of $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$.

[1]

- (c) At $(2, 1)$: in which unit direction does f increase fastest, and at what rate? Give a unit direction in which the directional derivative is 0. [1]

How to attempt this (the plan)

What it tests: $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ with a *unit* $\mathbf{u}$, and the two facts "steepest ascent $= \nabla f / \|\nabla f\|$ with rate $\|\nabla f\|$ " and "zero rate $\perp \nabla f$ ". **Plan:** partials; evaluate; normalise $\mathbf{v}$; dot; then read the answers off ∇f .

Worked solution

STEP 1 (a)

$$f_x = 3x^2 - 3y^2, \quad f_y = -6xy.$$

$$\nabla f(2, 1) = (3(4) - 3(1), -6(2)(1)) = (9, -12).$$

STEP 2 (b) $\mathbf{v}$ is not a unit vector: $\|\mathbf{v}\| = \sqrt{9 + 16} = 5$, so $\mathbf{u} = (\frac{3}{5}, \frac{4}{5})$.

$$D_{\mathbf{u}}f(2, 1) = \nabla f \cdot \mathbf{u} = 9 \cdot \frac{3}{5} + (-12) \cdot \frac{4}{5} = \frac{27 - 48}{5} = -\frac{21}{5} = -4.2.$$

Moving in the direction of $\mathbf{v}$, f *decreases* at rate 4.2 per unit length.

STEP 3 (c) $\|\nabla f(2, 1)\| = \sqrt{81 + 144} = \sqrt{225} = 15$.

$$\text{Steepest ascent direction: } \frac{\nabla f}{\|\nabla f\|} = \left(\frac{9}{15}, -\frac{12}{15}\right) = \left(\frac{3}{5}, -\frac{4}{5}\right), \quad \text{rate } 15.$$

Zero directional derivative $\iff \mathbf{u} \perp \nabla f$: take $\mathbf{u} = (\frac{4}{5}, \frac{3}{5})$ (check: $9 \cdot \frac{4}{5} - 12 \cdot \frac{3}{5} = \frac{36 - 36}{5} = 0$). The opposite vector $(-\frac{4}{5}, -\frac{3}{5})$ also works.

Answer. (a) $\nabla f(2, 1) = (9, -12)$. (b) $D_{\mathbf{u}}f = -\frac{21}{5}$. (c) fastest increase along $(\frac{3}{5}, -\frac{4}{5})$ at rate 15; zero rate along $\pm(\frac{4}{5}, \frac{3}{5})$.

Where marks are lost

Dotting with $\mathbf{v} = (3, 4)$ instead of the unit vector (gives -21 , wrong by a factor 5). Reporting the steepest-ascent *rate* as ∇f itself rather than its length. Sign of $f_y = -6xy$.

Answers at a glance

(a) $(9, -12)$ (b) $-21/5$ (c) $(\frac{3}{5}, -\frac{4}{5})$, rate 15; zero along $\pm(\frac{4}{5}, \frac{3}{5})$

Problem 48 · Normal vector and tangent plane of an implicitly defined surface

A surface in $\mathbb{R}^3$ is given by

$$F(x, y, z) = x^2 + 2y^2 + 3z^2 - 6 = 0.$$

- (a) Show that $P_0 = (1, 1, 1)$ lies on the surface and find a normal vector $\mathbf{N}$ at P_0 . [1.5]
 (b) Find the equation of the tangent plane at P_0 , and the parametric equation of the normal line. [1]

- (c) Find all points of the surface where the tangent plane is horizontal (parallel to the xy -plane). [0.5]

How to attempt this (the plan)

What it tests: " ∇F is normal to the level surface $F = 0$ ", the plane formula $\nabla F(P_0) \cdot (\mathbf{x} - P_0) = 0$, and reading a geometric condition ($\mathbf{N} \parallel \mathbf{k}$) as equations. **Plan:** substitute; three partials; plane; line $P_0 + t\mathbf{N}$; for (c) set $F_x = F_y = 0$ and use $F = 0$ to get z .

Worked solution

STEP 1 (a) $F(1, 1, 1) = 1 + 2 + 3 - 6 = 0$, so P_0 is on the surface. ✓

$$\nabla F = (2x, 4y, 6z), \quad \mathbf{N} = \nabla F(1, 1, 1) = (2, 4, 6) \text{ (or the simpler parallel vector } (1, 2, 3)).$$

STEP 2 (b) Tangent plane: $\mathbf{N} \cdot (\mathbf{x} - P_0) = 0$:

$$2(x - 1) + 4(y - 1) + 6(z - 1) = 0 \implies x + 2y + 3z = 6.$$

Check: P_0 satisfies it ($1 + 2 + 3 = 6$). ✓ Normal line:

$$(x, y, z) = (1, 1, 1) + t(1, 2, 3), \quad t \in \mathbb{R}.$$

STEP 3 (c) Horizontal tangent plane $\iff \nabla F \parallel (0, 0, 1) \iff F_x = F_y = 0$ and $F_z \neq 0$:

$$2x = 0, \quad 4y = 0 \implies x = y = 0.$$

On the surface: $3z^2 = 6 \implies z = \pm\sqrt{2}$ (there $F_z = 6z \neq 0$). The two points are $(0, 0, \sqrt{2})$ and $(0, 0, -\sqrt{2})$, with tangent planes $z = \pm\sqrt{2}$.

Answer. (a) $F(P_0) = 0$; $\mathbf{N} = (2, 4, 6) \parallel (1, 2, 3)$. (b) $x + 2y + 3z = 6$; line $(1, 1, 1) + t(1, 2, 3)$. (c) $(0, 0, \pm\sqrt{2})$.

Where marks are lost

Using ∇F as a *direction in* the plane instead of the normal to it. Writing the plane as $2x + 4y + 6z = 0$ (forgetting to shift to P_0 — the plane must pass through P_0). In (c) forgetting that the points must also satisfy $F = 0$.

Answers at a glance

(a) $\mathbf{N} = (2, 4, 6)$ (b) $x + 2y + 3z = 6$; $(1, 1, 1) + t(1, 2, 3)$ (c) $(0, 0, \pm\sqrt{2})$

Problem 49 · Linearization of a two-variable model and its error

A model's output is

$$f(w_1, w_2) = \sqrt{w_1} e^{w_2}.$$

- (a) Find the tangent-plane (linear) approximation $L(w_1, w_2)$ of f at $\mathbf{w}^{(0)} = (4, 0)$. [2]
 (b) Compute the actual value $f(4.4, -0.1)$ and the approximation $L(4.4, -0.1)$. [0.5]
 (c) Compute the absolute error $|f(4.4, -0.1) - L(4.4, -0.1)|$. [0.5]

(Use $\sqrt{4.4} = 2.0976$ and $e^{-0.1} = 0.90484$.)

How to attempt this (the plan)

What it tests: first-order Taylor in two variables, $L = f_0 + f_{w_1}\Delta w_1 + f_{w_2}\Delta w_2$, and honest evaluation of the error. **Plan:** value at the point; two partials at the point; assemble; substitute $\Delta w_1 = 0.4$, $\Delta w_2 = -0.1$.

Worked solution

STEP 1 (a) At the base point: $f(4, 0) = \sqrt{4}e^0 = 2$.

$$f_{w_1} = \frac{e^{w_2}}{2\sqrt{w_1}} \Rightarrow f_{w_1}(4, 0) = \frac{1}{2 \cdot 2} = \frac{1}{4}, \quad f_{w_2} = \sqrt{w_1}e^{w_2} \Rightarrow f_{w_2}(4, 0) = 2.$$

$$L(w_1, w_2) = 2 + \frac{1}{4}(w_1 - 4) + 2(w_2 - 0) = \frac{1}{4}w_1 + 2w_2 + 1.$$

STEP 2 (b) Actual: $f(4.4, -0.1) = \sqrt{4.4}e^{-0.1} = 2.0976 \times 0.90484 \approx 1.8980$. Approximate: $L(4.4, -0.1) = 2 + \frac{1}{4}(0.4) + 2(-0.1) = 2 + 0.1 - 0.2 = 1.9$.

STEP 3 (c) $|1.8980 - 1.9| \approx 0.0020$. The step is small (0.4 and -0.1), so the linear model is within about 0.1%; the error comes from the neglected second-order terms.

Answer. (a) $L = 1 + \frac{1}{4}w_1 + 2w_2$. (b) $f \approx 1.8980$, $L = 1.9$. (c) error ≈ 0.002 .

Where marks are lost

Differentiating $\sqrt{w_1}$ as $\frac{1}{\sqrt{w_1}}$ (missing the $\frac{1}{2}$). Plugging $(4.4, -0.1)$ into the partials instead of evaluating them at the base point $(4, 0)$. Reporting the error with the wrong sign or as a percentage without being asked.

Answers at a glance

(a) $L = 1 + \frac{1}{4}w_1 + 2w_2$ (b) 1.8980 vs 1.9 (c) ≈ 0.002

Topic 11

Taylor Expansion, Hessian & Critical-Point Classification

Taylor polynomials in one and two variables, the matrix form with the gradient and the Hessian, stationary points and their classification by the determinant test or by eigenvalues, the trap of a non-stationary point, and reading an expansion backwards.

Formulas you need for this topic

- One variable: $f(a+h) = f(a) + f'(a)h + \frac{1}{2!}f''(a)h^2 + \frac{1}{3!}f'''(a)h^3 + \dots$
- Two variables: $f(a+h, b+k) \approx f + f_x h + f_y k + \frac{1}{2}(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2)$, all derivatives at (a, b) .
- Matrix form: $f(\mathbf{x}_0 + \mathbf{h}) \approx f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^\top \mathbf{h} + \frac{1}{2}\mathbf{h}^\top H_f(\mathbf{x}_0)\mathbf{h}$. For a quadratic f this is exact.
- Stationary point: $\nabla f = \mathbf{0}$. Hessian $H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{bmatrix}$, $D = \det H = f_{xx}f_{yy} - f_{xy}^2$.
- $D > 0, f_{xx} > 0$: minimum. $D > 0, f_{xx} < 0$: maximum. $D < 0$: saddle. $D = 0$: test fails, argue directly.
- Eigenvalue version: both $\lambda > 0$ minimum, both < 0 maximum, opposite signs saddle. The test applies only at a stationary point.

Problem 50 · Second-order Taylor expansion of $e^x \sin y$ at the origin

Let $f(x, y) = e^x \sin y$.

- Compute ∇f and the Hessian H at $(0, 0)$.
- Write the second-order Taylor polynomial $Q(x, y)$ of f about $(0, 0)$.
- Use Q to approximate $f(0.1, 0.2)$, compare with the exact value, and state the sign and order of magnitude of the error. Explain, from the order of the first neglected term, why the error is essentially zero at the origin and grows as one moves away.

How to attempt this (the plan)

What it tests: the two-variable second-order Taylor formula $f(a+h, b+k) \approx f + f_x h + f_y k + \frac{1}{2}(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2)$ and reading off the gradient and Hessian. **Plan:** (a) compute the five partials, evaluate at the origin; (b) plug into the formula with $h = x, k = y$; (c) evaluate Q and the exact f at $(0.1, 0.2)$ and subtract.

Worked solution

STEP 1 (a) First and second partials of $f = e^x \sin y$:

$$f_x = e^x \sin y, \quad f_y = e^x \cos y, \quad f_{xx} = e^x \sin y, \quad f_{xy} = e^x \cos y, \quad f_{yy} = -e^x \sin y.$$

STEP 2 (a) Evaluate at $(0, 0)$: $f = 0, f_x = 0, f_y = 1, f_{xx} = 0, f_{xy} = 1, f_{yy} = 0$, so

$$\nabla f(0, 0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad H(0, 0) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

STEP 3 (b) With $(a, b) = (0, 0), h = x, k = y$:

$$Q(x, y) = 0 + (0 \cdot x + 1 \cdot y) + \frac{1}{2}(0 \cdot x^2 + 2 \cdot 1 \cdot xy + 0 \cdot y^2) = y + xy.$$

STEP 4 (c) Approximate value:

$$Q(0.1, 0.2) = 0.2 + (0.1)(0.2) = 0.22.$$

STEP 5 (c) Exact value:

$$f(0.1, 0.2) = e^{0.1} \sin(0.2) = 1.10517 \times 0.19867 = 0.21956.$$

STEP 6 (c) Error: $Q - f = 0.22 - 0.21956 = +4.4 \times 10^{-4}$, so Q slightly overestimates. The first neglected term is third order, $O(\|(x, y)\|^3)$: the error vanishes at the origin and grows like the cube of the distance from it, so at distance ≈ 0.22 it is only a few parts in 10^4 .

Answer. (a) $\nabla f(0, 0) = (0, 1)^\top$, $H(0, 0) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. (b) $Q(x, y) = y + xy$. (c) $Q(0.1, 0.2) = 0.22$, exact 0.21956, error $+4.4 \times 10^{-4}$ (third-order).

Where marks are lost

Forgetting the factor $\frac{1}{2}$ in front of the quadratic form, or forgetting the factor 2 on the cross term $f_{xy}hk$ (both errors cancel here only by luck; do not rely on it). Using degrees instead of radians for $\sin(0.2)$. Writing H with $f_{yy} = +1$: $\partial_y(e^x \cos y) = -e^x \sin y$, which is 0 at the origin.

Answers at a glance

(a) $\nabla f = (0, 1)^\top$, $H = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (b) $Q = y + xy$ (c) 0.22 vs 0.21956, error $+4.4 \times 10^{-4}$.

Problem 51 · Linear and quadratic estimates of $\sqrt{4.2}$

Let $f(x) = \sqrt{x}$, expanded about the point $a = 4$.

- Find the first-order (linear) polynomial $P_1(x)$ and the second-order (quadratic) polynomial $P_2(x)$ of f about $x = 4$.
- Use each to estimate $\sqrt{4.2}$.
- Compare the two errors against the true value $\sqrt{4.2} = 2.04939\dots$ and explain, using the sign of f'' , why the quadratic estimate is the better one.

How to attempt this (the plan)

What it tests: tangent-line versus quadratic Taylor approximation and the role of the second derivative. **Plan:** (a) compute $f(4)$, $f'(4)$, $f''(4)$ and write P_1 , P_2 ; (b) set $x - 4 = 0.2$; (c) subtract each estimate from 2.04939 and interpret the sign of $f'' < 0$.

Worked solution**STEP 1** (a) Derivatives of $f(x) = x^{1/2}$ at $a = 4$:

$$f(4) = 2, \quad f'(x) = \frac{1}{2}x^{-1/2} \Rightarrow f'(4) = \frac{1}{4}, \quad f''(x) = -\frac{1}{4}x^{-3/2} \Rightarrow f''(4) = -\frac{1}{32}.$$

STEP 2 (a) The quadratic coefficient is $\frac{f''(4)}{2!} = \frac{-1/32}{2} = -\frac{1}{64}$, so

$$P_1(x) = 2 + \frac{1}{4}(x - 4), \quad P_2(x) = 2 + \frac{1}{4}(x - 4) - \frac{1}{64}(x - 4)^2.$$

STEP 3 (b) At $x = 4.2$ the increment is $x - 4 = 0.2$:

$$P_1(4.2) = 2 + \frac{1}{4}(0.2) = 2.05.$$

$$P_2(4.2) = 2.05 - \frac{1}{64}(0.2)^2 = 2.05 - 0.000625 = 2.049375.$$

STEP 4 (c) Compare with $\sqrt{4.2} = 2.049390\dots$:

$$P_1(4.2) - \sqrt{4.2} = +6.1 \times 10^{-4}, \quad P_2(4.2) - \sqrt{4.2} = -1.5 \times 10^{-5}.$$

The quadratic error is about 40 times smaller.

STEP 5 (c) Why: P_1 is the tangent line at $x = 4$; it matches value and slope but ignores the downward curvature ($f'' < 0$), so it rides above the curve and drifts away as x moves from 4. P_2 adds the $-\frac{1}{64}(x - 4)^2$ term, which bends the approximation down to follow the curve; the leftover error is now third order, $\propto (x - 4)^3$.

Answer. (a) $P_1(x) = 2 + \frac{1}{4}(x - 4)$, $P_2(x) = 2 + \frac{1}{4}(x - 4) - \frac{1}{64}(x - 4)^2$. (b) $P_1(4.2) = 2.05$, $P_2(4.2) = 2.049375$. (c) errors $+6.1 \times 10^{-4}$ (linear) vs -1.5×10^{-5} (quadratic); the tangent line ignores $f'' < 0$.

Where marks are lost

Forgetting to halve $f''(4)$: the coefficient is $-\frac{1}{64}$, not $-\frac{1}{32}$. Sign error in $f''(x) = -\frac{1}{4}x^{-3/2}$. Evaluating $x^{-3/2}$ at 4 wrongly ($4^{-3/2} = \frac{1}{8}$). Plugging $x = 4.2$ instead of $x - 4 = 0.2$ into the polynomial.

Answers at a glance

(a) $P_1 = 2 + \frac{1}{4}(x - 4)$, $P_2 = P_1 - \frac{1}{64}(x - 4)^2$ (b) 2.05, 2.049375 (c) errors 6.1×10^{-4} vs 1.5×10^{-5} ; linear ignores curvature.

Problem 52 · One variable: MVT and a Taylor estimate

(a) For $f(x) = x^3$ on $[0, 2]$, find the point c guaranteed by the mean value theorem, i.e. the c with

$$f'(c) = \frac{f(2) - f(0)}{2 - 0}.$$

(b) Write the second-order Taylor polynomial $T_2(h)$ of $g(x) = \ln(1 + x)$ about $x = 0$.

(c) Use T_2 to estimate $\ln(1.1)$, and compute the absolute error given $\ln(1.1) = 0.09531$ (5 d.p.).

How to attempt this (the plan)

What it tests: the MVT as an equation you can solve; the Taylor coefficients $f^{(k)}(a)/k!$. **Plan:** (a) average slope, then solve $f'(c) = \text{that}$; (b) compute $g(0)$, $g'(0)$, $g''(0)$ and place them with $1, h, h^2/2$; (c) substitute $h = 0.1$.

Worked solution

STEP 1 (a) Average slope over $[0, 2]$:

$$\frac{f(2) - f(0)}{2 - 0} = \frac{8 - 0}{2} = 4.$$

STEP 2 (a) Solve $f'(c) = 4$:

$$f'(c) = 3c^2 = 4 \Rightarrow c^2 = \frac{4}{3} \Rightarrow c = \frac{2}{\sqrt{3}} \approx 1.155 \quad (\text{the root in } (0, 2)).$$

STEP 3 (b) Derivatives of g at 0:

$$g(x) = \ln(1+x), \quad g'(x) = \frac{1}{1+x}, \quad g''(x) = -\frac{1}{(1+x)^2},$$

$$g(0) = 0, \quad g'(0) = 1, \quad g''(0) = -1.$$

STEP 4 (b) Assemble the polynomial:

$$T_2(h) = g(0) + g'(0)h + \frac{g''(0)}{2}h^2 = h - \frac{h^2}{2}.$$

STEP 5 (c) Substitute $h = 0.1$:

$$T_2(0.1) = 0.1 - \frac{0.01}{2} = 0.1 - 0.005 = 0.095.$$

$$|\ln(1.1) - T_2(0.1)| = |0.09531 - 0.095| = 0.00031.$$

Compare with the first dropped term $h^3/3 = 0.00033$: the error is of the expected size.

Answer. (a) $c = 2/\sqrt{3} \approx 1.155$. (b) $T_2(h) = h - \frac{1}{2}h^2$. (c) Estimate 0.095; absolute error ≈ 0.00031 .

Where marks are lost

Forgetting the $\frac{1}{2}$ on the h^2 term (gives 0.09). Using $g''(0) = +1$. Reporting $c = -2/\sqrt{3}$, which lies outside $[0, 2]$.

Answers at a glance

(a) $2/\sqrt{3}$ (b) $h - h^2/2$ (c) 0.095, error 0.00031

Problem 53 · Second-order Taylor about a non-stationary point

Let $f(x, y) = x e^y + y^2$.

- Compute the gradient $\nabla f(x, y)$ and the Hessian $H_f(x, y)$.
- Obtain the second-order Taylor polynomial $T_2(x, y)$ of f about the point $(1, 0)$.
- With $h = (x - 1, y)^\top$, write the quadratic term of T_2 in the form $\frac{1}{2} h^\top H_f(1, 0) h$ and expand it.
- Can the Hessian-based second-derivative test be used to classify $(1, 0)$? Give a numerical reason.

How to attempt this (the plan)

What it tests: the full two-variable Taylor recipe, the $h^\top H h$ form, and the “not stationary” trap. **Plan:** partials symbolically; evaluate value, gradient and Hessian at $(1, 0)$; assemble $T_2 = f_0 + \nabla f^\top h + \frac{1}{2} h^\top H h$; for (d) check whether $\nabla f(1, 0) = 0$.

Worked solution

STEP 1 (a) First and second partials, symbolically:

$$\begin{aligned} f_x &= e^y, & f_y &= x e^y + 2y, \\ f_{xx} &= 0, & f_{xy} &= e^y, & f_{yy} &= x e^y + 2. \end{aligned}$$

$$\nabla f = \begin{bmatrix} e^y \\ x e^y + 2y \end{bmatrix}, \quad H_f = \begin{bmatrix} 0 & e^y \\ e^y & x e^y + 2 \end{bmatrix}.$$

STEP 2 (b) Evaluate at $(1, 0)$:

$$\begin{aligned} f(1, 0) &= 1 \cdot e^0 + 0 = 1, \\ \nabla f(1, 0) &= \begin{bmatrix} 1 \\ 1 + 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, & H_f(1, 0) &= \begin{bmatrix} 0 & 1 \\ 1 & 1 + 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix}. \end{aligned}$$

STEP 3 (b) Assemble with $h_1 = x - 1$, $h_2 = y$:

$$\begin{aligned} T_2 &= 1 + 1 \cdot h_1 + 1 \cdot h_2 + \frac{1}{2} (0 \cdot h_1^2 + 2 \cdot 1 \cdot h_1 h_2 + 3 h_2^2), \\ T_2(x, y) &= 1 + (x - 1) + y + (x - 1)y + \frac{3}{2} y^2. \end{aligned}$$

STEP 4 (c) Quadratic term in matrix form:

$$\begin{aligned} \frac{1}{2} h^\top H_f(1, 0) h &= \frac{1}{2} \begin{bmatrix} x - 1 & y \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x - 1 \\ y \end{bmatrix} = \frac{1}{2} \begin{bmatrix} x - 1 & y \end{bmatrix} \begin{bmatrix} y \\ (x - 1) + 3y \end{bmatrix}, \\ &= \frac{1}{2} ((x - 1)y + (x - 1)y + 3y^2) = (x - 1)y + \frac{3}{2} y^2, \end{aligned}$$

which matches the quadratic part of T_2 .

STEP 5 (d) Check the gradient:

$$\nabla f(1, 0) = (1, 1)^\top \neq (0, 0)^\top,$$

so $(1, 0)$ is not a stationary point. The second-derivative test classifies only stationary points, so it does not apply. (For the record, $D = \det H_f(1, 0) = 0 \cdot 3 - 1^2 = -1 < 0$, which says the quadratic term is indefinite, but that does not make $(1, 0)$ a saddle: the linear term $(x - 1) + y$ dominates near the point.)

Answer. (a) $\nabla f = (e^y, xe^y + 2y)^\top$, $H_f = \begin{bmatrix} 0 & e^y \\ e^y & xe^y + 2 \end{bmatrix}$. (b) $T_2 = 1 + (x - 1) + y + (x - 1)y + \frac{3}{2}y^2$. (c) $\frac{1}{2}h^\top Hh = (x - 1)y + \frac{3}{2}y^2$. (d) No: $\nabla f(1, 0) = (1, 1)^\top \neq 0$, the point is not stationary, so the test does not apply.

Where marks are lost

Writing $f_{xx} = e^y$ (it is 0: f is linear in x). Missing the factor 2 on the cross term, giving $\frac{1}{2}(x - 1)y$. Answering “saddle” in (d) because $D < 0$.

Answers at a glance

(a) $\nabla f = (e^y, xe^y + 2y)$, $H = \begin{bmatrix} 0 & e^y \\ e^y & xe^y + 2 \end{bmatrix}$ (b) $1 + (x - 1) + y + (x - 1)y + \frac{3}{2}y^2$ (c) $(x - 1)y + \frac{3}{2}y^2$ (d) not applicable

Problem 54 · A quadratic function: Taylor is exact

Consider $f(x, y) = 2x^2 - 2xy + y^2 - 4x + 2y + 5$.

- Find the gradient and the Hessian, and the stationary point of f .
- Compute f at the stationary point and classify it.
- Write the second-order Taylor expansion of f about the stationary point in the matrix form $f(\mathbf{x}_0 + h) = f(\mathbf{x}_0) + \frac{1}{2}h^\top Hh$, and expand it.
- Explain, by computing the third derivatives, why this expansion is exact for all (x, y) , and verify it by expanding your answer to (c).

How to attempt this (the plan)

What it tests: stationary point of a quadratic (a 2×2 linear system), constant Hessian, and the fact that for a quadratic the Taylor polynomial equals the function. **Plan:** partials; solve the two linear equations; D test; assemble T_2 with the gradient term zero; expand and compare with f .

Worked solution

STEP 1 (a) Gradient and Hessian:

$$f_x = 4x - 2y - 4, \quad f_y = -2x + 2y + 2,$$

$$f_{xx} = 4, \quad f_{xy} = -2, \quad f_{yy} = 2 \Rightarrow H = \begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix} \text{ (constant).}$$

STEP 2 (a) Set the gradient to zero:

$$-2x + 2y + 2 = 0 \Rightarrow y = x - 1,$$

$$4x - 2(x - 1) - 4 = 0 \Rightarrow 2x - 2 = 0 \Rightarrow x = 1, \quad y = 0.$$

STEP 3 (b) Value and classification:

$$f(1, 0) = 2(1) - 0 + 0 - 4 + 0 + 5 = 3,$$

$$D = f_{xx}f_{yy} - f_{xy}^2 = 4 \cdot 2 - (-2)^2 = 8 - 4 = 4 > 0, \quad f_{xx} = 4 > 0,$$

so $(1, 0)$ is a local (in fact global) minimum.

STEP 4 (c) The gradient is zero at $(1, 0)$, so with $h = (x - 1, y)^\top$:

$$f(x, y) = 3 + \frac{1}{2} \begin{bmatrix} x-1 & y \end{bmatrix} \begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} x-1 \\ y \end{bmatrix},$$

$$= 3 + \frac{1}{2} (4(x-1)^2 - 4(x-1)y + 2y^2) = 3 + 2(x-1)^2 - 2(x-1)y + y^2.$$

STEP 5 (d) All third derivatives of a quadratic are zero: $f_{xxx} = f_{xxy} = f_{xyy} = f_{yyy} = 0$, so every Taylor term beyond h^2 vanishes and $T_2 = f$ exactly. Verify by expanding (c):

$$2(x-1)^2 = 2x^2 - 4x + 2, \quad -2(x-1)y = -2xy + 2y,$$

$$3 + 2x^2 - 4x + 2 - 2xy + 2y + y^2 = 2x^2 - 2xy + y^2 - 4x + 2y + 5 = f(x, y). \quad \checkmark$$

Answer. (a) $\nabla f = (4x - 2y - 4, -2x + 2y + 2)^\top$, $H = \begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix}$, stationary point $(1, 0)$.
 (b) $f(1, 0) = 3$; $D = 4 > 0$, $f_{xx} > 0$: local (global) minimum. (c) $f(x, y) = 3 + 2(x-1)^2 - 2(x-1)y + y^2$. (d) Third derivatives all zero $\Rightarrow$ expansion exact; expanding (c) reproduces f term by term.

Where marks are lost

Solving the linear system wrongly (check by substituting back into both equations). Writing a gradient term in (c) even though it is zero at a stationary point. Forgetting the $\frac{1}{2}$ (gives $4(x-1)^2 \dots$, which does not expand back to f).

Answers at a glance

(a) $(1, 0)$, $H = \begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix}$ (b) 3, minimum (c) $3 + 2(x-1)^2 - 2(x-1)y + y^2$ (d) exact

Problem 55 · Four stationary points, all three types

Let $f(x, y) = x^3 - 3x + y^3 - 12y$.

- Find all stationary points of f .
- For each stationary point compute the Hessian matrix and its determinant.
- Classify each stationary point.
- Write the second-order Taylor expansion of f about the stationary point that is a local maximum.

How to attempt this (the plan)

What it tests: finding all stationary points (two cases in each variable), evaluating H at each, the D test, and T_2 at a stationary point. **Plan:** $f_x = 0$ gives $x = \pm 1$, $f_y = 0$ gives $y = \pm 2$, so four points; H is diagonal here; tabulate.

Worked solution

STEP 1 (a) Solve each partial separately:

$$f_x = 3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1,$$

$$f_y = 3y^2 - 12 = 0 \Rightarrow y^2 = 4 \Rightarrow y = \pm 2.$$

Every combination satisfies both equations, so there are four points: $(1, 2)$, $(1, -2)$, $(-1, 2)$, $(-1, -2)$.

STEP 2 (b) Second partials:

$$f_{xx} = 6x, \quad f_{xy} = 0, \quad f_{yy} = 6y \Rightarrow H = \begin{bmatrix} 6x & 0 \\ 0 & 6y \end{bmatrix}, \quad D = 36xy.$$

Point	H	D	f_{xx}
$(1, 2)$	$\begin{bmatrix} 6 & 0 \\ 0 & 12 \end{bmatrix}$	72	6
$(1, -2)$	$\begin{bmatrix} 6 & 0 \\ 0 & -12 \end{bmatrix}$	-72	6
$(-1, 2)$	$\begin{bmatrix} -6 & 0 \\ 0 & 12 \end{bmatrix}$	-72	-6
$(-1, -2)$	$\begin{bmatrix} -6 & 0 \\ 0 & -12 \end{bmatrix}$	72	-6

STEP 3 (c) Apply the D test:

$(1, 2) : D > 0, f_{xx} > 0 \Rightarrow$ local minimum; $(-1, -2) : D > 0, f_{xx} < 0 \Rightarrow$ local maximum;

$(1, -2)$ and $(-1, 2) : D < 0 \Rightarrow$ saddle points.

(Equivalently, the eigenvalues are the diagonal entries: $(6, 12)$ min; $(-6, -12)$ max; mixed signs saddle.)

STEP 4 (d) Value at the maximum:

$$f(-1, -2) = (-1) + 3 + (-8) + 24 = 18.$$

STEP 5 (d) With $h = (x + 1, y + 2)^\top$ and the gradient zero there:

$$T_2 = 18 + \frac{1}{2}(-6(x + 1)^2 + 0 - 12(y + 2)^2) = 18 - 3(x + 1)^2 - 6(y + 2)^2.$$

Both squares enter with negative signs, as a maximum should.

Answer. (a) $(1, 2)$, $(1, -2)$, $(-1, 2)$, $(-1, -2)$. (b) $H = \text{diag}(6x, 6y)$, $D = 72, -72, -72, 72$ respectively. (c) $(1, 2)$ minimum; $(-1, -2)$ maximum; $(1, -2)$ and $(-1, 2)$ saddles. (d) $T_2(x, y) = 18 - 3(x + 1)^2 - 6(y + 2)^2$.

Where marks are lost

Listing only two of the four points. Arithmetic in $f(-1, -2)$ (the sign of $-3x$ at $x = -1$ is $+3$). Writing $(x - 1)$ instead of $(x + 1)$ in h .

Answers at a glance

- (a) four points $(\pm 1, \pm 2)$ (b) $D = 36xy$ (c) min $(1, 2)$, max $(-1, -2)$, saddles $(1, -2)$, $(-1, 2)$
 (d) $18 - 3(x+1)^2 - 6(y+2)^2$

Problem 56 · A mixed term: determinant route and eigenvalue route

Let $f(x, y) = x^2 + xy + y^2 - 3x$.

- (a) Find the stationary point and the value of f there.
 (b) Compute the Hessian and classify the point using $D = f_{xx}f_{yy} - f_{xy}^2$.
 (c) Compute the eigenvalues of the Hessian and confirm the classification.
 (d) Using the Taylor expansion at the stationary point, compute the exact change in f for the step $h = (1, -1)^\top$, and check it by direct evaluation.

How to attempt this (the plan)

What it tests: a stationary point that needs two simultaneous equations, both classification routes, and using $\frac{1}{2}h^\top Hh$ as an actual number. **Plan:** solve $2x + y = 3$, $x + 2y = 0$; D ; characteristic equation of a 2×2 ; compute the quadratic form for the given h .

Worked solution

STEP 1 (a) Stationary point:

$$f_x = 2x + y - 3 = 0, \quad f_y = x + 2y = 0.$$

$$\text{From } f_y: x = -2y. \quad \text{Into } f_x: -4y + y - 3 = 0 \Rightarrow y = -1, x = 2.$$

$$f(2, -1) = 4 - 2 + 1 - 6 = -3.$$

STEP 2 (b) Hessian and D :

$$f_{xx} = 2, \quad f_{xy} = 1, \quad f_{yy} = 2 \Rightarrow H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

$$D = 2 \cdot 2 - 1^2 = 3 > 0, \quad f_{xx} = 2 > 0 \Rightarrow \text{local minimum.}$$

STEP 3 (c) Eigenvalues:

$$\det(H - \lambda I) = (2 - \lambda)^2 - 1 = 0 \Rightarrow 2 - \lambda = \pm 1 \Rightarrow \lambda = 1, 3.$$

Both positive $\Rightarrow H$ positive definite $\Rightarrow$ minimum. (Check: $\lambda_1 \lambda_2 = 3 = D$.)

STEP 4 (d) Since f is quadratic, T_2 is exact and the gradient term is zero at $(2, -1)$:

$$\Delta f = \frac{1}{2}h^\top Hh = \frac{1}{2} \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{2}(1 + 1) = 1.$$

STEP 5 (d) Direct check: the new point is $(3, -2)$, and

$$f(3, -2) = 9 - 6 + 4 - 9 = -2, \quad \Delta f = -2 - (-3) = 1. \checkmark$$

Answer. (a) Stationary point $(2, -1)$, $f(2, -1) = -3$. (b) $H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $D = 3 > 0$, $f_{xx} > 0$: local minimum. (c) $\lambda = 1, 3$, both positive: minimum confirmed. (d) $\Delta f = 1$; direct evaluation gives $f(3, -2) - f(2, -1) = -2 - (-3) = 1$.

Where marks are lost

Solving only one equation. Writing $f_{xy} = 2$ (the cross term is xy , so $f_{xy} = 1$). Forgetting the $\frac{1}{2}$ in (d) (gives 2, which the direct check exposes).

Answers at a glance

(a) $(2, -1)$, $f = -3$ (b) $D = 3$, minimum (c) $\lambda = 1, 3$ (d) $\Delta f = 1$

Problem 57 · Read a Taylor expansion backwards

(a) The second-order Taylor expansion of a function g about a stationary point is

$$T_2(x, y) = 5 - 2(x - 1)^2 + 6(x - 1)(y + 2) - 2(y + 2)^2.$$

Read off the stationary point, g at that point, and the Hessian matrix there. Classify the point.

(b) At a stationary point of another function the Hessian is $H = \begin{bmatrix} a & 2 \\ 2 & 1 \end{bmatrix}$. Find all values of a for which the point is (i) a local minimum, (ii) a saddle, (iii) undecided by the test.

How to attempt this (the plan)

What it tests: whether you know what each piece of T_2 is. **Plan:** match $T_2 = g_0 + \frac{1}{2}[f_{xx}h_1^2 + 2f_{xy}h_1h_2 + f_{yy}h_2^2]$ term by term, remembering the $\frac{1}{2}$ and the 2 on the cross term; for (b), $D = a - 4$.

Worked solution

STEP 1 (a) Here $h = (x - 1, y + 2)$, so the stationary point is $(1, -2)$ and the constant is $g(1, -2) = 5$.

STEP 2 (a) Compare coefficients with $\frac{1}{2}f_{xx}h_1^2 + f_{xy}h_1h_2 + \frac{1}{2}f_{yy}h_2^2$:

$$\frac{1}{2}f_{xx} = -2 \Rightarrow f_{xx} = -4, \quad f_{xy} = 6, \quad \frac{1}{2}f_{yy} = -2 \Rightarrow f_{yy} = -4.$$

$$H = \begin{bmatrix} -4 & 6 \\ 6 & -4 \end{bmatrix}.$$

STEP 3 (a) Classify:

$$D = (-4)(-4) - 36 = 16 - 36 = -20 < 0 \Rightarrow \text{saddle}.$$

(Eigenvalues: $-4 \pm 6 = 2, -10$, opposite signs.)

STEP 4 (b) Determinant in terms of a :

$$D = a \cdot 1 - 2^2 = a - 4.$$

(i) Minimum needs $D > 0$ and $f_{xx} = a > 0$: $a > 4$ (which already gives $a > 0$).

(ii) Saddle needs $D < 0$: $a < 4$.

(iii) $D = 0$: $a = 4$.

A maximum is impossible since $f_{yy} = 1 > 0$.

Answer. (a) Point $(1, -2)$; $g(1, -2) = 5$; $H = \begin{bmatrix} -4 & 6 \\ 6 & -4 \end{bmatrix}$; $D = -20$: saddle. (b) (i) $a > 4$; (ii) $a < 4$; (iii) $a = 4$.

Where marks are lost

Reading $f_{xx} = -2$ (forgetting the $\frac{1}{2}$). Reading $f_{xy} = 3$ or 12 (the cross coefficient equals f_{xy} exactly, because $\frac{1}{2} \cdot 2f_{xy} = f_{xy}$). Reading the point as $(-1, 2)$.

Answers at a glance

(a) $(1, -2), 5, H = \begin{bmatrix} -4 & 6 \\ 6 & -4 \end{bmatrix}$, saddle (b) $a > 4$ min, $a < 4$ saddle, $a = 4$ undecided

Topic 12

Automatic Differentiation & Backpropagation

The forward pass on a computation graph, the local derivative of every node, the backward pass with the fan-out (sum) rule, diagnosing a recorded pass, and sigmoid or tanh neurons with gradients and one gradient-descent step.

Formulas you need for this topic

- Forward pass: compute every intermediate node, top to bottom, and write the numbers down.
- Local derivative on each edge: $\partial(\text{child})/\partial(\text{parent})$, evaluated at the forward values.
- Backward pass from $\partial f/\partial f = 1$: $\frac{\partial f}{\partial x_i} = \sum_{j: x_i \in \text{Pa}(x_j)} \frac{\partial f}{\partial x_j} \frac{\partial x_j}{\partial x_i}$ — a node with several children *adds* the contributions.
- $\sigma'(z) = \sigma(z)(1 - \sigma(z)) = a(1 - a)$; $\tanh'(z) = 1 - \tanh^2 z$; $\text{ReLU}'(z) = 1$ for $z > 0$, else 0.
- Neuron $z = \mathbf{w}^\top \mathbf{x} + b$: $\delta^{(a)} = \delta^{(a)} \frac{\partial a}{\partial z}$, then $\frac{\partial \text{Loss}}{\partial \mathbf{w}} = \delta^{(z)} \mathbf{x}$, $\frac{\partial \text{Loss}}{\partial b} = \delta^{(z)}$, $\frac{\partial \text{Loss}}{\partial \mathbf{x}} = \delta^{(z)} \mathbf{w}$.
- One gradient-descent step: $\mathbf{w} \leftarrow \mathbf{w} - \eta \frac{\partial \text{Loss}}{\partial \mathbf{w}}$, $b \leftarrow b - \eta \frac{\partial \text{Loss}}{\partial b}$.

Problem 58 · Pure chain (warm-up)

Consider $f(x) = (2x + 1)^3$ with intermediate variables

$$a = 2x, \quad b = a + 1, \quad f = b^3.$$

Using reverse-mode automatic differentiation at $x = 1$:

- Perform the forward pass and report a, b, f .
- Write the local derivative of every node with respect to its parent, evaluated at $x = 1$.
- Perform the backward pass: compute $\partial f/\partial b$, $\partial f/\partial a$ and df/dx .
- Verify df/dx by direct differentiation.

How to attempt this (the plan)

What it tests: the three-column ledger with no branching. **Plan:** (1) forward, top to bottom; (2) one-line derivative per edge, plug in numbers; (3) start from $\bar{f} = 1$ and multiply backwards; (4) differentiate $(2x + 1)^3$ by the chain rule as a check.

Worked solution

STEP 1 (a) Forward pass at $x = 1$.

Node	Formula	Value
a	$2x$	$2(1) = 2$
b	$a + 1$	$2 + 1 = 3$
f	b^3	$3^3 = 27$

STEP 2 (b) Local derivatives at $x = 1$.

Edge	Formula	Value
$\partial a / \partial x$	2	2
$\partial b / \partial a$	1	1
$\partial f / \partial b$	$3b^2$	$3(3)^2 = 27$

STEP 3 (c) Backward pass, starting from $\bar{f} = 1$.

Bar	Formula	Value
$\bar{f}$	seed	1
$\bar{b}$	$\bar{f} \cdot \partial f / \partial b$	$1 \cdot 27 = 27$
$\bar{a}$	$\bar{b} \cdot \partial b / \partial a$	$27 \cdot 1 = 27$
$\bar{x}$	$\bar{a} \cdot \partial a / \partial x$	$27 \cdot 2 = 54$

STEP 4 (d) Cross-check by the chain rule:

$$f'(x) = 3(2x + 1)^2 \cdot 2 = 6(2x + 1)^2, \quad f'(1) = 6 \cdot 9 = 54. \checkmark$$

Answer. (a) $a = 2$, $b = 3$, $f = 27$. (b) $\partial a / \partial x = 2$, $\partial b / \partial a = 1$, $\partial f / \partial b = 27$. (c) $\partial f / \partial b = 27$, $\partial f / \partial a = 27$, $df/dx = 54$. (d) 54, matching.

Where marks are lost

Writing $3b^2$ without evaluating it to 27 in part (b). Starting the backward pass from $\bar{x}$ instead of $\bar{f}$. Doing part (d) only and presenting it as part (c).

Answers at a glance

(a) 2, 3, 27 (b) 2, 1, 27 (c) 27, 27, 54 (d) 54

Problem 59 · Fan-out in the middle, with a reciprocal

Consider $f(x) = (2x)^2 + \frac{1}{2x}$ for $x > 0$, with

$$a = 2x, \quad b = a^2, \quad c = \frac{1}{a}, \quad f = b + c.$$

Node a feeds two children, b and c . At $x = 1$:

- Forward pass: report a, b, c, f .
- Local derivatives of every node with respect to each parent, at $x = 1$.
- Backward pass: $\partial f / \partial b$, $\partial f / \partial c$, $\partial f / \partial a$, and df/dx .

How to attempt this (the plan)

What it tests: the add rule at an internal node, and the sign of the derivative of $1/a$. **Plan:** forward; five local slopes (watch $-1/a^2$); bars of b, c ; then $\bar{a}$ is the sum of two terms with opposite signs; multiply once more for $\bar{x}$.

Worked solution

STEP 1 (a) Forward pass at $x = 1$.

Node	Formula	Value
a	$2x$	2
b	a^2	$2^2 = 4$
c	$1/a$	$1/2 = 0.5$
f	$b + c$	$4 + 0.5 = 4.5$

STEP 2 (b) Local derivatives at $x = 1$ (five edges).

Edge	Formula	Value
$\partial a / \partial x$	2	2
$\partial b / \partial a$	$2a$	4
$\partial c / \partial a$	$-1/a^2$	$-\frac{1}{4}$
$\partial f / \partial b$	1	1
$\partial f / \partial c$	1	1

STEP 3 (c) Backward pass from $\bar{f} = 1$.

Bar	Formula	Value
$\bar{f}$	seed	1
$\bar{b}$	$\bar{f} \cdot \partial f / \partial b$	$1 \cdot 1 = 1$
$\bar{c}$	$\bar{f} \cdot \partial f / \partial c$	$1 \cdot 1 = 1$
$\bar{a}$	$\bar{b} \cdot \partial b / \partial a + \bar{c} \cdot \partial c / \partial a$	$1 \cdot 4 + 1 \cdot (-\frac{1}{4}) = \frac{15}{4} = 3.75$
$\bar{x}$	$\bar{a} \cdot \partial a / \partial x$	$3.75 \cdot 2 = 7.5$

Node a has two children, so both contributions are added.**STEP 4** (c) Cross-check:

$$f = 4x^2 + \frac{1}{2}x^{-1}, \quad f' = 8x - \frac{1}{2x^2} = 8 - 0.5 = 7.5. \checkmark$$

Answer. (a) $a = 2$, $b = 4$, $c = 0.5$, $f = 4.5$. (b) 2, 4, $-\frac{1}{4}$, 1, 1. (c) $\partial f / \partial b = 1$, $\partial f / \partial c = 1$, $\partial f / \partial a = 3.75$, $df/dx = 7.5$.

Where marks are lost

Sign of $\partial c / \partial a$ (using $+\frac{1}{4}$ gives $\bar{a} = 4.25$, $df/dx = 8.5$). Dropping the c path (gives $\bar{a} = 4$, $df/dx = 8$).

Answers at a glance

(a) 2, 4, 0.5, 4.5 (b) 2, 4, $-\frac{1}{4}$, 1, 1 (c) 1, 1, 3.75, 7.5

Problem 60 · Diagnose a recorded backward pass

For the computation graph of the previous problem at $x = 1$, three students recorded the following final answers for df/dx . The correct value is 7.5.

	Student P	Student Q	Student R
$\partial f/\partial a$ reported	4	4.25	3.75
df/dx reported	8	8.5	3.75

For each student, identify the single mistake (which edge or which rule) that produces exactly their numbers.

How to attempt this (the plan)

What it tests: whether you can read a ledger backwards and locate a slip. **Plan:** compare each reported $\bar{a}$ with the correct sum $4 - \frac{1}{4}$, then compare each df/dx with $\bar{a} \cdot 2$.

Worked solution

STEP 1 **P** Correct $\bar{a} = 4 + (-\frac{1}{4}) = 3.75$. Reported $\bar{a} = 4$ means the c -path contribution $\bar{c} \cdot \partial c/\partial a = -\frac{1}{4}$ was omitted. Then $df/dx = 4 \cdot 2 = 8$ is consistent.

P forgot the second child of a (the edge $a \rightarrow c$); the add rule was not applied.

STEP 2 **Q** Reported $\bar{a} = 4.25 = 4 + \frac{1}{4}$: the c -path was included but with $\partial c/\partial a = +\frac{1}{4}$. Then $4.25 \cdot 2 = 8.5$ is consistent.

Q used the wrong sign for $\partial(1/a)/\partial a$; it is $-1/a^2 = -\frac{1}{4}$.

STEP 3 **R** $\bar{a} = 3.75$ is correct. Reported $df/dx = 3.75 = \bar{a} \cdot 1$, so the final edge $x \rightarrow a$ was multiplied by 1 instead of $\partial a/\partial x = 2$.

R stopped at $\bar{a}$ (or used $\partial a/\partial x = 1$); the last multiplication by 2 was missed.

STEP 4 **check** Each diagnosis reproduces the student's pair of numbers exactly:

Student	Slip	$\bar{a}$	df/dx
P	drop edge $a \rightarrow c$	4	$4 \cdot 2 = 8$ ✓
Q	$\partial c/\partial a = +\frac{1}{4}$	4.25	$4.25 \cdot 2 = 8.5$ ✓
R	$\partial a/\partial x = 1$	3.75	$3.75 \cdot 1 = 3.75$ ✓

Answer. P: missed the edge $a \rightarrow c$ (no add rule at a). Q: sign of $-1/a^2$. R: missed the last factor $\partial a/\partial x = 2$.

Where marks are lost

Naming a mistake that does not reproduce the student's numbers exactly. Each diagnosis must be checked arithmetically: compute $\bar{a}$ and $\bar{a} \cdot \partial a/\partial x$ under the proposed slip and confirm both numbers match.

Answers at a glance

P: missed edge $a \rightarrow c$. Q: sign of $-1/a^2$. R: missed $\partial a / \partial x = 2$.

Problem 61 · Sigmoid neuron with squared loss and one GD step

A sigmoid neuron computes $z = w^\top x + b$, $a = \sigma(z)$, and is trained with $\text{Loss} = (y - a)^2$. For one example,

$$w = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad b = 0, \quad y = 0.$$

- Compute z , a and Loss.
- Compute $\delta^{(a)} = \partial \text{Loss} / \partial a$ and $\delta^{(z)} = \partial \text{Loss} / \partial z$.
- Compute $\partial \text{Loss} / \partial w$, $\partial \text{Loss} / \partial b$ and $\partial \text{Loss} / \partial x$.
- Update w and b by one gradient-descent step with $\eta = 2$, and compute the new z . (You may use $\sigma(0) = 0.5$.)

How to attempt this (the plan)

What it tests: the neuron ledger end to end. **Plan:** forward three numbers; $\delta^{(a)} = 2(a - y)$ (mind the sign); $\delta^{(z)} = \delta^{(a)} a(1 - a)$; weights use x , input uses w , bias uses 1; subtract $\eta \times \text{gradient}$; recompute z to check the direction.

Worked solution

STEP 1 (a) Forward pass.

Node	Formula	Value
z	$w^\top x + b$	$(1)(1) + (1)(-1) + 0 = 0$
a	$\sigma(z)$	$\sigma(0) = 0.5$
Loss	$(y - a)^2$	$(0 - 0.5)^2 = 0.25$

STEP 2 (b) Error signals.

$$\delta^{(a)} = \frac{\partial}{\partial a} (y - a)^2 = 2(y - a)(-1) = 2(a - y) = 2(0.5 - 0) = 1$$

$$\sigma'(z) = a(1 - a) = 0.5 \times 0.5 = 0.25$$

$$\delta^{(z)} = \delta^{(a)} \cdot \sigma'(z) = 1 \times 0.25 = 0.25$$

STEP 3 (c) Gradients (weights use x , bias uses 1, input uses w).

$$\frac{\partial \text{Loss}}{\partial w} = \delta^{(z)} x = 0.25 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.25 \\ -0.25 \end{bmatrix}$$

$$\frac{\partial \text{Loss}}{\partial b} = \delta^{(z)} \cdot 1 = 0.25$$

$$\frac{\partial \text{Loss}}{\partial x} = \delta^{(z)} w = 0.25 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.25 \\ 0.25 \end{bmatrix}$$

STEP 4 (d) Gradient-descent step with $\eta = 2$ (minus sign!).

$$w_{\text{new}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} 0.25 \\ -0.25 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 1.5 \end{bmatrix}$$

$$b_{\text{new}} = 0 - 2(0.25) = -0.5$$

$$z_{\text{new}} = (0.5)(1) + (1.5)(-1) - 0.5 = -1.5$$

So a falls from 0.5 towards $y = 0$; the step helps.

Answer. (a) $z = 0$, $a = 0.5$, Loss = 0.25. (b) $\delta^{(a)} = 1$, $\delta^{(z)} = 0.25$. (c) $\partial \text{Loss} / \partial w = (0.25, -0.25)^\top$, $\partial \text{Loss} / \partial b = 0.25$, $\partial \text{Loss} / \partial x = (0.25, 0.25)^\top$. (d) $w_{\text{new}} = (0.5, 1.5)^\top$, $b_{\text{new}} = -0.5$, $z_{\text{new}} = -1.5$.

Where marks are lost

$\delta^{(a)} = -1$ (sign). Swapping x and w in (c). Using $+\eta$. Halving everything by silently using $\frac{1}{2}(y - a)^2$.

Answers at a glance

(a) 0, 0.5, 0.25 (b) 1, 0.25 (c) $(0.25, -0.25)^\top$, 0.25, $(0.25, 0.25)^\top$ (d) $(0.5, 1.5)^\top$, -0.5, $z_{\text{new}} = -1.5$

Problem 62 · Tanh neuron with an incoming error signal

A hidden neuron computes $z = w^\top x + b$ and $h = \tanh(z)$. For one example,

$$w = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \quad x = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad b = 0,$$

and the error signal arriving from the next layer is $\delta^{(h)} = \partial \text{Loss} / \partial h = 5$.

- Show that $\frac{d}{dz} \tanh z = 1 - \tanh^2 z$, starting from $\tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}}$.
- Compute z , h , and $\partial h / \partial z$.
- Compute $\delta^{(z)}$, $\partial \text{Loss} / \partial w$, $\partial \text{Loss} / \partial b$.
- Compute $\partial \text{Loss} / \partial x$, the signal passed to the previous layer.

How to attempt this (the plan)

What it tests: that changing the activation changes exactly one local slope and nothing else; a short derivation. **Plan:** quotient rule once; forward; multiply $\delta^{(h)}$ by the tanh slope; then the same three gradients as always.

Worked solution

STEP 1 (a) Write $N = e^z - e^{-z}$ and $D = e^z + e^{-z}$, so $\tanh z = N/D$. Note $N' = e^z + e^{-z} = D$ and $D' = e^z - e^{-z} = N$.

$$\frac{d}{dz} \frac{N}{D} = \frac{N'D - ND'}{D^2} = \frac{D \cdot D - N \cdot N}{D^2} = 1 - \frac{N^2}{D^2} = 1 - \tanh^2 z.$$

STEP 2 (b) Forward pass.

Node	Formula	Value
z	$w^\top x + b$	$(3)(1) + (-1)(3) + 0 = 0$
h	$\tanh z$	$\tanh 0 = 0$
$\partial h / \partial z$	$1 - h^2$	$1 - 0 = 1$

STEP 3 (c) Backward to the parameters.

$$\delta^{(z)} = \delta^{(h)} \cdot \frac{\partial h}{\partial z} = 5 \times 1 = 5$$

$$\frac{\partial \text{Loss}}{\partial w} = \delta^{(z)} x = 5 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 15 \end{bmatrix}, \quad \frac{\partial \text{Loss}}{\partial b} = \delta^{(z)} = 5$$

STEP 4 (d) Backward to the input.

$$\frac{\partial \text{Loss}}{\partial x} = \delta^{(z)} w = 5 \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 15 \\ -5 \end{bmatrix}$$

Answer. (a) $\tanh'(z) = 1 - \tanh^2 z = 1 - h^2$. (b) $z = 0$, $h = 0$, $\partial h / \partial z = 1$. (c) $\delta^{(z)} = 5$, $\partial \text{Loss} / \partial w = (5, 15)^\top$, $\partial \text{Loss} / \partial b = 5$. (d) $\partial \text{Loss} / \partial x = (15, -5)^\top$.

Where marks are lost

Using $h(1 - h)$ (the sigmoid slope) for $\tanh$. Swapping x and w in (c)/(d). Forgetting that $\delta^{(h)}$ is given and trying to differentiate a loss that is not specified.

Answers at a glance

(a) $1 - h^2$ (b) 0, 0, 1 (c) 5, $(5, 15)^\top$, 5 (d) $(15, -5)^\top$